

My Walk Through Combinatorics

MAT344 Portfolio

Mohsin Muzammil

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CHAPTER 1

THE PIGEONHOLE PRINCIPLE

1.1 The Statement

Result: The Pigeonhole Principle

Let n and k be positive integers such that $n > k$. Suppose we want to distribute n identical balls into k identical boxes. Then there will be at least one box in which we place at least two balls.

Proof

Suppose for contradiction that each of the m boxes has at most 1 ball. Then all the boxes together can hold at most k balls. As $k < n$, this contradicts the fact that we distributed n balls. $\square$

Result: The General Pigeonhole Principle

Let n, m , and r be positive integers so that $n > rm$, and let us distribute n identical balls into m identical boxes. Then there will be at least one box into which we place at least $r + 1$ balls.

Proof

Suppose for contradiction that each of the m boxes has at most r balls. Then all the boxes together can hold at most rm balls. As $rm < n$, this contradicts the fact that we distributed n balls. $\square$

1.2 Variations

Problem

If n or more pigeons are distributed among $k > 0$ pigeonholes, then at least one pigeonhole contains at least $\lceil \frac{n}{k} \rceil$ pigeons.

Proof

Suppose for contradiction that each pigeonhole contains at most $\lceil \frac{n}{k} \rceil - 1$ pigeons. Then the total number of pigeons is at most $k(\lceil \frac{n}{k} \rceil - 1) < k(\frac{n}{k}) = n$. This contradicts the fact that there were n pigeons distributed. $\square$

Problem: The Pigeonhole Principle for Real Numbers

Prove that if the sum of k real numbers is n , then there must be one of them which is greater than or equal to $\frac{n}{k}$.

Proof

Suppose for contradiction that each of the k real numbers is less than $\frac{n}{k}$. Then the sum of the k real numbers is less than $k \cdot \frac{n}{k} = n$. This contradicts the fact that the sum of the k real numbers is n . $\square$

1.3 Problems

Problem: 1967

Prove that the sequence 1967, 19671967, 196719671967, ... contains an element that is divisible by 1969.

Proof

Define the sequence $\{a_n\}$ where $a_1 = 1967, a_2 = 19671967, a_3 = 196719671967, \dots$ and suppose for contradiction that no element of the sequence is divisible by 1969.

Take the first 1969 elements and divide each of them by 1969. As none of the elements in the sequence are divisible by 1969, it follows that the possible values for the remainders are $\{1, 2, \dots, 1968\}$. Since there are 1969 numbers and only 1968 possible remainders, it follows, by the Pigeonhole Principle, that two elements of the sequence, a_i and a_j (where $i < j$), have the same remainder when divided by 1969. It follows that their difference, $a_j - a_i$, is divisible by 1969.

Observe that $a_j - a_i$ consists of $j - i$ 1967s followed by $4i$ 0s. In other words,

$$a_j - a_i = a_{j-i} \cdot 10^{4i}$$

As 10^{4i} is relatively prime to 1969, it must be the case that a_{j-i} is divisible by 1969. This contradicts the assumption that no element of the sequence is divisible by 1969.

Therefore the sequence 1967, 19671967, 196719671967, ... contains an element that is divisible by 1969, as required. $\square$

Remark. We ended up proving an even stronger statement. Namely, one of the first 1969 elements of the sequence is divisible by 1969.

Remark. The 1968 possible values for the remainders were the pigeonholes and the first 1969 elements of the sequence were the pigeons. As there were more pigeons than pigeonholes, the Pigeonhole Principle applied. An interesting point to note is that the application of the Pigeonhole Principle only helped us conclude that $a_j - a_i$ is divisible by 1969, which is not actually an element of the sequence.

Problem

Prove that among any seven real numbers $y_1, y_2, \dots, y_7$ there is a pair, y_i and y_j , such that

$$0 \leq \frac{y_i - y_j}{1 + y_i y_j} \leq \frac{1}{\sqrt{3}}$$

Proof

Let $y_1, y_2, \dots, y_7$ be seven real numbers. Define the pigeons to be $x_k = \arctan(y_k)$ for $k \in \{1, 2, \dots, 7\}$. Notice that for all $k \in \{1, 2, \dots, 7\}$, we have $x_k \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

Let the pigeonholes be the following six intervals:

$$\left(-\frac{\pi}{2}, -\frac{\pi}{3}\right], \left(-\frac{\pi}{3}, -\frac{\pi}{6}\right], \left(-\frac{\pi}{6}, 0\right], \left(0, \frac{\pi}{6}\right], \left(\frac{\pi}{6}, \frac{\pi}{3}\right], \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$$

(Notice that we have divided the interval $(-\frac{\pi}{2}, \frac{\pi}{2})$ into six smaller intervals, each having length $\frac{\pi}{6}$).

By the Pigeonhole Principle, some pigeonhole contains two numbers, x_i and x_j (where $x_j \leq x_i$).

Hence,

$$\begin{aligned} 0 &\leq x_i - x_j \leq \frac{\pi}{6} \\ \tan(0) &\leq \tan(x_i - x_j) \leq \tan\left(\frac{\pi}{6}\right) \end{aligned}$$

Now using the facts that $\tan(a - b) = \frac{\tan(a) - \tan(b)}{1 + \tan(a)\tan(b)}$ and $\tan(x_k) = \tan(\arctan(y_k)) = y_k$ we get,

$$\begin{aligned} 0 &\leq \frac{\tan(x_i) - \tan(x_j)}{1 + \tan(x_i)\tan(x_j)} \leq \frac{1}{\sqrt{3}} \\ 0 &\leq \frac{y_i - y_j}{1 + y_i y_j} \leq \frac{1}{\sqrt{3}} \end{aligned}$$

as required. □

Problem

Prove that there exists a positive integer n so that $44^n - 1$ is divisible by 7.

Proof

Let n be a positive integer and consider the remainder when $44^n - 1$ is divided by 7. There are only seven possible values for this remainder. Therefore, by the Pigeonhole Principle, there exists positive integers $k_1 < k_2$ such that both $44^{k_1} - 1$ and $44^{k_2} - 1$ have the same remainder r when divided by 7. In other terms,

$$44^{k_1} - 1 = 7m_1 + r$$

$$44^{k_2} - 1 = 7m_2 + r$$

for some positive integers $m_1 < m_2$. Subtracting the first equation from the second gives

$$44^{k_2} - 44^{k_1} = 7m_2 - 7m_1$$

$$44^{k_1}(44^{k_2-k_1} - 1) = 7(m_2 - m_1)$$

As the RHS is divisible by 7, so must be the LHS. Since 44^{k_1} is relatively prime to 7, we must have that $(44^{k_2-k_1} - 1)$ is divisible by 7 (where $k_2 - k_1$ is a positive integer), as required. $\square$

1.3.1 Subset Sum

Problem

Prove that if seven distinct numbers are selected from $\{1, 2, \dots, 11\}$, then two of them sum to 12.

Proof

Let the pigeons be the seven distinct numbers selected and let the pigeonholes be the following six sets: $\{1, 11\}, \{2, 10\}, \{3, 9\}, \{4, 8\}, \{5, 7\}, \{6\}$. By the Pigeonhole Principle, some pigeonhole contains two numbers. These two numbers sum to 12, by way of construction of the pigeonholes. $\square$

Remark. Seven is the best possible as it is possible to select six numbers from $\{1, 2, \dots, 11\}$ so that no two selected numbers sum to 12. One example of six such numbers is 1, 2, 3, 4, 5, 6.

Problem: Generalization

Prove that if $n + 1$ distinct numbers are selected from $\{1, 2, \dots, 2n - 1\}$, then two of them sum to $2n$.

Proof

Let the pigeons be the $n + 1$ distinct numbers selected and let the pigeonholes be the following n sets: $\{1, 2n - 1\}, \{2, 2n - 2\}, \{3, 2n - 3\}, \dots, \{\lceil \frac{2n-1}{2} \rceil\}$. Since $\lceil \frac{2n-1}{2} \rceil = \lceil n - \frac{1}{2} \rceil = n < n + 1$, it follows, by the Pigeonhole Principle, that some pigeonhole contains two numbers. These two numbers sum to $2n$, by way of construction of the pigeonholes. $\square$

Remark. $n + 1$ is the best possible as it is possible to select n numbers from $\{1, 2, \dots, 2n - 1\}$ so that no two selected numbers sum to $2n$. One example of n such numbers is $1, 2, 3, \dots, \lceil \frac{2n-1}{2} \rceil = n$.

Generalize. Formulate and prove similar statements for subsets of numbers selected from $\{1, 2, \dots, 2n\}$.

Problem

Prove that if six distinct numbers are selected from $\{1, 2, \dots, 10\}$, then two of them sum to 11.

Proof

Let the pigeons be the six distinct numbers selected and let the pigeonholes be the following five sets: $\{1, 10\}, \{2, 9\}, \{3, 8\}, \{4, 7\}, \{5, 6\}$. By the Pigeonhole Principle, some pigeonhole contains two numbers. These two numbers sum to 11, by way of construction of the pigeonholes. $\square$

Remark. Six is the best possible as it is possible to select five numbers from $\{1, 2, \dots, 10\}$ so that no two selected numbers sum to 11. One example of five such numbers is 1, 2, 3, 4, 5.

Problem

Prove that if $n + 1$ distinct numbers are selected from $\{1, 2, \dots, 2n\}$, then two of them sum to $2n + 1$.

Proof

Let the pigeons be the $n + 1$ distinct numbers selected and let the pigeonholes be the following n sets: $\{1, 2n\}, \{2, 2n - 1\}, \{3, 2n - 2\}, \dots, \{n, n + 1\}$. Since $n < n + 1$, it follows, by the Pigeonhole Principle, that some pigeonhole contains two numbers. These two numbers sum to $2n + 1$, by way of construction of the pigeonholes. $\square$

Remark. $n + 1$ is the best possible as it is possible to select n numbers from $\{1, 2, \dots, 2n\}$ so that no two selected numbers sum to $2n + 1$. One example of n such numbers is 1, 2, 3, $\dots$, n .

1.3.2 Same Sum Subsets

Problem

For a subset $X \subseteq \{1, 2, 3, \dots, 9\}$, define $\phi(X) = \sum_{x \in X} x$. Prove that among any 26 subsets of $\{1, 2, 3, \dots, 9\}$ each having size at most 3, there are subsets A and B such that $\phi(A) = \phi(B)$.

Proof

Let $A \subseteq \{1, 2, 3, \dots, 9\}$ with $|A| \leq 3$. The smallest possible value of $\phi(A) = 0$ (when $A = \{\}$) and the largest possible value of $\phi(A) = 24$ (when $A = \{7, 8, 9\}$). Since there are 25 possible values for $\phi(A)$ and 26 subsets are selected, it follows, by the Pigeonhole Principle, that two of the selected subsets have the same sum. $\square$

Problem: Generalization

Prove that among any $3n - 1$ subsets of $\{1, 2, \dots, n\}$ ($n \geq 3$) each having size at most 3, there are subsets A and B such that $\phi(A) = \phi(B)$.

Proof

Let $A \subseteq \{1, 2, 3, \dots, n\}$ with $|A| \leq 3$. The smallest possible value of $\phi(A) = 0$ (when $A = \{\}$) and the largest possible value of $\phi(A) = n + (n - 1) + (n - 2) = 3n - 3$ (when $A = \{n - 2, n - 1, n\}$). Since there are $3n - 2$ possible values for $\phi(A)$ and $3n - 1$ subsets are selected, it follows, by the Pigeonhole Principle, that two of the selected subsets have the same sum. $\square$

Problem: Further Generalization

Prove that among any $k \cdot n - \frac{k(k-1)}{2} + 2$ subsets of $\{1, 2, \dots, n\}$ ($n \geq k$) each having size at most k , there are subsets A and B such that $\phi(A) = \phi(B)$.

Proof

Let $A \subseteq \{1, 2, 3, \dots, n\}$ with $|A| \leq k$. The smallest possible value of $\phi(A) = 0$ (when $A = \{\}$) and the largest possible value of $\phi(A) = n + (n - 1) + (n - 2) + \dots + (n - k + 1) = k \cdot n - \frac{k(k-1)}{2}$ (when $A = \{n, n - 1, n - 2, \dots, n - k + 1\}$). Since there are $k \cdot n - \frac{k(k-1)}{2} + 1$ possible values for $\phi(A)$ and $k \cdot n - \frac{k(k-1)}{2} + 2$ subsets are selected, it follows, by the Pigeonhole Principle, that two of the selected subsets have the same sum. $\square$

Problem

How many subsets of $\{1, 2, \dots, 10\}$, each containing at least three and at most five elements, must be selected in order to guarantee that the selection contains subsets A , B , and C such that $\phi(A) = \phi(B) = \phi(C)$.

Solution

Let $A \subseteq \{1, 2, \dots, 10\}$ with $3 \leq |A| \leq 5$. The smallest possible value of $\phi(A) = 6$ (when $A = \{1, 2, 3\}$) and the largest possible value of $\phi(A) = 40$ (when $A = \{6, 7, 8, 9, 10\}$). Since there are 35 possible values for $\phi(A)$, select $2 \cdot 35 + 1 = 71$ subsets, then it follows, by the Generalized Pigeonhole Principle, that three of the selected subsets have the same sum.

Problem: Generalization

How many subsets of $\{1, 2, \dots, n\}$, each containing at least three and at most five elements, must be selected in order to guarantee that the selection contains subsets A , B , and C such that $\phi(A) = \phi(B) = \phi(C)$.

Solution

Let $A \subseteq \{1, 2, \dots, n\}$ with $3 \leq |A| \leq 5$. The smallest possible value of $\phi(A) = 6$ (when $A = \{1, 2, 3\}$) and the largest possible value of $\phi(A) = n + (n - 1) + (n - 2) + (n - 3) + (n - 4) = 5n - 10$ (when $A = \{n, n - 1, n - 2, n - 3, n - 4\}$). Since there are $5n - 10 - 6 + 1 = 5n - 15$ possible values for $\phi(A)$, select $2 \cdot (5n - 15) + 1 = 10n - 29$ subsets, then it follows, by the Generalized Pigeonhole Principle, that three of the selected subsets have the same sum.

Problem: (IMO 1972)

Prove that from ten distinct two-digit numbers, one can always choose two disjoint nonempty subsets, so that their elements have the same sum.

Proof

Let S be a set of ten distinct two-digit numbers. Then S has $2^{10} - 1 = 1023$ non-empty subsets (pigeons). Let $A \subseteq S$ then $10 \leq \phi(A) \leq 945$, and so there are 936 possible sums for the subsets (pigeonholes).

As there are more subsets than possible sums, it follows, by the Pigeonhole Principle, that two nonempty subsets, A_1 and A_2 , have the same sum. If $A_1 \cap A_2 = \{\}$ then we are done. Otherwise, we can remove the common elements (which reduces the totals by the same amount) to get two disjoint subsets with the same sum. $\square$

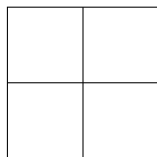
1.3.3 Points in Square

Problem

Prove that if five points are selected from the interior of a 1×1 square, then there are two points within $\frac{\sqrt{2}}{2}$ of each other.

Proof

Let the pigeons be the five selected points and let the pigeonholes be the following four sub-squares of the 1×1 square:



By the Pigeonhole Principle, some pigeonhole contains two points. The distance between these two points is at most the length of a diagonal of a sub-square, which is $\frac{\sqrt{2}}{2}$. $\square$

Problem: Generalization

Let $n \geq 1$. Prove that if $n^2 + 1$ points are selected from the interior of a 1×1 square, then there are two points within $\frac{\sqrt{2}}{n}$ of each other.

Proof

Divide the 1×1 square into n^2 sub-squares of equal area.

Let the pigeons be the $n^2 + 1$ selected points and let the pigeonholes be the n^2 sub-squares of the 1×1 square. By the Pigeonhole Principle, some pigeonhole contains two points. The distance between these two points is at most the length of a diagonal of a sub-square, which is $\frac{\sqrt{2}}{n}$. $\square$

Going Beyond.

Two natural questions follow:

- If $n \geq 2$ points are selected from the interior of a 1×1 square, how close can we guarantee two of them to be?
- Given a distance $d > 0$, how many points should we select from the interior of a 1×1 square so that two of them are within a distance d of each other?

We will answer these questions by dividing the 1×1 square into k^2 sub-squares of equal area.

Claim. If $n \geq 2$ points are selected from the interior of a 1×1 square, then two of them will be within $\frac{\sqrt{2}}{k}$ of each other where k is the largest positive integer such that $\frac{n-1}{k^2} \geq 1$.

Examples.

- If $n = 2$ points are selected then compute $\frac{1}{k^2}$ for $k \in \{1, 2, \dots\}$. In this case, $k = 1$ is the largest positive integer so that $\frac{n-1}{k^2} \geq 1$. So, the two points will be within $\frac{\sqrt{2}}{1} = \sqrt{2}$ of each other.

$\vdots$

- If $n = 4$ points are selected then compute $\frac{3}{k^2}$ for $k \in \{1, 2, \dots\}$. In this case, $k = 1$ is the largest positive integer so that $\frac{n-1}{k^2} \geq 1$. So, two of the four points will be within $\frac{\sqrt{2}}{1} = \sqrt{2}$ of each other.
- If $n = 5$ points are selected then compute $\frac{4}{k^2}$ for $k \in \{1, 2, \dots\}$. In this case, $k = 2$ is the largest positive integer so that $\frac{n-1}{k^2} \geq 1$. So, two of the five points will be within $\frac{\sqrt{2}}{2}$ of each other.

$\vdots$

- If $n = 9$ points are selected then compute $\frac{8}{k^2}$ for $k \in \{1, 2, \dots\}$. In this case, $k = 2$ is the largest positive integer so that $\frac{n-1}{k^2} \geq 1$. So, two of the nine points will be within $\frac{\sqrt{2}}{2}$ of each other.
- If $n = 10$ points are selected then compute $\frac{9}{k^2}$ for $k \in \{1, 2, \dots\}$. In this case, $k = 3$ is the largest positive integer so that $\frac{n-1}{k^2} \geq 1$. So, two of the ten points will be within $\frac{\sqrt{2}}{3}$ of each other.

$\vdots$

- If $n = 16$ points are selected then compute $\frac{15}{k^2}$ for $k \in \{1, 2, \dots\}$. In this case, $k = 3$ is the largest positive integer so that $\frac{n-1}{k^2} \geq 1$. So, two of the sixteen points will be within $\frac{\sqrt{2}}{3}$ of each other.
- If $n = 17$ points are selected then compute $\frac{16}{k^2}$ for $k \in \{1, 2, \dots\}$. In this case, $k = 4$ is the largest positive integer so that $\frac{n-1}{k^2} \geq 1$. So, two of the seventeen points will be within $\frac{\sqrt{2}}{4}$ of each other.

$\vdots$

Proof

Suppose for contradiction that $n \geq 2$ points are selected from the interior of a 1×1 square such that no two of them are within $\frac{\sqrt{2}}{k}$ of each other where k is the largest positive integer such that $\frac{n-1}{k^2} \geq 1$. Divide the 1×1 square into k^2 sub-squares of equal area. Let the pigeons be the n selected points and let the k^2 sub-squares be the pigeonholes. By the Pigeonhole Principle, at least one of these sub-squares contains at least $\lceil \frac{n}{k^2} \rceil$ points. The distance between these $\lceil \frac{n}{k^2} \rceil$ points is at most the length of a diagonal of a sub-square, which is $\frac{\sqrt{2}}{k}$. This contradicts the assumption that no two points are within $\frac{\sqrt{2}}{k}$. $\square$

Claim. If given a distance $d > 0$, then selecting $k^2 + 1$ points, where $k = \lceil \frac{\sqrt{2}}{d} \rceil$, from the interior of a 1×1 square will result in two of the points being within a distance d of each other.

Examples.

- If $d = \sqrt{2}$ then compute $k = \lceil \frac{\sqrt{2}}{\sqrt{2}} \rceil = 1$ and so select $1^2 + 1 = 2$ points.
- If $d = 1$ then compute $k = \lceil \frac{\sqrt{2}}{1} \rceil = 2$ and so select $2^2 + 1 = 5$ points.
- If $d = 0.72$ then compute $k = \lceil \frac{\sqrt{2}}{0.72} \rceil = 2$ and so select $2^2 + 1 = 5$ points.
- If $d = 0.68$ then compute $k = \lceil \frac{\sqrt{2}}{0.68} \rceil = 3$ and so select $3^2 + 1 = 10$ points.
- If $d = 0.5$ then compute $k = \lceil \frac{\sqrt{2}}{0.5} \rceil = 3$ and so select $3^2 + 1 = 10$ points.
- If $d = 0.45$ then compute $k = \lceil \frac{\sqrt{2}}{0.45} \rceil = 4$ and so select $4^2 + 1 = 17$ points.

Proof

Let $d > 0$ be a given distance, and let $k = \lceil \frac{\sqrt{2}}{d} \rceil$. Select $k^2 + 1$ points and suppose for contradiction that no two points are within a distance of d of each other. Consider dividing the 1×1 square into k^2 sub-squares of equal area.

Let the pigeons be the $k^2 + 1$ selected points and let the k^2 sub-squares be the pigeonholes. It follows, by the Pigeonhole Principle, that some pigeonhole contains two points. The distance between these two points is at most the length of a diagonal of a sub-square, which is $\frac{\sqrt{2}}{k}$. Since no two points are within a distance of d of each other, we must have

$$d < \frac{\sqrt{2}}{k} \iff k < \frac{\sqrt{2}}{d}$$

which contradicts the fact that $k = \lceil \frac{\sqrt{2}}{d} \rceil$.

Therefore, it must be the case that if given a distance $d > 0$, then selecting $k^2 + 1$ points, where $k = \lceil \frac{\sqrt{2}}{d} \rceil$, from the interior of a 1×1 square will result in two of the points being within a distance d of each other. $\square$

Improvement.

In my previous attempt of this problem I came up with the following results

- If $n \geq 2$ points are selected from the interior of a 1×1 square, then two of them will be within $\frac{\sqrt{2}}{2^k}$ of each other where k is the largest non-negative integer such that $\frac{n-1}{4^k} \geq 1$.
- If $4^n + 1$ points are selected from the interior of a 1×1 square, then two of them will be within $\frac{\sqrt{2}}{2^n}$ of each other
- If given a distance $d > 0$, then selecting $4^k + 1$ points, where $k = \lceil \log_2 \left(\frac{\sqrt{2}}{d} \right) \rceil$, from the interior of a 1×1 square will result in two of the points being within a distance d of each other.

In short, my idea was to divide the 1×1 square into 4^k sub-squares.

I ended up realizing that the 1×1 square could be divided into k^2 sub-squares, and so I went back and refined my results. This led to better and simpler looking conclusions.

CHAPTER 2

INDUCTION

2.1 Bounding the Fibonacci Sequence

Problem

Define $\{a_n\}_{n=0}^{\infty}$ where $a_0 = 0$, $a_1 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for all $n \geq 2$. Prove that $a_n \leq 2^n$ for all $n \geq 0$.

Proof

We proceed by strong induction. For the base cases we have,

- if $n = 0$ then $a_0 = 0$ (given) and $a_0 = 0 \leq 1 = 2^0$.
- if $n = 1$ then $a_1 = 1$ (given) and $a_1 = 1 \leq 2 = 2^1$.

Now suppose for all $n \in \{0, 1, 2, \dots, k\}$ where $k \geq 1$, we have that $a_n \leq 2^n$. Consider $n = k + 1$,

$$\begin{aligned} a_{k+1} &= a_k + a_{k-1} \\ &\leq 2^k + 2^{k-1} && \text{(by I.H.)} \\ &= 2 \cdot 2^{k-1} + 2^{k-1} \\ &= 3 \cdot 2^{k-1} \\ &\leq 4 \cdot 2^{k-1} \\ &= 2^{k+1} \end{aligned}$$

which shows $a_{k+1} \leq 2^{k+1}$.

Therefore, by the Principle of Mathematical Induction, $a_n \leq 2^n$ for all $n \geq 0$. □

Problem: Tightest Upper Bound

Find the smallest value M such that $a_n \leq M^n$ for all $n \geq 0$.

We will experiment with a couple of different values and try to make an observation.

Let us try $M = 1.9$:

For the base cases we have,

- if $n = 0$ then $a_0 = 0$ (given) and $a_0 = 0 \leq 1 = 1.9^0$.
- if $n = 1$ then $a_1 = 1$ (given) and $a_1 = 1 \leq 1.9 = 1.9^1$.

Now suppose for all $n \in \{0, 1, 2, \dots, k\}$ where $k \geq 1$, we have that $a_n \leq 1.9^n$. Consider $n = k + 1$,

$$\begin{aligned}
 a_{k+1} &= a_k + a_{k-1} \\
 &\leq 1.9^k + 1.9^{k-1} && \text{(by I.H.)} \\
 &= 1.9 \cdot 1.9^{k-1} + 1.9^{k-1} \\
 &= 2.9 \cdot 1.9^{k-1} \\
 &\leq 1.9^2 \cdot 1.9^{k-1} && (1.9 + 1 \leq 1.9^2) \\
 &= 1.9^{k+1}
 \end{aligned}$$

which shows $a_{k+1} \leq 1.9^{k+1}$.

Therefore, by the Principle of Mathematical Induction, $a_n \leq 1.9^n$ for all $n \geq 0$.

Let us try $M = 1.8$:

For the base cases we have,

- if $n = 0$ then $a_0 = 0$ (given) and $a_0 = 0 \leq 1 = 1.8^0$.
- if $n = 1$ then $a_1 = 1$ (given) and $a_1 = 1 \leq 1.8 = 1.8^1$.

Now suppose for all $n \in \{0, 1, 2, \dots, k\}$ where $k \geq 1$, we have that $a_n \leq 1.8^n$. Consider $n = k + 1$,

$$\begin{aligned}
 a_{k+1} &= a_k + a_{k-1} \\
 &\leq 1.8^k + 1.8^{k-1} && \text{(by I.H.)} \\
 &= 1.8 \cdot 1.8^{k-1} + 1.8^{k-1} \\
 &= 2.8 \cdot 1.8^{k-1} \\
 &\leq 1.8^2 \cdot 1.8^{k-1} && (1.8 + 1 \leq 1.8^2) \\
 &= 1.8^{k+1}
 \end{aligned}$$

which shows $a_{k+1} \leq 1.8^{k+1}$.

Therefore, by the Principle of Mathematical Induction, $a_n \leq 1.8^n$ for all $n \geq 0$.

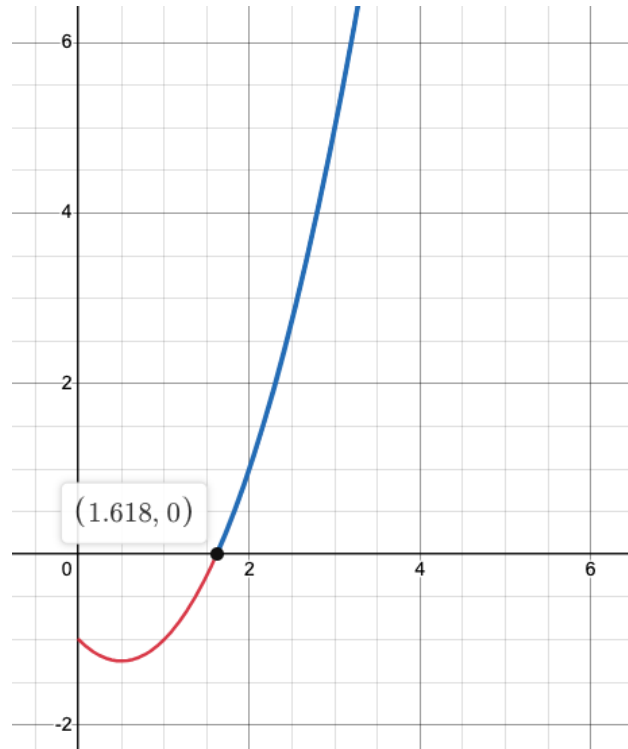
Notice that for the proof to work, we must have $M + 1 \leq M^2$. Let us analyze this inequality further.

$$M + 1 \leq M^2 \iff 0 \leq M^2 - M - 1$$

By the quadratic formula, the roots of the function $f(M) = M^2 - M - 1$ are

$$M = \frac{1 \pm \sqrt{1 - 4(1)(-1)}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

and from the diagram



we see that

$$M^2 - M - 1 \geq 0 \implies M \geq \frac{1 + \sqrt{5}}{2} \approx 1.618$$

(Note that $M^2 - M - 1 \geq 0 \implies M \leq \frac{1 - \sqrt{5}}{2} \approx -0.618$ is also true, but this solution is non-permissible as $n \geq 0$.)

Therefore, the smallest value of M for which $a_n \leq M^n$ for all $n \geq 0$ is

$$M = \frac{1 + \sqrt{5}}{2}$$

as required.

2.2 Flaw in Induction Proof

Result

All people in MAT344 have the same name.

Proof

We proceed by induction on the number of people.

For the base case, if there is only one person in MAT344, then of course all people in MAT344 have the same name.

Now suppose that any n people in MAT344 have the same name (where $n \geq 1$), we will prove that any $n + 1$ people have the same name.

Line the $n + 1$ people up (in any way). The first n people have the same name (by I.H.). The last n people have the same name (by I.H.).

Therefore, the first person and the last person must have the same name. So that means all $n + 1$ people have the same name.

Therefore, by the Principle of Mathematical Induction, all people in MAT344 have the same name. $\square$

Where is the flaw in the proof?

Incorrect Attempt. We cannot use the induction hypothesis the second time. The induction hypothesis covers the first n people, not the last n people (as that is a different set of people.)

This attempt is incorrect because the induction hypothesis states "for some $n \geq 1$, any n people in MAT344 have the same name" so it does not matter which n people we choose, as long as there are n people then they all have the same name. However, this points us to where the actual flaw is...

Another Attempt. The argument does not hold for $n + 1 = 2$ people. If we line the 2 people up, the first 1 "people" have the same name (by I.H.). The last 1 "people" have the same name (by I.H.).

But this time there is no overlap between the first n and the last n people. Hence, the two people may or may not have the same name. In summary, the link between $n = 1$ and $n = 2$ cannot be established as the induction step only works for $n \geq 2$.

2.3 Blue Eyed Islanders

Problem

There is an island where n people live, who all have blue eyes. They can all see the colour of each others' eyes, but not their own. They cannot communicate with each other, there are no mirrors, and everyone is a perfect logician. The rule of the island is that if you ever discover the colour of your eyes, you must leave the island on the first night you discover this. One day, a visitor comes to the island, and says "It's nice to see that there is someone with blue eyes on this island!". Describe what happens next.

Let us consider some cases so that we can make a conjecture.

- If the island only had one person, then that person would realize immediately that the visitor was referring to them since they see no one else on the island with blue eyes. So, that person would leave on the first night after hearing the visitor's comment.
- If the island only had two people, then each blue-eyed person would see one blue-eyed individual. Each knows that "if that person is the only one with blue eyes, then they will leave the island tonight". The next morning when they see each other, they would realize that the only reason the other didn't leave is because the other person also sees a person with blue eyes, and since there is no one else on the island, they both realize it must be themselves. Hence, both will leave together on the second night after hearing the visitor's comment.
- If the island only had three people, then each blue-eyed person would see two blue-eyed individuals. All three of them know that "if there are only two people with blue eyes, then both will leave the island tomorrow night". One the third morning when they all see each other, they would realize that the only reason the other two didn't leave is because they see another person with blue eyes, and since there is no one else on the island, they all realize that it must be themselves. Hence, all three will leave together on the third night after hearing the visitor's comment.

This pattern continues and so we claim that if there are n people on the island, then all n of them will leave the island together on the n^{th} day after hearing the visitor's comment.

Claim: All n people will leave the island together on the n^{th} night after hearing the visitor's comment.

Proof

We proceed by induction on the number of people.

Let $P(k)$ be the statement "If there were only k people on the island with blue eyes then all k of them would leave the island on the k^{th} night after hearing the visitor's comment".

For the base case, $P(1)$, if there was only one person on the island with blue eyes, then after hearing the visitor's comment, they would realize that the visitor is referring to them since they see no one else on the island with blue eyes, and would leave on first night after hearing the visitor's comment. Hence $P(1)$ holds.

Now suppose that for some $k \geq 1$ that $P(k)$ holds.

Consider $P(k+1)$ people. As all $k+1$ people are perfect logicians, all of them know that if there were only k people on the island with blue eyes then all k of them would leave the island on the k^{th} night after hearing the visitor's comment.

On the $k+1^{\text{th}}$ morning, when each sees that the other k people have not yet left, it means that they see another person with blue-eyes, and since there is no one else on the island, they all realize that it must be themselves. So, all $k+1$ people realize this simultaneously, and all leave on the $k+1^{\text{th}}$ night after hearing the visitor's comment. Hence $P(k+1)$ holds.

Therefore, by the Principle of Mathematical Induction, all n people will leave the island together on the n^{th} night after hearing the visitor's comment. $\square$

CHAPTER 3

ELEMENTARY COUNTING PROBLEMS

3.1 Rule of Sum

Result: Rule of Sum

If there are n_1 choices for action one, n_2 choices for action two, $\dots$, n_k choices for action k where the k actions cannot happen at the same time, then there are $n_1 + n_2 + \dots + n_k$ ways to choose one of these actions.

Problem

There are 5 action movies, 2 thriller movies, and 3 comedy movies playing at the local movie theatre. If Alice goes to the theatre to watch a single movie, how many choices does she have for which movie to watch?

Solution

Since Alice can watch one of the 5 action movies or one of the 2 thriller movies or one of the 3 comedy movies and the choices are mutually exclusive, it follows, by the Rule of Sum, that she has $5 + 2 + 3 = 10$ different choices for which movie to watch.

3.2 Rule of Product

Result: Rule of Product

If a task can be broken down into k independent steps where there are

- n_1 ways to complete step one
- n_2 ways to complete step two
- $\vdots$
- n_k ways to complete step k

Then the number of ways to complete the entire task is $n_1 \cdot n_2 \cdot \dots \cdot n_k$.

Problem

How many different sandwiches are possible if selection is made from either white or brown bread, ham, chicken or roast-beef, and mustard or mayonnaise?

Solution

The task of creating a sandwich can be broken down into three steps:

- Step 1: Pick either white or brown bread.
- Step 2: Pick either ham, chicken, or roast-beef.
- Step 3: Pick either mustard or mayonnaise.

Therefore, by the Rule of Product, $2 \times 3 \times 2 = 12$ different sandwiches are possible.

Problem

How many positive divisors does 3600 have?

Solution

The prime factorization of 3600 is $2^4 \cdot 3^2 \cdot 5^2$. Hence, any positive divisor of 3600 must have the form $2^a 3^b 5^c$ where a , b , and c are integers satisfying $0 \leq a \leq 4$, $0 \leq b \leq 2$, and $0 \leq c \leq 2$. There are 5 possibilities for a , 3 possibilities for b , and 3 possibilities for c . Therefore, by the Rule of Product, there are $5 \times 3 \times 3 = 45$ positive divisors of 3600.

Problem: Rule of Sum and Rule of Product

Postal Codes in Canada are 6 characters in length, starting with letters and alternating numbers and letters. How many Canadian Postal Codes contain at least one vowel?

Solution: Direct

Consider cases depending on the number of vowels present:

- One Vowel:

$$\underbrace{5}_{\text{vowel}} \underbrace{10}_{0-9} \underbrace{21}_{\text{no vowel}} \underbrace{10}_{0-9} \underbrace{21}_{\text{no vowel}} \underbrace{10}_{0-9}$$

but the vowel can be in any one of the three letter positions which multiplies the number of possibilities by 3 to give $3 \cdot (5 \times 10^3 \times 21^2) = 6615000$

- Two Vowels:

$$\underbrace{5}_{\text{vowel}} \underbrace{10}_{0-9} \underbrace{5}_{\text{vowel}} \underbrace{10}_{0-9} \underbrace{21}_{\text{no vowel}} \underbrace{10}_{0-9}$$

but the two vowels can be in any of the three letter positions which multiplies the number of possibilities by 3 to give $3 \cdot (5^2 \times 10^3 \times 21) = 1575000$

- Three Vowels:

$$\underbrace{5}_{\text{vowel}} \underbrace{10}_{0-9} \underbrace{5}_{\text{vowel}} \underbrace{10}_{0-9} \underbrace{5}_{\text{vowel}} \underbrace{10}_{0-9}$$

so the number of possibilities is $5^3 \times 10^3 = 125000$

Therefore, there are $6615000 + 1575000 + 125000 = 8315000$ Canadian Postal Codes that contain at least one vowel.

Solution: Indirect

Without any restrictions, there are

$$\underbrace{26}_{\text{letter}} \underbrace{10}_{0-9} \underbrace{26}_{\text{letter}} \underbrace{10}_{0-9} \underbrace{26}_{\text{letter}} \underbrace{10}_{0-9} \implies 26^3 \times 10^3 = 17576000$$

Canadian Postal Codes. Moreover, there are

$$\underbrace{21}_{\text{no vowel}} \underbrace{10}_{0-9} \underbrace{21}_{\text{no vowel}} \underbrace{10}_{0-9} \underbrace{21}_{\text{no vowel}} \underbrace{10}_{0-9} \implies 21^3 \times 10^3 = 9261000$$

Canadian Postal Codes that contain zero vowels.

Therefore, there are $17576000 - 9261000 = 8315000$ Canadian Postal Codes that contain at least one vowel.

3.3 Permutations without Repetition

Result

The arrangement of different objects into a linear order using each object exactly once is called a permutation of these objects. The number

$$n \cdot (n - 1) \cdot (n - 2) \cdot \dots \cdot 2 \cdot 1$$

of all permutations of n objects is called n factorial, and is denoted by $n!$ (we define $0! = 1$).

Things we can answer with $n!$:

- The number of ways to shuffle a deck of n cards is $n!$.
- The number of ways to rearrange n distinct letters is $n!$.
- The number of bijective functions from $[n]$ to $[n]$ is $n!$.
- The number of ways to arrange n people in a line is $n!$.
- In general, the number of ways to arrange n distinct objects in a linear order, using each object exactly once, is $n!$.

Problem

In how many ways could ten people line up for a photo if the shortest person and tallest person are

- a) side-by-side?
- b) not side-by-side?

Solution

- a) If we treat the shortest person and tallest person as a single unit, then there are $9!$ ways to linearly order nine distinct objects. However, the shortest person and tallest person can swap places which multiplies the number possibilities by $2!$. Therefore, there are $9! \times 2$ ways.
- b) Without any restrictions, there are $10!$ ways for the ten people to line up for a photo. In part a), we determined there are $9! \times 2$ ways where the shortest person and tallest person are side-by-side. Therefore, the number of ways where the shortest person and tallest person are not side-by-side is $10! - (9! \times 2)$.

Problem

- a) In how many ways can a photographer arrange six women and five men if the women must sit in the first row and the men must sit in the second row?
- b) How many different arrangements are there if the tallest man and the two tallest women must be at the centre of their respective rows?

Solution

- a) The photographer can arrange the six women in the first row in $6!$ ways and can arrange the five men in the second row in $5!$ ways. Therefore, by the Rule of Product, there are $6! \times 5!$ ways.
- b) The photographer can begin by putting the two tallest women in the centre of the first row in $2!$ ways. Then the photographer can arrange the remaining four women in $4!$ ways. Similarly, the photographer can begin by putting the tallest man in the centre of the second row and then arrange the remaining four men in $4!$ ways. Therefore, by the Rule of Product, there are $(2 \times 4!) \cdot 4! = 1152$ ways.

3.4 Permutations with Repetition

A multiset is a collection of objects, possibly with repeats.

Result

A linear order that contains all the elements of a multiset exactly once is called a multiset permutation. Let $n, k, a_1, a_2, \dots, a_k$ be nonnegative integers satisfying $a_1 + a_2 + \dots + a_k = n$. Consider a multiset of n objects, in which a_i objects are of type i , for all $i \in [k]$. Then the number of ways to linearly order these objects is

$$\frac{n!}{a_1! \cdot a_2! \cdot \dots \cdot a_k!}$$

Problem

How many different arrangements are there for the word MISSISSAUGA if

- a) There are no restrictions?
- b) It must start with M?
- c) It must end with A?
- d) It must start with M and end with A?
- e) It must start with M or end with A?

Solution

a) There are eleven letters in MISSISSAUGA which can be arranged in $11!$ ways, however, there are many arrangements which look the same because there are

- 2 Is
- 4 Ss
- 2 As

Hence, there are

$$\frac{11!}{2! \cdot 4! \cdot 2!}$$

different arrangements for the word MISSISSAUGA.

b) Fix M in the first position to satisfy the condition. Now there are

$$\frac{10!}{2! \cdot 4! \cdot 2!}$$

ways to arrange the letters ISSISSAUGA.

c) Fix A in the last position to satisfy the condition. Now there are

$$\frac{10!}{2! \cdot 4!}$$

ways to arrange the letters MISSISSAUG.

d) Fix M in the first position and A in the last position to satisfy the condition. Now there are

$$\frac{9!}{2! \cdot 4!}$$

ways to arrange the letters ISSISSAUG.

e) Simply adding the number of ways the word can start with M or end with A overcounts since there are words that both start with M and end with A that are being double counted. Hence, there are

$$\underbrace{\frac{10!}{2! \cdot 4! \cdot 2!}}_{\text{Starts with M}} + \underbrace{\frac{10!}{2! \cdot 4!}}_{\text{Ends with A}} - \underbrace{\frac{9!}{2! \cdot 4!}}_{\text{Starts with M and Ends with A}}$$

ways for the word to start with M or end with A.

Problem

Consider a $m \times n$ grid. How many paths are there from $(0, 0)$ to (m, n) if the only steps allowed are $(0, 1)$ (NORTH) and $(1, 0)$ (EAST)?

Solution

Any valid path will consist of m EAST steps and n NORTH steps. Therefore, the question can be rephrased as "how many different arrangements are there of the string $\underbrace{EE \dots E}_m \underbrace{NN \dots N}_n$ ".

Hence, there are

$$\frac{(m+n)!}{m! \cdot n!}$$

paths.

Problem

How many different five digit even numbers less than 60,000 can be formed using the digits 2, 2, 5, 6, 7?

Solution

For the number to be less than 60,000, it must start with 2 or 5. For the number to be even, it must end with either 2 or 6. This leads to the following cases:

- The number begins with 2 and ends with 2. The problem is to now arrange the digits 5, 6, 7 which can be done in $3! = 6$ ways.
- The number begins with 2 and ends with 6. The problem is to now arrange the digits 2, 5, 7 which can be done in $3! = 6$ ways.
- The number begins with 5 and ends with 2. The problem is to now arrange the digits 2, 6, 7 which can be done in $3! = 6$ ways.
- The number begins with 5 and ends with 6. The problem is to now arrange the digits 2, 2, 7 which can be done in $\frac{3!}{2!} = 3$ ways.

Therefore, there are $6 + 6 + 6 + 3 = 21$ such numbers.

3.5 Words with Repetition

Result

Let Σ be an n -element set of symbols. The number of k -digit strings over Σ is n^k .

Things we can answer with n^k :

- The number of possible passwords of length k , where each character can be chosen from a set of n possible characters.
- The number of possible outcomes of rolling a n -sided die k times, where each roll can result in any one of the n possible outcomes.
- The number of functions from $[k]$ to $[n]$.
- The number of possible outcomes of a k -question multiple choice test, where each question has n possible answers.
- The number of ways to distribute k distinct objects among n distinct bins.

Problem

A multiple choice test contains 12 questions, each with 5 possible answers. How many possible answer combinations are there?

Solution

Let our alphabet be $\Sigma = \{A, B, C, D, E\}$ which corresponds to the options for a question. Then a submission for the multiple choice test consists of forming a 12-digit string from the 5-element alphabet Σ . Therefore, there are 5^{12} possible answer combinations.

Another way to get the same result is to notice that there are 5 ways to answer the first question, 5 ways to answer the second question, ..., 5 ways to answer the twelfth question. Therefore, by the Rule of Product, there are 5^{12} possible answer combinations.

3.6 Words without Repetition

Result

Let n and k be positive integers satisfying $n \geq k$. Then the number of k -digit strings over an n -element alphabet in which no letter is used more than once is

$$(n)_k = n \cdot (n-1) \cdot (n-2) \cdot \dots \cdot (n-k+1) = \frac{n!}{(n-k)!}$$

Things we can answer with $(n)_k$:

- The number of lists of length k from n distinct objects.
- The number of injective functions from $[k]$ to $[n]$ (assuming $k \leq n$, otherwise if $k > n$ then there are 0 such functions).

The only difference between $n!$ and $(n)_k$ is that in $(n)_k$ we do not necessarily use all n objects, we stop after selecting k of them.

Problem

From a group of seven people, how many ways could a President and Vice President be chosen?

Solution

Let our alphabet be $\Sigma = \{p_1, p_2, \dots, p_7\}$ where p_i corresponds to person i . There are $(7)_2$ 2-digit strings over Σ in which no person is used more than once.

Another way to get the same result is to notice that there are seven choices for the President and after the President is picked, we have six choices for the Vice President. Therefore, by the Rule of Product, there are 7×6 ways a President and Vice President be chosen from a group of seven people.

3.7 Combinations without Repetition

Result

The number of k -element subsets of $[n]$ is denoted $\binom{n}{k}$ and is read “ n choose k ”. For all nonnegative integers $k \leq n$, the equality

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{(n)_k}{k!}$$

holds.

Things we can answer with $\binom{n}{k}$:

- The number of ways to choose a committee of k people from a group of n individuals.
- The number of ways to select k balls from an urn containing n balls without replacement.

The difference between $(n)_k$ and $\binom{n}{k}$ is that in $(n)_k$ order matters whereas in $\binom{n}{k}$ the order of the elements selected does not matter.

Problem

A company has 20 employees, 12 males and 8 females. How many ways are there to form a committee of 5 employees that contains at least one male and at least one female?

Solution: 1

To form a committee of 5 employees that consists of at least one male and at least one female, we can

- choose 1 male and 4 females in $\binom{12}{1}\binom{8}{4}$ ways.
- choose 2 males and 3 females in $\binom{12}{2}\binom{8}{3}$ ways.
- choose 3 males and 2 females in $\binom{12}{3}\binom{8}{2}$ ways.
- choose 4 males and 1 female in $\binom{12}{4}\binom{8}{1}$ ways.

Therefore, there are $\binom{12}{1}\binom{8}{4} + \binom{12}{2}\binom{8}{3} + \binom{12}{3}\binom{8}{2} + \binom{12}{4}\binom{8}{1} = 14656$ ways to form a committee of 5 employees that contains at least one male and at least one female.

Solution: 2

Without any restrictions whatsoever, there are $\binom{20}{5}$ ways to form a committee of 5 employees.

There are $\binom{12}{5}$ ways to form a committee of 5 employees that consists of only males. Similarly, there are $\binom{8}{5}$ ways to form a committee of 5 employees that consists of only females.

Therefore, there are $\binom{20}{5} - \binom{12}{5} - \binom{8}{5} = 14656$ ways to form a committee of 5 employees that contains at least one male and at least one female.

Problem

Alice is writing a multiple choice exam consisting of 12 problems. She is instructed to answer 7 problems, but she must answer at least 3 of the first 5 problems of the exam. In how many ways can she finish the exam?

Solution

Since Alice must answer at least 3 of the first 5 problems, she can finish the exam by

- answering 3 of the first 5 problems in $\binom{5}{3}$ ways and then answering 4 of the remaining 7 problems in $\binom{7}{4}$ ways.
- answering 4 of the first 5 problems in $\binom{5}{4}$ ways and then answering 3 of the remaining 7 problems in $\binom{7}{3}$ ways.
- answering all 5 of the first 5 problems in $\binom{5}{5}$ ways and then answering 2 of the remaining 7 problems in $\binom{7}{2}$ ways.

Therefore, there are $\binom{5}{3}\binom{7}{4} + \binom{5}{4}\binom{7}{3} + \binom{5}{5}\binom{7}{2}$ ways for Alice to finish the exam.

3.8 Combinations with Repetition

Result

The number of k -element multisets whose elements all belong to $[n]$ is

$$\binom{n+k-1}{k}$$

Say we have a set S with n elements. We wish to count the number of subsets of S with k elements where elements can be repeated (i.e. with replacement).

We can specify a multisubset of size k of S by specifying the multiplicity of each element in S .

To do this, we will have n bins, one for each of the n elements in S , and we will distribute k balls. If bin i has m_i balls then that indicates that element i from S is chosen m_i times. The number of ways to distribute k balls into n bins corresponds to the number of rearrangements of the string consisting of k stars (representing the balls), and $n-1$ bars (representing the walls between the bins). There are

$$\frac{(n-1+k)!}{k!(n-1)!} = \binom{n+k-1}{k}$$

Things we can answer with $\binom{n+k-1}{k}$:

- The number of ways to put k identical balls into buckets $B_1, \dots, B_n$.
- The number of ways to distribute k candy bars to n people.

- The number of ways to buy k sodas from a vending machine with n varieties.
- The number of ways to choose k scoops of ice cream from n flavors.
- The number of nonnegative integer solutions to $x_1 + x_2 + \dots + x_n = k$.

Problem

Charlie is participating in a lottery game where five numbers are drawn out of $[90]$ such that the numbers drawn are put back into the basket right after being selected. To win the jackpot, Charlie must play the same multiset of numbers as the one drawn (regardless of the order in which the numbers were drawn). How many lottery tickets should Charlie buy to guarantee a win?

Solution

For Charlie to guarantee a win, he must buy a ticket for each possible 5-element multisubset of $[90]$. By the above reasoning, there are $\binom{90+5-1}{4} = \binom{94}{5}$ such subsets. Therefore, Charlie must buy $\binom{94}{5}$ tickets (and fill them out right) to guarantee a win.

3.9 Summary

Consider an n -element set S . We can construct:

- lists from S (order matters)
- sets from S (order doesn't matter)

We can count:

- lists of length k :
 - $n \cdot (n-1) \cdot \dots \cdot (n-k+1) = \frac{n!}{(n-k)!}$, if no repetitions allowed
 - n^k , if repetitions are allowed
- sets of size k :
 - $\binom{n}{k} = \frac{n!}{k!(n-k)!}$, if no repetitions allowed
 - $\binom{n+k-1}{k}$, if repetitions are allowed

3.10 Problems

Problem

A host invites n couples to a party. She wants to ask a subset of the $2n$ guests to give a speech, but she does not want to ask both members of any couple to give speeches. In how many ways can she proceed?

Solution: 1

For each couple, the host can make three choices:

- the couple does not speak
- the husband speaks
- the wife speaks

Therefore, the host has $\underbrace{3 \cdot 3 \cdot \dots \cdot 3}_{n \text{ of them}} = 3^n$ ways to select a subset of speakers.

Solution: 2

The host can begin by selecting k couples to give a speech, and then select one member from each of the k selected couples in $\binom{2}{1}$ ways. Summing over the possible values of k gives

$$\sum_{k=0}^n \binom{n}{k} \binom{2}{1}^k = \sum_{k=0}^n \binom{n}{k} 2^k 1^{n-k} = 3^n$$

Therefore, the host has 3^n ways to select a subset of speakers.

Problem

A track and field championship has participants from 49 countries. The flag of each participating country consists of three horizontal stripes of different colors. However, no flag contains colors other than red, white, blue, and green. Is it true that there are three participating countries with identical flags?

Solution

TRUE. There are $4 \cdot 3 \cdot 2 = 24$ different 3-colour flags using four colours. Let these be the pigeonholes and let the pigeons be the 49 flags of the participating countries. Since $2 \cdot 24 = 48 < 49$, it follows, by the Pigeonhole Principle, that there are three participating countries with identical flags.

Problem

How many functions $f : [n] \rightarrow [n]$ are there for which there exists exactly one $i \in [n]$ satisfying $f(i) = i$.

Solution

Begin by picking an element x_i from the domain that will get mapped to itself (i.e. $f(x_i) = x_i$). There are n ways to pick this element.

Now consider the remaining $n - 1$ domain elements. Since none of them can get mapped to themselves, it follows that we have $n - 1$ choices for what each one maps to.

Therefore there are $n \cdot (n - 1)^{n-1}$ functions $f : [n] \rightarrow [n]$ for which there exists exactly one $i \in [n]$ satisfying $f(i) = i$.

Problem

Prove that the number of subsets of $[n]$ that have an odd number of elements is 2^{n-1} .

Proof

To specify a subset of $[n]$ with an odd number of elements, we can instead specify an arbitrary subset of $[n - 1]$. Indeed, consider an arbitrary subset A of $[n - 1]$. If $|A|$ is odd then we will not include n in A . Otherwise $|A|$ is even, in which case we will include n in A .

(The main point is that once we have a subset of $[n - 1]$, we know exactly what to do with the element n .)

Since $[n - 1]$ has 2^{n-1} subsets, it follows that $[n]$ has 2^{n-1} subsets with an odd number of elements. $\square$

CHAPTER 4

THE BINOMIAL THEOREM

Result: The Binomial Theorem

For all nonnegative integers n ,

$$(x + y)^n = \sum_{k=0}^n \binom{n}{k} x^k y^{n-k} = \sum_{k=0}^n \binom{n}{k} x^{n-k} y^k$$

Problem

Expand and simplify $(2x - 4)^5$.

Solution

$$\begin{aligned}(2x - 4)^5 &= \underbrace{(2x)}_x + \underbrace{(-4)}_y \bigg)^5 \\&= \sum_{k=0}^5 \binom{5}{k} (2x)^k (-4)^{5-k} \\&= \binom{5}{0} (2x)^0 (-4)^5 + \binom{5}{1} (2x)^1 (-4)^4 + \binom{5}{2} (2x)^2 (-4)^3 \\&\quad + \binom{5}{3} (2x)^3 (-4)^2 + \binom{5}{4} (2x)^4 (-4)^1 + \binom{5}{5} (2x)^5 (-4)^0 \\&= -1024 + 2560x - 2560x^2 + 1280x^3 - 320x^4 + 32x^5 \\&= 32x^5 - 320x^4 + 1280x^3 - 2560x^2 + 2560x - 1024\end{aligned}$$

Soon we will see that the Binomial Theorem can also be used to prove identities about binomial coefficients.

We can define the binomial coefficient $\binom{m}{k}$ for all real numbers m as follows,

Result

Let m be any real number, and let k be a nonnegative integer. Then $\binom{m}{0} = 1$, and

$$\binom{m}{k} = \frac{m(m-1)\dots(m-k+1)}{k!}$$

if $k > 0$.

which allows us to generalize the Binomial Theorem as follows,

Result: The Generalized Binomial Theorem

Let m be any real number. Then

$$(1+x)^m = \sum_{n \geq 0} \binom{m}{n} x^n$$

where the sum is taken over all nonnegative integers n .

Problem

Let m be a positive integer.

- Compute the power series form of $(1+x)^m$.
- Compute the power series form of $(1+x)^{-m}$.
- Compute the power series form of $(1-x)^{-m}$.

Solution

a) By the Generalized Binomial Theorem,

$$\begin{aligned} (1+x)^m &= \sum_{n \geq 0} \binom{m}{n} x^n \\ &= \binom{m}{0} x^0 + \binom{m}{1} x^1 + \dots + \binom{m}{m} x^m + \binom{m}{m+1} x^{m+1} + \binom{m}{m+2} x^{m+2} + \dots \\ &= \binom{m}{0} x^0 + \binom{m}{1} x^1 + \dots + \binom{m}{m} x^m + 0 + 0 + \dots \\ &= \sum_{n=0}^m \binom{m}{n} x^n \end{aligned}$$

Hence, if m is a positive integer then we only get a sum of $m+1$ elements for $(1+x)^m$ since $\binom{m}{n} = 0$

if $n > m$.

b) By the Generalized Binomial Theorem,

$$(1+x)^{-m} = \sum_{n \geq 0} \binom{-m}{n} x^n$$

Notice that

$$\begin{aligned} \binom{-m}{n} &= \frac{(-m)(-m-1)(-m-2)\dots(-m-n+1)}{n!} \\ &= \frac{(-1)^n(m)(m+1)(m+2)\dots(m+n-1)}{n!} \\ &= (-1)^n \frac{(m+n-1)(m+n-2)\dots(m+1)m}{n!} \\ &= (-1)^n \binom{m+n-1}{n} \end{aligned}$$

Therefore,

$$(1+x)^{-m} = \sum_{n \geq 0} \binom{m+n-1}{n} (-1)^n x^n$$

c) Replace x with $-x$ in the previous result to get

$$(1-x)^{-m} = (1+(-x))^{-m} = \sum_{n \geq 0} \binom{m+n-1}{n} (-1)^n (-x)^n = \sum_{n \geq 0} \binom{m+n-1}{n} x^n$$

Problem

Find the power series expansion of $\sqrt{1-4x}$.

Solution

By the Generalized Binomial Theorem,

$$\sqrt{1-4x} = (1+(-4x))^{\frac{1}{2}} = \sum_{n \geq 0} \binom{\frac{1}{2}}{n} (-4x)^n = \sum_{n \geq 0} (-1)^n \cdot 2^{2n} \cdot \binom{\frac{1}{2}}{n} x^n$$

Notice that $\binom{\frac{1}{2}}{0} = 1$, $\binom{\frac{1}{2}}{1} = \frac{1}{2}$, and for $n \geq 2$ we have

$$\begin{aligned} \binom{\frac{1}{2}}{n} &= \frac{\left(\frac{1}{2}\right)\left(\frac{1}{2}-1\right)\left(\frac{1}{2}-2\right)\dots\left(\frac{1}{2}-n+1\right)}{n!} \\ &= \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)\dots\left(-\frac{2n-3}{2}\right)}{n!} \\ &= (-1)^{n-1} \cdot \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3)}{2^n \cdot n!} \end{aligned}$$

Hence,

$$\begin{aligned}
 \sqrt{1-4x} &= \sum_{n \geq 0} (-1)^n \cdot 2^{2n} \cdot \binom{\frac{1}{2}}{n} x^n \\
 &= 1 - 2x + \sum_{n \geq 2} (-1)^n \cdot 2^{2n} \cdot \binom{\frac{1}{2}}{n} x^n \\
 &= 1 - 2x + \sum_{n \geq 2} (-1)^n \cdot 2^{2n} \cdot (-1)^{n-1} \cdot \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3)}{2^n \cdot n!} x^n \\
 &= 1 - 2x + \sum_{n \geq 2} (-1)^{2n-1} \cdot 2^n \cdot \frac{1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3)}{n!} x^n \\
 &= 1 - 2x - \sum_{n \geq 2} 2 \cdot \frac{2^{n-1} (1 \cdot 2 \cdot 3 \cdot \dots \cdot (n-1)) (1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3))}{(n-1)! \cdot n!} x^n \\
 &= 1 - 2x - \sum_{n \geq 2} \frac{2}{n} \cdot \frac{(2 \cdot 4 \cdot 6 \cdot \dots \cdot (2n-2)) (1 \cdot 3 \cdot 5 \cdot \dots \cdot (2n-3))}{(n-1)! \cdot (n-1)!} x^n \\
 &= 1 - 2x - \sum_{n \geq 2} \frac{2}{n} \cdot \frac{1 \cdot 2 \cdot 3 \cdot \dots \cdot (2n-2)}{(n-1)! \cdot (n-1)!} x^n \\
 &= 1 - 2x - \sum_{n \geq 2} \frac{2}{n} \cdot \frac{(2n-2)!}{(n-1)! \cdot (n-1)!} x^n \\
 &= 1 - 2x - \sum_{n \geq 2} \frac{2}{n} \cdot \binom{2n-2}{n-1} x^n
 \end{aligned}$$

Therefore,

$$\sqrt{1-4x} = 1 - 2x - 2 \sum_{n \geq 2} \frac{1}{n} \binom{2n-2}{n-1} x^n$$

Try to remember the above example as we will use it later on to derive a closed formula for a very special sequence of numbers.

CHAPTER 5

COMBINATORIAL PROOFS

5.1 Template for Combinatorial Proofs

To give a combinatorial proof for $A = B$,

1. Find a counting problem that you can answer in two ways.
2. Explain why one answer to the counting problem is A .
3. Explain why the other answer to the counting problem is B .

Since both A and B are answers to the same problem, we must have $A = B$.

Here is a problem one would typically solve using the Principle of Mathematical Induction, however, using a combinatorial argument leads to a much cleaner and faster solution.

Problem

Prove that for all positive integers n , the inequality $\binom{2n}{n} < 4^n$ holds.

Proof

Begin by rewriting the problem as $\binom{2n}{n} < 2^{2n}$.

The left-hand side is the number of n -element subsets of $[2n]$, while the right-hand side is the number of all subsets of $[2n]$. □

5.2 Proving Identities using The Binomial Theorem

Result

For all positive integers n ,

$$\sum_{k=0}^n (-1)^k \cdot \binom{n}{k} = 0$$

Proof

Apply the Binomial Theorem with $x = -1$ and $y = 1$ to get

$$0 = (-1 + 1)^n = \sum_{k=0}^n \binom{n}{k} (-1)^k$$

□

Remark. Consider subsets of the set $[n]$ where $n \geq 1$. The above result says that the number of even sized subsets is equal to the number of odd sized subsets. Moreover, the next result shows that the number of even sized subsets plus the number of odd sized subsets is 2^n . It follows that the number of even sized subsets and the number of odd sized subsets is both 2^{n-1} .

Result

For all nonnegative integers n ,

$$2^n = \sum_{k=0}^n \binom{n}{k}$$

Proof: Using Combinatorial Argument

Both sides count the number of subsets of $[n]$. The LHS counts directly, while the RHS counts the number number of subsets of size k and then sums over all possible values of k . □

Proof: Using Binomial Theorem

Apply the Binomial Theorem with $x = 1$ and $y = 1$ to get

$$2^n = (1 + 1)^n = \sum_{k=0}^n \binom{n}{k}$$

□

Problem

Prove for all integers $n \geq 1$,

$$2^{n-1} \cdot n = \sum_{k=1}^n k \binom{n}{k}$$

Proof: Using Combinatorial Argument

We will count the number of ways to choose a committee among n people, and then to choose a president from the committee.

One way to do this is to first choose a k -member committee in $\binom{n}{k}$ ways, then choose its president in $\binom{k}{1}$ ways, and finally sum over all possible values for k to get a total of $\sum_{k=1}^n k \binom{n}{k}$ many ways.

Another way is to first choose the president in $\binom{n}{1}$ ways and then choose a subset of the remaining $n - 1$ people in 2^{n-1} ways to get a total of $n \cdot 2^{n-1}$ many ways.

Since both ways are counting the number of ways to choose a committee among n people, and then to choose a president from the committee, we must have

$$2^{n-1} \cdot n = \sum_{k=1}^n k \binom{n}{k}$$

as required. □

Proof: Using Binomial Theorem

Apply the Binomial Theorem with $y = 1$ to get the identity

$$(x + 1)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

Now differentiate both sides with respect to x to get

$$n(x + 1)^{n-1} = \sum_{k=1}^n \binom{n}{k} k \cdot x^{k-1}$$

Finally, substitute $x = 1$ to get

$$2^{n-1} \cdot n = \sum_{k=1}^n k \binom{n}{k}$$

□

Problem

Prove for all integers $n \geq 2$,

$$2^{n-2} \cdot n \cdot (n-1) = \sum_{k=2}^n k(k-1) \binom{n}{k}$$

Proof: Using Combinatorial Argument

We will count the number of ways to choose a committee among n people, and then to choose a president and vice-president from the committee.

One way to do this is to first choose a k -member committee in $\binom{n}{k}$ ways, then choose its president in $\binom{k}{1}$ ways and its vice-president in $\binom{k-1}{1}$ ways, and finally sum over all possible values of k to get a total of $\sum_{k=2}^n k(k-1) \binom{n}{k}$ many ways.

Another way is to first choose the president in $\binom{n}{1}$ ways and the vice-president in $\binom{n-1}{1}$ ways, and then choose a subset of the remaining $n-2$ people in 2^{n-2} ways to get a total of $n \cdot (n-1) \cdot 2^{n-1}$ many ways.

Since both ways are counting the number of ways to choose a committee among n people, and then to choose a president and vice-president from the committee, we must have

$$2^{n-1} \cdot n \cdot (n-1) = \sum_{k=2}^n k(k-1) \binom{n}{k}$$

as required. □

Proof: Using Binomial Theorem

Apply the Binomial Theorem with $y = 1$ to get the identity

$$(x+1)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

Now differentiate both sides with respect to x twice to get

$$n(n-1)(x+1)^{n-2} = \sum_{k=2}^n \binom{n}{k} k(k-1) \cdot x^{k-2}$$

Finally, substitute $x = 1$ to get

$$2^{n-1} \cdot n \cdot (n-1) = \sum_{k=2}^n k(k-1) \binom{n}{k}$$

□

Result: Generalization

For all nonnegative integers $n \geq m$,

$$2^{n-m}(n)_m = \sum_{k=m}^n (k)_m \binom{n}{k}$$

Proof: Using Combinatorial Argument

We will count the number of ways to choose a committee among n people, and then to choose m committee members for m different positions.

One way to do this is to first choose a k -member committee in $\binom{n}{k}$ ways, then choose m members to fill m different positions in $(k)_m$ ways, and finally sum over all possible values for k to get a total of $\sum_{k=m}^n (k)_m \binom{n}{k}$ many ways.

Another way is to first choose m members to fill m different positions in $(n)_m$ ways, and then choose a subset of the remaining $n - m$ people in 2^{n-m} ways to get a total of $(n)_m 2^{n-m}$ many ways.

Since both ways are counting the number of ways to choose a committee among n people, and then to choose m committee members for m different positions, we must have

$$2^{n-m}(n)_m = \sum_{k=m}^n (k)_m \binom{n}{k}$$

as required. □

Proof: Using Binomial Theorem

Apply the Binomial Theorem with $y = 1$ to get the identity

$$(x + 1)^n = \sum_{k=0}^n \binom{n}{k} x^k$$

Now differentiate both sides m times with respect to x to get

$$(n)_m (x + 1)^{n-m} = \sum_{k=m}^n \binom{n}{k} (k)_m \cdot x^{k-m}$$

Finally, substitute $x = 1$ to get

$$2^{n-m}(n)_m = \sum_{k=m}^n (k)_m \binom{n}{k}$$

□

5.3 Sum of the Natural Numbers

Problem

Give a combinatorial proof for the identity

$$1 + 2 + \dots + n = \binom{n+1}{2}$$

Proof: 1

Consider $n + 1$ people in a room where everyone shakes hands with everyone else. We will count the total number of handshakes in two ways.

One way is to notice that there is a handshake for every possible pair of people, and there are $\binom{n+1}{2}$ possible pairs of people.

Another way is to form a line $P_1, P_2, \dots, P_{n+1}$ with the $n + 1$ people and then

- let P_1 shake hands with $P_2, P_3, \dots, P_{n+1}$. A total of n handshakes occurred.
- let P_2 shake hands with $P_3, P_4, \dots, P_{n+1}$. A total of $n - 1$ handshakes occurred.
- let P_3 shake hands with $P_4, P_5, \dots, P_{n+1}$. A total of $n - 2$ handshakes occurred.

$\vdots$

- let P_n shake hands with P_{n+1} . A total of 1 handshake occurred.

Hence, the total number of handshakes is

$$n + (n - 1) + (n - 2) + \dots + 1$$

Since both ways are counting the number of handshakes in a room of $n + 1$ people where everyone shakes hands with everyone else, we must have

$$1 + 2 + \dots + n = \binom{n+1}{2}$$

as required. □

Remark. If everyone shakes hands with everyone else, then each person has shook n hands, and there are $n + 1$ people. However, this counts every handshake twice. Hence, the number of handshakes in a room of $n + 1$ people where everyone shakes hands with everyone else is also $\frac{n(n+1)}{2}$.

Remark. The proof is equivalent to counting the number of edges in the complete graph on $n + 1$ vertices, K_{n+1} .

Proof: 2

We will count the number of 2-element subsets of the set $[n + 1]$ in two ways.

One way to do this, by definition, is $\binom{n+1}{2}$, which is also the RHS of the identity.

Another way is to consider a 2-element subset, $\{a, b\}$ where $a < b$, of $[n + 1]$. We are going to count the number of subsets that can be formed for each distinct value of b . The smallest value b can take on is 2, and the largest value b can take on is $n + 1$.

- When $b = 2$, we have $\{a, 2\}$. There is only one choice for a (which is 1). Thus, there are 1 subsets when $b = 2$.
- When $b = 3$, we have $\{a, 3\}$. There are two choices for a (which are 1, 2). Thus, there are 2 subsets when $b = 3$.
- When $b = 4$, we have $\{a, 4\}$. There are three choices for a (which are 1, 2, 3). Thus, there are 3 subsets when $b = 4$.
- $\vdots$
- When $b = n + 1$, we have $\{a, n + 1\}$. There are n choices for a (which are 1, 2, 3, ..., n). Thus, there are n subsets when $b = n + 1$.

Since the above cases are disjoint and exhaust all possible cases, the number of 2-element subsets of the set $[n + 1]$ is

$$1 + 2 + 3 + \dots + n$$

which is also the LHS of the identity.

Since both ways are counting the number of 2-element subsets of the set $[n + 1]$, we must have

$$1 + 2 + \dots + n = \binom{n+1}{2}$$

as required. □

Remark. The above proof can be generalized as follows. Suppose we want to choose $r + 1$ things from the set $[n + 1]$.

One on hand, there are $\binom{n+1}{r+1}$ ways to that.

On the other hand, consider a $r + 1$ -subset of $[n + 1]$, say $\{a_1, a_2, \dots, a_{r+1}\}$, and fix a_1 to be the largest. Then $r + 1 \leq a_1 \leq n + 1$ and there are $\binom{a_1-1}{r}$ ways of choosing the rest.

This gives rise to the following identity

$$\binom{r}{r} + \binom{r+1}{r} + \dots + \binom{n}{r} = \binom{n+1}{r+1}$$

which is a famous result known as the Hockey Stick Identity.

Result: Generalization

For all nonnegative integers k and n ,

$$\binom{r}{r} + \binom{r+1}{r} + \dots + \binom{n}{r} = \binom{n+1}{r+1}$$

Proof: Hockey Stick Identity

We will count the number of $r+1$ -element subsets of the set $[n+1]$ in two ways.

One way to do this, by definition, is $\binom{n+1}{r+1}$, which is also the RHS of the identity.

Another way is to consider a $r+1$ -element subset, $\{x_1, x_2, \dots, x_{r+1}\}$ where x_1 is the largest element in the set, of $[n+1]$. We are going to count the number of subsets that can be formed for each distinct value of x_1 . The smallest value x_1 can take on is $r+1$, and the largest value x_1 can take on is $n+1$.

- When $x_1 = r+1$, we have $\{r+1, x_2, x_3, \dots, x_{r+1}\}$. There are $\binom{(r+1)-1}{r}$ ways of choosing the rest. Thus, there are $\binom{r}{r}$ subsets when $x_1 = r+1$.
- When $x_1 = r+2$, we have $\{r+2, x_2, x_3, \dots, x_{r+1}\}$. There are $\binom{(r+2)-1}{r}$ ways of choosing the rest. Thus, there are $\binom{r+1}{r}$ subsets when $x_1 = r+2$.
- When $x_1 = r+3$, we have $\{r+3, x_2, x_3, \dots, x_{r+1}\}$. There are $\binom{(r+3)-1}{r}$ ways of choosing the rest. Thus, there are $\binom{r+2}{r}$ subsets when $x_1 = r+3$.

$\vdots$

- When $x_1 = n+1$, we have $\{n+1, x_2, x_3, \dots, x_{r+1}\}$. There are $\binom{(n+1)-1}{r}$ ways of choosing the rest. Thus, there are $\binom{n}{r}$ subsets when $x_1 = n+1$.

Since the above cases are disjoint and exhaust all possible cases, the number of $r+1$ -element subsets of the set $[n+1]$ is

$$\binom{r}{r} + \binom{r+1}{r} + \dots + \binom{n}{r}$$

which is also the LHS of the identity.

Since both ways are counting the number of $r+1$ -element subsets of the set $[n+1]$, we must have

$$\binom{r}{r} + \binom{r+1}{r} + \dots + \binom{n}{r} = \binom{n+1}{r+1}$$

as required. □

5.4 Generalization of Vandermonde's Identity

Problem

Give a combinatorial proof for the identity

$$\binom{n_1 + n_2 + \dots + n_k}{m} = \sum_{a_1 + a_2 + \dots + a_k = m} \binom{n_1}{a_1} \binom{n_2}{a_2} \cdot \dots \cdot \binom{n_k}{a_k}$$

Proof

Consider counting the number of ways to choose m elements the set $[n_1 + n_2 + \dots + n_k]$.

One on hand, there is $\binom{n_1 + n_2 + \dots + n_k}{m}$ ways to do this.

On the other hand, we can choose a_1 elements from the n_1 -element set, a_2 elements from the n_2 -element set, $\dots$, a_k elements from the n_k -element set, until a total of m elements have been chosen from the k sets. There are $\sum_{a_1 + a_2 + \dots + a_k = m} \binom{n_1}{a_1} \binom{n_2}{a_2} \cdot \dots \cdot \binom{n_k}{a_k}$ ways to do this.

Since both ways count the number of ways to select m elements from an $(n_1 + n_2 + \dots + n_k)$ -element set, we must have

$$\binom{n_1 + n_2 + \dots + n_k}{m} = \sum_{a_1 + a_2 + \dots + a_k = m} \binom{n_1}{a_1} \binom{n_2}{a_2} \cdot \dots \cdot \binom{n_k}{a_k}$$

as required. □

Here is an easy way to remember:

Suppose a university has k departments where department i has n_i faculty members (where $i \in [k]$). Consider counting the number of ways to form a committee of m faculty members.

- One answer is to simply pick m faculty from from a pool of $n_1 + n_2 + \dots + n_k$ faculty, which can be done in $\binom{n_1 + n_2 + \dots + n_k}{m}$ ways.
- Another way is to pick a_1 faculty from department 1, a_2 faculty from department 2, $\dots$, a_k faculty from department k such that a total of m faculty members have been chosen. This can be done in $\sum_{a_1 + a_2 + \dots + a_k = m} \binom{n_1}{a_1} \binom{n_2}{a_2} \cdot \dots \cdot \binom{n_k}{a_k}$ ways.

As both ways count the same objects, we must have

$$\binom{n_1 + n_2 + \dots + n_k}{m} = \sum_{a_1 + a_2 + \dots + a_k = m} \binom{n_1}{a_1} \binom{n_2}{a_2} \cdot \dots \cdot \binom{n_k}{a_k}$$

A special case of the previous identity occurs when $k = 2$ (e.g. when a university has only two departments).

Result: Vandermonde's Identity

For all positive integers n , m , and k ,

$$\binom{n+m}{k} = \sum_{i=0}^k \binom{n}{i} \binom{m}{k-i}$$

Proof

Consider counting the number of ways of choosing a k -member committee from a pool of n men and m women.

On one hand, we can simply choose k people from the pool of $n + m$ people, which can be done in $\binom{n+m}{k}$ ways.

On the other hand, we can sum over all possible values of i where the committee consists of i men and $k - i$ women. This can be done in $\sum_{i=0}^k \binom{n}{i} \binom{m}{k-i}$ ways.

Since both ways are counting the number of ways of choosing a k -member committee from a pool of n men and m women, we must have

$$\binom{n+m}{k} = \sum_{i=0}^k \binom{n}{i} \binom{m}{k-i}$$

□

A special case of Vandermonde's identity occurs when $m = n$ and $k = n$,

$$\binom{2n}{n} = \sum_{i=0}^n \binom{n}{i} \binom{n}{n-i} = \sum_{i=0}^n \binom{n}{i} \binom{n}{n-(n-i)} = \sum_{i=0}^n \binom{n}{i}^2$$

We now give a cool combinatorial proof of this identity using lattice paths.

Problem

Give a combinatorial proof for the following identity by counting lattice paths

$$\sum_{i=0}^n \binom{n}{i}^2 = \binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n}^2 = \binom{2n}{n}$$

Proof

We will count the number of lattice paths from $(0, 0)$ to (n, n) in two ways.

One way to do this is by noticing that we must travel a total of $2n$ steps where n of them must be NORTH steps. Therefore, the number of lattice paths from $(0, 0)$ to (n, n) is

$$\binom{2n}{n}$$

which is also the RHS of the identity.

Another way to form lattice paths from $(0, 0)$ to (n, n) is to consider cases depending on where the path crosses the line $x + y = n$. That is, any lattice path from $(0, 0)$ to (n, n) must pass through exactly one of the points: $(0, n), (1, n - 1), (2, n - 2), \dots, (n, 0)$.

- The number of paths passing through $(0, n)$ is $\binom{n}{0}$ (we need to travel a total of n steps and 0 of them are EAST steps). Now the number of paths from $(0, n)$ to (n, n) is also $\binom{n}{0}$ (we need to travel a total of n steps and 0 of them are NORTH steps). Thus, there are $\binom{n}{0} \cdot \binom{n}{0}$ paths from $(0, 0)$ to (n, n) that pass through the point $(0, n)$.
- The number of paths passing through $(1, n - 1)$ is $\binom{n}{1}$ (we need to travel a total of n steps and 1 of them is EAST steps). Now the number of paths from $(1, n - 1)$ to (n, n) is also $\binom{n}{1}$ (we need to travel a total of n steps and 1 of them is NORTH steps). Thus, there are $\binom{n}{1} \cdot \binom{n}{1}$ paths from $(0, 0)$ to (n, n) that pass through the point $(1, n - 1)$.

$\vdots$

- The number of paths passing through $(n, 0)$ is $\binom{n}{n}$ (we need to travel a total of n steps and n of them are EAST steps). Now the number of paths from $(n, 0)$ to (n, n) is also $\binom{n}{n}$ (we need to travel a total of n steps and n of them are NORTH steps). Thus, there are $\binom{n}{n} \cdot \binom{n}{n}$ paths from $(0, 0)$ to (n, n) that pass through the point $(n, 0)$.

Therefore, the number of lattice paths from $(0, 0)$ to (n, n) is

$$\binom{n}{0}^2 + \binom{n}{1}^2 + \dots + \binom{n}{n}^2$$

which is also the LHS of the identity. □

5.5 More Problems

Problem

Give a combinatorial proof for the identity

$$\binom{2n+2}{n+1} = \binom{2n}{n+1} + 2\binom{2n}{n} + \binom{2n}{n-1}$$

Proof

Consider forming a committee of $n+1$ people from a pool of $2n+2$ people.

One way to do this is to simply pick $n+1$ people from the pool of $2n+2$ people. This can be done in $\binom{2n+2}{n+1}$ ways, which is also the LHS of the identity.

Another way is to consider four cases based on two people, say Alice and Bob:

- Neither Alice nor Bob are apart of the committee. In this case, we pick $n+1$ people from the remaining $2n$ people. This can be done in $\binom{2n}{n+1}$ ways.
- Alice is apart of the committee (but not Bob). To complete the committee, we pick n people from the remaining $2n$ people. This can be done in $\binom{2n}{n}$ ways.
- Bob is apart of the committee (but not Alice). To complete the committee, we pick n people from the remaining $2n$ people. This can be done in $\binom{2n}{n}$ ways.
- Both Alice and Bob are apart of the committee. To complete the committee, we pick $n-1$ people from the remaining $2n$ people. This can be done in $\binom{2n}{n-1}$ ways.

Since the above cases are disjoint and exhaust all possible cases, the numbers of ways to form a committee of $n+1$ people from a pool of $2n+2$ people is

$$\binom{2n}{n+1} + 2\binom{2n}{n} + \binom{2n}{n-1}$$

which is also the RHS of the identity.

Since both ways are counting the number of ways to form a committee of $n+1$ people from a pool of $2n+2$ people, we must have

$$\binom{2n+2}{n+1} = \binom{2n}{n+1} + 2\binom{2n}{n} + \binom{2n}{n-1}$$

as required. □

We now give two algebraic proofs of the same result.

Proof: Algebraic Proof 1

We will show $LHS = RHS$ by applying Pascal's identity $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ three times.

Applying it once gives

$$\binom{2n}{n-1} + \binom{2n}{n} = \binom{2n+1}{n}$$

applying it a second time gives

$$\binom{2n}{n} + \binom{2n}{n+1} = \binom{2n+1}{n+1}$$

and applying it a third time gives

$$\binom{2n+1}{n} + \binom{2n+1}{n+1} = \binom{2n+2}{n+1}$$

Hence,

$$\begin{aligned} \binom{2n}{n+1} + 2\binom{2n}{n} + \binom{2n}{n-1} &= \binom{2n}{n-1} + \binom{2n}{n} + \binom{2n}{n} + \binom{2n}{n+1} \\ &= \binom{2n+1}{n} + \binom{2n+1}{n+1} \\ &= \binom{2n+2}{n+1} \end{aligned}$$

as required. □

Proof: Algebraic Proof 2

We will show $LHS = RHS$. Indeed,

$$\begin{aligned} RHS &= \binom{2n}{n+1} + 2\binom{2n}{n} + \binom{2n}{n-1} &&= \frac{2 \cdot (n+1) \cdot (2n+1) \cdot (2n)!}{(n+1) \cdot n! \cdot (n+1)!} \\ &= \frac{(2n)!}{(n+1)! \cdot (n-1)!} + 2 \cdot \frac{(2n)!}{n! \cdot n!} + \frac{(2n)!}{(n-1)! \cdot (n+1)!} &&= \frac{(2n+2) \cdot (2n+1) \cdot (2n)!}{(n+1)! \cdot (n+1)!} \\ &= 2 \cdot \frac{n \cdot (2n)!}{(n+1)! \cdot n!} + 2 \cdot \frac{(n+1) \cdot (2n)!}{(n+1)! \cdot n!} &&= \frac{(2n+2)!}{(n+1)! \cdot (n+1)!} \\ &= \frac{2n \cdot (2n)! + 2 \cdot (n+1) \cdot (2n)!}{(n+1)! \cdot n!} &&= \frac{(2n+2)!}{(n+1)! \cdot (2n+2 - (n+1))!} \\ &= \frac{2 \cdot (2n)! \cdot [n + (n+1)]}{(n+1)! \cdot n!} &&= \binom{2n+2}{n+1} \\ &= \frac{2 \cdot (2n+1) \cdot (2n)!}{(n+1)! \cdot n!} &&= LHS \end{aligned}$$

as required. □

Problem

Give a combinatorial proof for the identity

$$\binom{2n}{2} = 2 \cdot \binom{n}{2} + n^2$$

Proof

We count the number of 2-element subsets of the set $[2n]$ in two ways.

One way to do this is by selecting 2 elements from the set $[2n]$. This is by definition $\binom{2n}{2}$, which is also the LHS of the identity.

Another way to select 2 elements from $2n$ elements is to create two groups of size n : $A = [n]$ and $B = \{n+1, n+2, \dots, 2n\}$. Now we can pick both elements from A , pick both elements from B , or pick one element from A and one element from B . This can be done in $\binom{n}{2} + \binom{n}{2} + \binom{n}{1} \cdot \binom{n}{1} = 2 \cdot \binom{n}{2} + n^2$ ways, which is also the RHS of the identity.

Since both ways are counting the number of 2-element subsets of the set $[2n]$, we must have

$$\binom{2n}{2} = 2 \cdot \binom{n}{2} + n^2$$

as required. □

Here is an easy way to remember:

- LHS: Pick 2 people from from a pool of n men and n women.
- RHS: To pick 2 people from from a pool of n men and n women, we can either pick 2 men, 2 women, or 1 man and 1 woman.

Proof: Algebraic Proof

We will show $LHS = RHS$. Indeed,

$$\begin{aligned} LHS &= 2 \cdot \binom{n}{2} + n^2 &&= \frac{(2n) \cdot (2n-1) \cdot (2n)!}{2 \cdot (2n)!} \\ &= 2 \cdot \frac{n!}{2! \cdot (n-2)!} + n^2 &&= \frac{(2n)!}{2! \cdot (2n-2)!} \\ &= \frac{n \cdot (n-1) \cdot n!}{n!} + n^2 &&= \binom{2n}{2} \\ &= 2n^2 - n &&= RHS \end{aligned}$$

□

Problem

Prove that

$$\binom{2n}{n} = 2 \binom{2n-1}{n-1}$$

Proof

We count the number of ways that we can select n people from a group of $2n$ people.

One way to do this, by definition, is to select n people from a set of $2n$ people, which can be done in $\binom{2n}{n}$. This is the LHS of the identity.

Another way is to consider two cases on a specific person, say Alice:

- We pick Alice as one of the n people selected. Then we need to pick $n - 1$ more people from a remaining pool of $2n - 1$ people, which can be done in $\binom{2n-1}{n-1}$ ways.
- We do not pick Alice as one of the n people selected. In this case we need to exclude $n - 1$ more people from a remaining pool $2n - 1$ people, which can be done in $\binom{2n-1}{n-1}$ ways.
(Notice that after excluding n people we must select the remaining n people)

Since the above cases are disjoint and exhaust all possible cases, the number of ways to select n people from a pool of $2n$ people is

$$2 \binom{2n-1}{n-1}$$

which is the RHS of the identity.

Since both ways are counting the number of ways to select n people from a pool of $2n$ people, we must have

$$\binom{2n}{n} = 2 \binom{2n-1}{n-1}$$

as required. □

Problem

Give a combinatorial proof of the identity

$$k \binom{n}{k} = n \binom{n-1}{k-1}$$

Proof

Consider forming a team consisting of k people from a pool of n people and appointing a manager.

One way to do this is to first select a manager which can be done in $\binom{n}{1}$ ways. Then select the remaining $k-1$ team members from the remaining pool of $n-1$ people. This can be done in $\binom{n-1}{k-1}$ ways. Overall, we have $n \binom{n-1}{k-1}$ ways to do this.

Another way is to first select k people that will make up the team, which can be done in $\binom{n}{k}$ ways. Then from the k selected people, appoint one of them to be the manager. This can be done in $\binom{k}{1}$ ways. Overall, we have $k \binom{n}{k}$ ways to do this.

Since both ways are counting the number of ways to create a team consisting of k people from a pool of n people, and then appointing one of them as the manager, we must have

$$k \binom{n}{k} = n \binom{n-1}{k-1}$$

as required. □

Proof: Algebraic Proof

We will show that $LHS = RHS$. Indeed,

$$\begin{aligned} LHS &= n \binom{n-1}{k-1} \\ &= n \cdot \frac{(n-1)!}{(k-1)! \cdot (n-1-(k-1))!} \\ &= n \cdot \frac{(n-1)!}{(k-1)! \cdot (n-k)!} \\ &= k \cdot \frac{n \cdot (n-1)!}{k \cdot (k-1)! \cdot (n-k)!} \\ &= k \cdot \frac{n!}{k! \cdot (n-k)!} \\ &= k \binom{n}{k} = RHS \end{aligned}$$

□

Problem

If $n \geq 3$ is an integer, give a combinatorial proof for the identity

$$\binom{n}{3} = \sum_{k=2}^{n-1} (k-1)(n-k)$$

Proof

We will count the number of 3-element subsets of the set $[n]$ in two ways.

One way to do this is by selecting 3 elements from the set $[n]$. This is by definition $\binom{n}{3}$, which is also the LHS of the identity.

Another way to form 3-element subsets from the elements in $[n]$, is to consider cases depending on the middle number of the subset. Suppose $\{a, b, c\}$ is a 3-element subset of $[n]$ where the elements are in increasing order. We are going to count the number of subsets that can be formed for each distinct value of b . The smallest value b can take on is 2, and the largest value b can take on is $n-1$.

- When $b = 2$, we have $\{a, 2, c\}$. There is only one choice for a (which is 1) and we have $n-2$ choices for c (which are $3, 4, 5, \dots, n$). Thus, there are $1 \cdot (n-2)$ subsets when $b = 2$.
- When $b = 3$, we have $\{a, 3, c\}$. There are two choices for a (which are 1, 2) and we have $n-3$ choices for c (which are $4, 5, 6, \dots, n$). Thus, there are $2 \cdot (n-3)$ subsets when $b = 3$.
- When $b = 4$, we have $\{a, 4, c\}$. There are three choices for a (which are 1, 2, 3) and we have $n-4$ choices for c (which are $5, 6, 7, \dots, n$). Thus, there are $3 \cdot (n-4)$ subsets when $b = 4$.

$\vdots$

- When $b = n-1$, we have $\{a, n-1, c\}$. There are $n-2$ choices for a (which are $1, 2, 3, \dots, n-2$) and we have only 1 choice for c (which is n). Thus, there are $(n-2) \cdot 1$ subsets when $b = n-1$.

Note that all cases are disjoint since the b values are always different. Furthermore, the cases are exhaustive and cover all 3-element subsets of $[n]$. Therefore, the number of 3-element subsets of $[n]$ are

$$\begin{aligned} & 1 \cdot (n-2) + 2 \cdot (n-3) + 3 \cdot (n-4) + \dots + (n-2) \cdot 1 = \\ & = (2-1) \cdot (n-2) + (3-1) \cdot (n-3) + (4-1) \cdot (n-4) + \dots + ((n-1)-1) \cdot (n-(n-1)) \\ & = \sum_{k=2}^{n-1} (k-1)(n-k) \end{aligned}$$

which is also the RHS of the identity. □

CHAPTER 6

COMPOSITIONS, SET PARTITIONS, INTEGER PARTITIONS

In this chapter we consider distribution problems. In how many ways can we distribute n objects into k boxes?

6.1 Compositions

Consider the case when the n objects are identical and the k boxes are distinct.

A sequence of integers $(a_1, a_2, \dots, a_k)$ where $a_i \geq 0$ for all $i \in [k]$ such that $a_1 + a_2 + \dots + a_k = n$ is called a weak composition of n into k parts. If, in addition, the $a_i > 0$ for all $i \in [k]$, then the sequence $(a_1, a_2, \dots, a_k)$ is called a (strong) composition of n into k parts.

Result: Weak Compositions

For all positive integers n and k , the number of weak compositions of n into k parts is

$$\binom{n+k-1}{k-1} = \binom{n+k-1}{n}$$

As an example, consider the case when $n = 5$ (five identical objects) and $k = 4$ (four distinct boxes). A possible distribution is

$$\underbrace{\star}_{\text{1st Box}} \mid \underbrace{\star\star\star}_{\text{2nd Box}} \mid \underbrace{}_{\text{3rd Box}} \mid \underbrace{\star}_{\text{4th Box}}$$

where the first box has 1 object, the second box has 3 objects, the third box has 0 objects, and the fourth box has 1 object.

To count the number of such configurations, we simply need to count the number of ways to choose $k-1$ positions, out of a total of $n+k-1$ positions, for the dividers, which is $\binom{n+k-1}{k-1}$.

Or put another way, we simply need to count the number of ways to permute the string consisting of n $\star$'s and $k-1$ $|$'s, which is $\frac{(n+k-1)!}{n!(k-1)!}$.

The previous scenario can also be phrased as follows:

Let n, k be positive integers. There are $\binom{n+k-1}{k-1}$ many distinct integer-valued vectors $(x_1, x_2, \dots, x_k)$ satisfying the equation

$$x_1 + x_2 + \dots + x_k = n$$

where $x_i \geq 0$ for all $i \in [k]$.

(Here the n represents the identical objects and each x_i represents the distinct boxes).

Suppose we impose the condition that every box must get at least one object (for this to make sense we must have $n \geq k$). Then we are counting the number of distinct integer-valued vectors $(x_1, x_2, \dots, x_k)$ satisfying the equation

$$x_1 + x_2 + \dots + x_k = n$$

where $x_i \geq 1$ for all $i \in [k]$.

In this case we can begin by distributing 1 object to every box leaving $n - k$ objects left to distribute. Distributing $n - k$ identical objects into k distinct boxes can be done in $\binom{(n-k)+(k-1)}{k-1} = \binom{n-1}{k-1}$ ways.

Result: Compositions

For all positive integers n and k , the number of (strong) compositions of n into k parts is

$$\binom{n-1}{k-1}$$

We can generalize this. Suppose we impose the condition that every box must get at least m objects (for this to make sense we must have $n \geq mk$). Then we are counting the number of distinct integer-valued vectors $(x_1, x_2, \dots, x_k)$ satisfying the equation

$$x_1 + x_2 + \dots + x_k = n$$

where $x_i \geq m$ for all $i \in [k]$.

In this case we can begin by distributing m objects to every box leaving $n - mk$ objects left to distribute. Distributing $n - mk$ identical objects into k distinct boxes can be done in

$$\binom{(n-mk)+(k-1)}{k-1} = \binom{n-(m-1)k-1}{k-1}$$

ways.

Now consider the number of compositions of n into any number of parts. This only applies to compositions, not weak compositions (because if 0 is allowed to be a part then it can be appended infinitely). The number of all compositions of n into any number of parts is

$$\sum_{k=1}^n \binom{n-1}{k-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} = 2^{n-1}$$

Problem

What is the number of all compositions of n in which the first part is not 2?

Solution

The number of all compositions of n is 2^{n-1} . The number of compositions of n in which the first part is 2 is equal to the number of compositions of $n - 2$, which is 2^{n-3} . Therefore, the number of all compositions of n in which the first part is not 2 is $2^{n-1} - 2^{n-3} = 3 \cdot 2^{n-3}$.

Problem

What is the number of compositions of n into four parts, each of which is at least 2?

Solution

First distribute 2 to each of the four parts, leaving $n - 8$ to distribute. We can distribute $n - 8$ identical objects into 4 distinct boxes in $\binom{(n-8)+(4-1)}{4-1} = \binom{n-5}{3}$ ways.

Problem

What is the number of weak compositions of 10 into five parts so that exactly two parts are 0?

Solution

Begin by picking 2 boxes that will have 0 objects. This can be done in $\binom{5}{2}$ ways. Now the remaining three boxes must form a composition of 10 into three parts, which can be done in $\binom{10-1}{3-1} = \binom{9}{2}$ ways. Therefore, the number of weak compositions of 10 into five parts so that two parts are 0 is $\binom{5}{2} \cdot \binom{9}{2}$.

6.2 Set Partitions

Consider the case when the n objects are distinct and the k boxes are identical.

We can label the distinct objects by the numbers 1 through n and our goal is to partition the set $[n]$ into k nonempty subsets.

The number of set partitions of $[n]$ into k nonempty blocks is denoted by $S(n, k)$. The numbers $S(n, k)$ are called the Stirling numbers of the second kind.

Since there is no closed formula for $S(n, k)$, let us prove some identities for some special cases.

- $S(0, 0) = 1$ since there is only one way to distribute zero objects into zero boxes, namely by not doing anything.
- If $n \geq 1$, then $S(n, 0) = 0$ since there are no ways to partition $[n]$ into 0 nonempty subsets.
- If $n < k$, then $S(n, k) = 0$ since we cannot partition $[n]$ into k non-empty subsets if there are more boxes than objects.
- If $n \geq 1$, then $S(n, 1) = 1$ since there is only one way to partition $[n]$ into one box, namely by putting all the objects into the one box.
- If $n \geq 1$, then $S(n, n) = 1$ since there is only one way to partition $[n]$ into n boxes, namely by putting one object in each box.
- If $n \geq 1$, then $S(n, n-1) = \binom{n}{2}$. By the Pigeonhole Principle, any partition of $[n]$ into $n-1$ boxes will result in a box having at least 2 objects. Furthermore, since we require that none of the boxes can be empty, it follows that every box has one object, except for one box which has 2 objects. Therefore, we simply need to choose which two objects go into the same box, which can be done in $\binom{n}{2}$ ways.
- If $n \geq 2$, then $S(n, n-2) = \binom{n}{3} + 3\binom{n}{4}$. A set partition of $[n]$ into $n-2$ bins results in two cases:
 - There is a bin with 3 elements. We can choose the elements of that bin in $\binom{n}{3}$.
 - There are two bins each with two objects. We can choose the elements of the first bin in $\binom{n}{2}$ ways and the elements of the second bin $\binom{n-2}{2}$ ways. However, using this method, we are distinguishing between the two bins which overcounts by a factor of 2. Therefore, we have $\frac{1}{2}\binom{n}{2}\binom{n-2}{2} = 3\binom{n}{4}$.

Overall, we get that $S(n, n-2) = \binom{n}{3} + 3\binom{n}{4}$.

- If $n \geq k$, then $S(n, k) = S(n-1, k-1) + k \cdot S(n-1, k)$. Count the number of ways to partition $[n]$ into k nonempty parts. One answer is, by definition, $S(n, k)$. Another way is to consider an arbitrary partition of $[n]$ into k nonempty subsets and focus on a particular element, say n .
 - If n is in a singleton $\{n\}$, then the rest of the partition is a partition of $[n-1]$ into $k-1$ subsets, which is given by $S(n-1, k-1)$.
 - If n is in a subset of size larger than 1 (i.e. with other elements), then we can partition $[n-1]$ into k subsets in $S(n-1, k)$ many ways, and then insert n into any one of these k subsets in k many ways. This gives $kS(n-1, k)$ possibilities.

Overall, we get that the number of ways to partition $[n]$ into k nonempty parts is also $S(n-1, k-1) + k \cdot S(n-1, k)$.

6.2.1 Counting Surjective Functions

Consider the case when the n objects are distinct and the k boxes are distinct.

We can label the distinct objects by the numbers 1 through n and the distinct boxes by the numbers 1 through k . As before, our goal is to partition the set $[n]$ into k nonempty subsets.

We can first partition $[n]$ into k identical boxes in $S(n, k)$ ways, and then we can label the k boxes with labels $1, 2, \dots, k$ in $k!$ ways. Therefore the number of ways to distribute n distinct objects into k distinct boxes is $k! \cdot S(n, k)$.

Result: Number of Surjective Functions

The number of surjective functions $f : [n] \rightarrow [k]$ is $k! \cdot S(n, k)$.

Proof

Given a surjective function $f : [n] \rightarrow [k]$, we can form a partition of $[n]$ by grouping those elements which are mapped to the same output in $[k]$. There are $S(n, k)$ such groupings. Now specify which subset is sent to which element of $[k]$ in $k!$ many ways. Thus, $k! \cdot S(n, k)$ gives the number of surjective functions $f : [n] \rightarrow [k]$. $\square$

So counting surjective functions is the same as distributing distinct objects into distinct bins.

Problem

Find a formula for $S(n, 2)$.

Solution

A set partition of $[n]$ into two nonempty subsets A and B will result in $B = A^c$.

Hence, we simply need to specify the subset A . There are 2^n possible subsets of $[n]$, but $A \neq \{\}$ and $A \neq [n]$ (because A and B must be nonempty subsets). Hence, A can be any one of the $2^n - 2$ subsets of $[n]$. However, using this method we are distinguishing between A and A^c and so $S(n, 2) = \frac{1}{2}(2^n - 2) = 2^{n-1} - 1$.

Now we will work towards finding a formula for $S(n, 3)$, but we first need to establish an identity.

Problem

Prove the identity

$$6S(n, 3) + 6S(n, 2) + 3S(n, 1) = 3^n$$

Proof

Consider the problem of counting the number of functions $f : [n] \rightarrow [3]$.

On one hand, to specify such a function, we need to determine $\{f(1), f(2), \dots, f(n)\}$. For each input $i \in [n]$ there are 3 possibilities for its image $f(i) \in [3]$. This gives $3 \cdot 3 \cdot \dots \cdot 3 = 3^n$ functions in total, which is also the RHS of the identity.

On the other hand, a function $f : [n] \rightarrow [3]$ can be counted according to the size of its image.

- Count the number of functions $f : [n] \rightarrow [3]$ such that $|Im(f)| = 3$. Since $|Im(f)| = 3 = |[3]|$, it follows that we are counting the number of surjective functions $f : [n] \rightarrow [3]$, which is $3! \cdot S(n, 3) = 6S(n, 3)$.
- Count the number of functions $f : [n] \rightarrow [3]$ such that $|Im(f)| = 2$. First we can pick 2 values that the function will hit, which can be done in $\binom{3}{2}$ ways. Then we can partition the inputs $[n]$ into two nonempty subsets, which can be done in $S(n, 2)$ ways. Finally, we can assign which subset gets mapped to which output value in $2!$ many ways. Overall, we get $2! \cdot \binom{3}{2} \cdot S(n, 2) = 6S(n, 2)$ functions $f : [n] \rightarrow [3]$ such that $|Im(f)| = 2$.
- Count the number of functions $f : [n] \rightarrow [3]$ such that $|Im(f)| = 1$. First we can pick the 1 value that the function will hit, which can be done in $\binom{3}{1}$ ways. Then we can partition the inputs $[n]$ into a single nonempty subset, which can be done in $S(n, 1)$ ways. Finally, we can assign the subset to the output value in $1!$ many ways. Overall, we get $1! \cdot \binom{3}{1} \cdot S(n, 1) = 3S(n, 1)$ functions $f : [n] \rightarrow [3]$ such that $|Im(f)| = 1$.

Therefore, the number of functions $f : [n] \rightarrow [3]$ is also $6S(n, 3) + 6S(n, 2) + 3S(n, 1)$. □

Problem

Find a formula for $S(n, 3)$.

Solution

In the previous problem we established that $6S(n, 3) + 6S(n, 2) + 3S(n, 1) = 3^n$.

Using the fact that $S(n, 2) = 2^{n-1} - 1$ and $S(n, 1) = 1$, we get that

$$\begin{aligned} S(n, 3) &= \frac{1}{6} (3^n - 6S(n, 2) - 3S(n, 1)) \\ &= \frac{1}{6} (3^n - 6(2^{n-1} - 1) - 3 \cdot 1) \\ &= \frac{1}{6} \cdot 3^n - 2^{n-1} + \frac{1}{2} \\ &= \frac{3^{n-1} + 1}{2} - 2^{n-1} \end{aligned}$$

Therefore, $S(n, 3) = \frac{3^{n-1} + 1}{2} - 2^{n-1}$.

Problem: Generalization

Count the number of functions $f : [n] \rightarrow [k]$ in two ways to derive an identity.

Solution

On one hand, to specify such a function, we need to determine $\{f(1), f(2), \dots, f(n)\}$. For each input $i \in [n]$ there are k possibilities for its image $f(i) \in [k]$. This gives $k \cdot k \cdot \dots \cdot k = k^n$ functions in total. On the other hand, a function $f : [n] \rightarrow [k]$ can be counted according to the size of its image, $1 \leq |Im(f)| \leq k$. Suppose f is a function such that $|Im(f)| = i$ (i.e. f hits i values in $[k]$). There are $\binom{k}{i}$ many ways of choosing these i output values. Then we can partition the inputs $[n]$ into i nonempty subsets in $S(n, i)$ ways (inputs in the same subset will have the same image). Finally, we can assign which subset gets mapped to which output value in $i!$ many ways. Overall, we get $i! \cdot \binom{k}{i} \cdot S(n, i)$ functions $[n] \rightarrow [k]$ such that $|Im(f)| = i$.

Summing over the possible values of i gives,

$$\sum_{i=1}^k i! \binom{k}{i} S(n, i) = \sum_{i=1}^k (k)_i S(n, i)$$

functions in total.

Therefore, we have

$$\sum_{i=1}^k (k)_i S(n, i) = k^n$$

as a result.

Notice that we can change the upper bound on the summation to n . This is because for $i > k$, one of the factors in $k(k-1)(k-2) \dots (k-i+1)$ will be zero, making the entire term zero. Hence we can rewrite the identity as

$$\sum_{i=1}^n (k)_i S(n, i) = k^n$$

Now this identity holds for any positive integer k , but it turns out that it holds for any real number k

Result

For all real numbers x and all non-negative integers n ,

$$\sum_{k=0}^n (x)_k S(n, k) = x^n$$

Problem

Prove that for all positive integers n

$$\sum_{k=1}^n (-1)^k k! S(n, k) = (-1)^n$$

Explain what this identity means without resorting to mathematical formulas.

Proof

Let $x = -1$ in the identity we derived. Then we get

$$\sum_{k=1}^n (-1)^k k! S(n, k) = (-1)^n$$

Notice that we are summing over all the possible sizes for the image. As this sum is ± 1 , it follows that the sum of the numbers of surjective functions with an even-sized image (which corresponds to the terms with an even value of k) differs by ± 1 from the sum of the numbers of surjective functions with an odd-sized image (which corresponds to the terms with an odd value of k). Therefore, the number of even-sized image surjective functions differs by ± 1 from the number of odd-sized image surjective functions. $\square$

6.2.2 Bell Numbers

The number of set partitions of $[n]$ into any number of nonempty parts is denoted by $B(n)$ where

$$B(n) = \sum_{k=0}^n S(n, k)$$

The numbers $B(n)$ are called the Bell numbers (we set $B(0) = 1$).

Problem

Prove that for all non-negative integers n ,

$$B(n+1) = \sum_{k=0}^n \binom{n}{k} B(k)$$

Proof

Count the number of ways to partition $[n+1]$ into any number of nonempty parts. One answer is, by definition, $B(n+1)$. Another way is to consider an arbitrary partition of $[n+1]$ and focus on a particular element, say $n+1$. There are $\binom{n}{k}$ ways to pick k elements that will not be in the same box as $n+1$. These k elements can be partitioned in anyway whatsoever in $B(k)$ ways. This gives $\binom{n}{k} B(k)$ partitions of $[n+1]$ where k elements are not in the same box as $n+1$. Summing over the possible values of k gives that the number of ways to partition $[n+1]$ into any number of nonempty parts is also

$$\sum_{k=0}^n \binom{n}{k} B(k)$$

□

Alternatively, we could have picked k elements which do go into the same box as $n+1$ in $\binom{n}{k}$ ways. Then the remaining $n-k$ elements can be partitioned in anyway whatsoever in $B(n-k)$ ways. Summing over the possible values of k gives

$$\sum_{k=0}^n \binom{n}{k} B(n-k) = \sum_{k=0}^n \binom{n}{n-k} B(n-k) = \sum_{k=0}^n \binom{n}{k} B(k)$$

as expected.

6.3 Integer Partitions

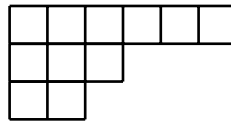
Consider the case when the n objects are identical and the k boxes are identical.

An integer partition of n is a way of writing n as a sum of positive integers, where the order of the integers does not matter.

Let $a_1 \geq a_2 \geq \dots \geq a_k \geq 1$ be integers such that $a_1 + a_2 + \dots + a_k = n$. Then the sequence $(a_1, a_2, \dots, a_k)$ is called a partition of the integer n . The number of partitions of n into exactly k parts is denoted by $p_k(n)$. The number of partitions of n into any number of parts is denoted by $p(n)$.

A nice way to visualize integer partitions is using a Ferrers shape. A Ferrers shape of a partition $p = (a_1, a_2, \dots, a_k)$ is a set of n square boxes such that in the i^{th} row we have a_i boxes and all rows start at the same vertical line.

For example, the Ferrers shape of $p = (6, 3, 2)$ is



The Durfee square of a partition p is the largest square that fits in the top left corner of the Ferrers shape of p .

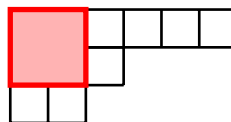
Problem

If we know the parts of a partition p , how can we figure out the side length of its Durfee square without drawing the Ferrers shape of p ?

Solution

Let $p = (p_1, p_2, \dots, p_k)$. Then the side length of the Durfee square is the largest i such that $p_i \geq i$.

For example, the Durfee square of $p = (6, 3, 2)$ has side length 2:



A Farooq rectangle of a partition p is the largest area rectangle that fits in the top left corner of the Ferrers shape of p .

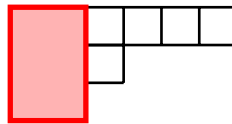
Problem

If we know the parts of a partition p , how can we figure out the dimensions of its Farooq rectangle without drawing the Ferrers shape of p ?

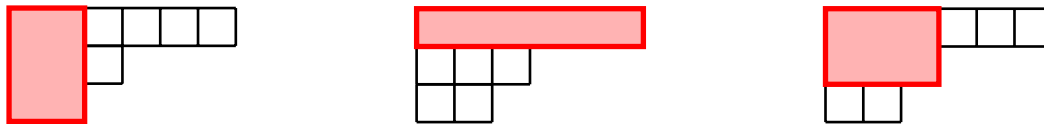
Solution

Let $p = (p_1, p_2, \dots, p_k)$. Then the dimensions of a Farooq rectangle is (i, p_i) such that $i \cdot p_i$ is a maximum.

For example, a Farooq rectangle of $p = (6, 3, 2)$ has area 6:



Unlike Durfee squares, Farooq rectangles are not unique. For example, the partition $p = (6, 3, 2)$ has three Farooq rectangles of maximum area 6:



Problem

Under what conditions is a Farooq rectangle equal to the Durfee square?

Solution

One condition under which a Farooq rectangle equals the Durfee square is when we have a partition of the form $p = (a_1, a_2, \dots, a_k)$ where $a_1 = a_2 = \dots = a_k$ and $k = a_1$. So in other words, all the parts must be of the same size, say k , and there must be exactly k parts.

For example, consider $p = (1)$:



Here a Farooq rectangle is equal to the Durfee square (both are 1×1 and have area 1). However, if we add another block to get $p = (2)$ or $p = (1, 1)$ then a Farooq rectangle is no longer equal to the Durfee square:

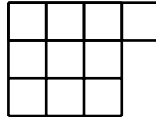


But when $p = (2, 2)$ then a Farooq rectangle is equal to the Durfee square:



Hence we see that when all the parts of the partition are of same size, say k , and there are exactly k parts then a Farooq rectangle is equal to the Durfee square.

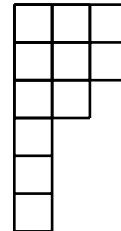
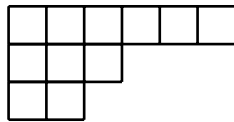
Remark. The above condition is not the only condition under which a Farooq rectangle is equal to the Durfee square. Indeed, consider the partition $p = (4, 3, 3)$



Here a Farooq rectangle is equal to the Durfee square, but not all the parts are of the same size.

If we reflect the Ferrers shape of a partition p with respect to its main diagonal then we get the Ferrers shape of the conjugate partition of p .

For example, the Ferrers shape of $p = (6, 3, 2)$ and the Ferrers shape of the conjugate partition of p is $p' = (3, 3, 2, 1, 1, 1)$ are



A partition of n is called self-conjugate if it is equal to its conjugate. (e.g. the partition $(6, 3, 2)$ is not self-conjugate)

It is clear from the visual definition that the Durfee square of a partition and its conjugate partition have the same size and that a Farooq rectangle of a partition and its conjugate partition have the same size.

CHAPTER 7

A CLOSER LOOK AT PERMUTATIONS

So far we have viewed permutations as linear arrangements of different objects. However, we can also view permutations as bijective functions from a set of objects to itself.

We can display permutations in multiple ways. Consider the set $[8]$ and define the bijective function $f : [8] \rightarrow [8]$ by $f(1) = 2, f(2) = 4, f(3) = 7, f(4) = 1, f(5) = 3, f(6) = 8, f(7) = 5, f(8) = 6$.

- One-line notation: The permutation above can be written compactly as 24713856.
- Canonical cycle notation: The permutation above can be written compactly as $(412)(753)(86)$.
(Each cycle should be written with its largest element first, and the cycles will be written in increasing order of their first elements.)

Result: Permutations of a Given Type

If an n -permutation p has a_i cycles of length i , for $i \in [n]$ then we say that $(a_1, a_2, \dots, a_n)$ is the type or cycle type of p . The number of permutations of type $(a_1, a_2, \dots, a_n)$ is

$$\frac{n!}{a_1! a_2! \dots a_n! \cdot 1^{a_1} 2^{a_2} \dots n^{a_n}}$$

The idea is to write down the elements of $[n]$ in some linear order in $n!$ ways. Next, place parentheses going left to right according to the required cycle lengths (i.e. first a_1 pairs of parentheses creating a_1 1-cycles, then a_2 pairs of parentheses creating a_2 2-cycles, and so on). There is only one way to do this. Now we need to account for over counting.

Consider the a_i cycles of length i . These can be permuted among themselves in $a_i!$ ways without changing the permutation. Hence, we divide by $a_i!$ for all $i \in [n]$. Moreover, the elements within each cycle of length i can be shifted i times without changing the cycle. As there are a_i cycles of length i we divide by i^{a_i} for all $i \in [n]$.

7.1 Cycles in Permutations

Consider the problem of counting the number of permutations of $[n]$ which consist of k cycles.

The number of n -permutations with k cycles is denoted by $c(n, k)$. The numbers $c(n, k)$ are called the signless Stirling numbers of the first kind. The numbers $s(n, k) = (-1)^{n-k}c(n, k)$ are called the Stirling numbers of the first kind.

Problem

Compute $c(6, 3)$.

Solution

In order to decompose a permutation of $[6]$ into three cycles, the possible cycle configurations are:

- 4-cycle, 1-cycle, 1-cycle which corresponds to the type $(a_1, a_2, a_3, a_4, a_5, a_6) = (2, 0, 0, 1, 0, 0)$.

Hence there are

$$\frac{6!}{2! \cdot 1! \cdot 1! \cdot 4^1} = 90$$

permutations of type $(2, 0, 0, 1, 0, 0)$.

- 3-cycle, 2-cycle, 1-cycle which corresponds to the type $(a_1, a_2, a_3, a_4, a_5, a_6) = (1, 1, 1, 0, 0, 0)$.

Hence there are

$$\frac{6!}{1! \cdot 1! \cdot 1! \cdot 1^1 \cdot 2^1 \cdot 3^1} = 120$$

permutations of type $(2, 0, 0, 1, 0, 0)$.

- 2-cycle, 2-cycle, 2-cycle which corresponds to the type $(a_1, a_2, a_3, a_4, a_5, a_6) = (0, 3, 0, 0, 0, 0)$.

Hence there are

$$\frac{6!}{3! \cdot 2^3} = 15$$

permutations of type $(2, 0, 0, 1, 0, 0)$.

Therefore, $c(6, 3) = 90 + 120 + 15 = 225$.

Since there is no closed formula for $c(n, k)$, let us prove some identities for some special cases.

- $c(0, 0) = 1$ since there is only one way for a permutation of zero objects to form zero cycles, namely by doing nothing.
- If $n \geq 1$, then $c(n, 0) = 0$ since a nonempty permutation cannot have zero cycles.
- If $n < k$, then $c(n, k) = 0$ since a permutation cannot have more cycles than number of elements.
- If $n \geq 1$, then $c(n, 1) = (n-1)!$. Consider n -permutations that have a single cycle of length n . The elements in the cycle can be ordered in $\frac{n!}{n} = (n-1)!$ ways (as we need to account for cyclic symmetry).
- If $n \geq 1$, then $c(n, n) = 1$ since there is only one way for an n -permutation to have n cycles, which is the identity permutation.

- If $n \geq 2$, then $c(n, n-1) = \binom{n}{2}$. Consider n -permutations that have $n-1$ cycles. By the Pigeonhole Principle, there is a cycle of length 2. In fact, every cycle is of length 1, except for one cycle which is of length 2. Therefore, we simply need to choose which two objects go into the 2-cycle, which can be done in $\binom{n}{2}$ ways (note, in general, we also need to consider permuting the 2 elements which can be done in $\frac{2!}{2} = 1$ way).
- If $n \geq k$, then $c(n, k) = c(n-1, k-1) + (n-1) \cdot c(n-1, k)$. Count the number of n -permutations that have k cycles. One answer is, by definition, $c(n, k)$. Another way is to focus on a particular element, say n . There are two possibilities:
 - If n is in a cycle by itself, then we can form $k-1$ more cycles by breaking up the remaining $n-1$ elements into $k-1$ cycles, which is given by $c(n-1, k-1)$.
 - If n is in a cycle of size larger than 1 (i.e. with other elements), then we can form k cycles using $[n-1]$ in $c(n-1, k)$ ways. Next we can insert n into one of these k cycles. There are $n-1$ possible locations for where n can go:

$$(a_1 \square a_2 \square) (a_3 \square a_4 \square a_5 \square) \dots (a_{n-1} \square)$$

which gives $(n-1) \cdot c(n-1, k)$ possibilities.

(Note we do not need to insert n at the beginning of a cycle since that case is covered by inserting n at the end of a cycle, that is, $(na_1a_2) = (a_1a_2n)$).

Overall, we get that the number of n -permutations with k cycles is also $c(n-1, k-1) + (n-1) \cdot c(n-1, k)$.

- If $n \geq 0$ then $\sum_{k=0}^n c(n, k) = n!$. Count the number of permutations of $[n]$. One answer is, by definition, $n!$. Another way is to consider cases depending on the number of cycles present in the permutation (which can be between 1 and n inclusive).

Problem

Prove that the inequality $S(n, k) \leq c(n, k)$ holds.

Proof

Both $S(n, k)$ and $c(n, k)$ involve a partition $[n]$, but $c(n, k)$ also takes into account the order of elements within a box, whereas $S(n, k)$ does not. Therefore, $S(n, k) \leq c(n, k)$. $\square$

Problem

Find an explicit formula for

- a) $c(n, n-2)$
 b) $c(n, n-3)$

Solution: a)

Consider an n -permutation that has $n-2$ cycles. There are two cases to consider:

- There is one 3-cycle and $n-3$ 1-cycles. We can choose the elements of the 3-cycle in $\binom{n}{3}$ ways and order the elements of the 3-cycle in $\frac{3!}{3}$ ways.
- There are two 2-cycles and $n-4$ 1-cycles. We can choose the elements of the first 2-cycle in $\binom{n}{2}$ ways and choose the elements of the second 2-cycle in $\binom{n-2}{2}$ ways. Since the order of the two 2-cycles cycles does not matter we have $\frac{1}{2!} \binom{n}{2} \binom{n-2}{2}$.

Overall, we get that

$$c(n, n-2) = 2 \binom{n}{3} + \frac{1}{2} \binom{n}{2} \binom{n-2}{2} = \frac{3n^4 - 10n^3 + 9n^2 - 2n}{24}$$

Solution: b)

Consider an n -permutation that has $n-3$ cycles. There are three cases to consider:

- There is one 4-cycle and $n-4$ 1-cycles. We can choose the elements of the 4-cycle in $\binom{n}{4}$ ways and order the elements of the 4-cycle in $\frac{4!}{4}$ ways.
- There is one 3-cycle, one 2-cycle, and $n-5$ 1-cycles. We can choose the elements of the 3-cycle in $\binom{n}{3}$ ways and order the elements of the 3-cycle in $\frac{3!}{3}$ ways. Finally, we can choose the elements of the 2-cycle in $\binom{n-3}{2}$ ways.
- There are three 2-cycles and $n-6$ 1-cycles. We can choose the elements of the first 2-cycle in $\binom{n}{2}$ ways, choose the elements of the second 2-cycle in $\binom{n-2}{2}$ ways, and choose the elements of the third 2-cycle in $\binom{n-4}{2}$ ways. Since the order of the three 2-cycles cycles does not matter we have $\frac{1}{3!} \binom{n}{2} \binom{n-2}{2} \binom{n-4}{2}$.

Overall, we get that

$$c(n, n-3) = 6 \binom{n}{4} + 2 \binom{n}{3} \binom{n-3}{2} + \frac{1}{6} \binom{n}{2} \binom{n-2}{2} \binom{n-4}{2} = \frac{n^6 - 7n^5 + 17n^4 - 17n^3 + 6n^2}{48}$$

Problem

Let $r(n)$ be the number of n -permutations whose square is the identity permutation. Find a recursive formula for the number $r(n)$.

Solution

Consider an n -permutation whose square is the identity permutation. All cycles present must be either 1-cycles or 2-cycles. Focus on the n^{th} element.

- If n forms a 1-cycle then the remaining $n - 1$ entries consist of only 1-cycles and 2-cycles in $r(n - 1)$ ways.
- If n forms a 2-cycle then we can pick the other entry of the 2-cycle in $\binom{n-1}{1}$ ways. Now remaining $n - 2$ entries consist of only 1-cycles and 2-cycles in $r(n - 2)$ ways. Overall, we have $(n - 1)r(n - 2)$ possibilities when n is part of a 2-cycle.

Therefore, for $n \geq 2$, the number of n -permutations whose square is the identity permutation is

$$r(n) = r(n - 1) + (n - 1)r(n - 2)$$

Problem

Let $t(n)$ be the number of n -permutations whose cube is the identity permutation. Find a recursive formula for the number $t(n)$.

Solution

Consider an n -permutation whose cube is the identity permutation. All cycles present must be either 1-cycles or 3-cycles. Focus on the n^{th} element.

- If n forms a 1-cycle then the remaining $n - 1$ entries consist of only 1-cycles and 3-cycles in $t(n - 1)$ ways.
- If n forms a 3-cycle then we can pick the other two entries of the 3-cycle in $\binom{n-1}{2}$ ways. We can order the 3-cycle in $\frac{3!}{3} = 2$ ways. Now remaining $n - 3$ entries consist of only 1-cycles and 3-cycles in $t(n - 3)$ ways. Overall, we have $2\binom{n-1}{2}t(n - 3) = (n - 1)(n - 2)t(n - 3)$ possibilities when n is part of a 3-cycle.

Therefore, for $n \geq 3$, the number of n -permutations whose cube is the identity permutation is

$$t(n) = t(n - 1) + [(n - 1)(n - 2)]t(n - 3)$$

Problem: Generalization

Let $f(n)$ be the number of n -permutations whose k^{th} power (where $k \geq 2$) is the identity permutation. Find a recursive formula for the number $f(n)$.

Solution

Consider an n -permutation whose k^{th} power is the identity permutation. All cycles present must be either 1-cycles or k -cycles. Focus on the n^{th} element.

- If n forms a 1-cycle then the remaining $n - 1$ entries consist of only 1-cycles and k -cycles in $f(n - 1)$ ways.
- If n forms a k -cycle then we can pick the other $k - 1$ entries of the k -cycle in $\binom{n-1}{k-1}$ ways. We can order the k -cycle in $\frac{k!}{k} = (k - 1)!$ ways. Now remaining $n - k$ entries consist of only 1-cycles and k -cycles in $f(n - k)$ ways. Overall, we have $(k - 1)! \binom{n-1}{k-1} f(n - k) = (n - 1)(n - 2) \dots (n - k + 1)$ possibilities when n is part of a k -cycle.

Therefore, the number of n -permutations whose k^{th} power is the identity permutation is

$$f(n) = f(n - 1) + [(n - 1)(n - 2) \dots (n - k + 1)]f(n - k)$$

if $n \geq k$.

7.2 Inversions in Permutations

Let $p = p_1 p_2 \dots p_n$ be a permutation. An inversion of p is a pair of entries (p_i, p_j) so that $i < j$ but $p_i > p_j$. Let us call a permutation even (odd) if it has an even (odd) number of inversions.

- If $p = 462513$ then p has 10 inversions:

$$(4, 2), (4, 1), (4, 3), (6, 2), (6, 5), (6, 1), (6, 3), (2, 1), (5, 1), (5, 3)$$

Hence, p is an even permutation.

- If $p = 123456$ then p has 0 inversions. Hence, p is an even permutation.
- If $p = 654321$ then p has 15 inversions:

$$(6, 5), (6, 4), (6, 3), (6, 2), (6, 1), (5, 4), (5, 3), (5, 2), (5, 1), (4, 3), (4, 2), (4, 1), (3, 2), (3, 1), (2, 1)$$

Hence, p is an odd permutation.

Inversions can be viewed as a measure of how “out of order” a permutation is.

Hence, the minimum number of inversions for a permutation p is 0 when the permutation is “fully in order” (e.g. $1\,2\,3\,\dots\,n$).

On the other hand, the maximum number of inversions for a permutation p is $\sum_{i=0}^{n-1} i = \frac{n(n-1)}{2} = \binom{n}{2}$ when the permutation is “fully out of order” (e.g. $n\,(n-1)\,(n-2)\,\dots\,2\,1$).

Problem

Let $I(n, k)$ be the number of n -permutations that have k inversions. Prove that $I(n, k) = I\left(n, \binom{n}{2} - k\right)$.

Proof

Let A be the set of n -permutations with k inversions and let B be the set of n -permutations with $\binom{n}{2} - k$ inversions.

Define a function $f : A \rightarrow B$ by $f(p) = p'$ where the permutation p' is obtained by writing the permutation p in reverse order.

For a given n -permutation $p = p_1 p_2 \dots p_n$, the pair (p_i, p_j) is an inversion if and only if (p_i, p_j) is not an inversion in p' . Hence, p has k inversions if and only if p' has $\binom{n}{2} - k$ inversions.

Clearly f is a bijection as f is its own inverse. It now follows that $|A| = |B|$. In other words, the number of n -permutations that have k inversions is equal to the number of n -permutations that have $\binom{n}{2} - k$ inversions. Therefore, $I(n, k) = I\left(n, \binom{n}{2} - k\right)$. $\square$

Problem

Prove that

$$\sum_{k=0}^{\binom{n}{2}} I(n, k) x^k = (1+x)(1+x+x^2) \dots (1+x+\dots+x^{n-1})$$

Proof

The coefficients of x^k on the LHS count the number of n -permutations with k inversions. We will show that the coefficients of x^k on the RHS also counts the number of n -permutations with k inversions.

Consider a construction of an n -permutation:

- Element 1 can only be placed in one location and there are 0 inversions (x^0).
- Element 2 can be placed in two locations. If 2 is placed after 1 then the number of inversions does not change (x^0). If 2 is placed before 1 then the number of inversions increases by 1 (x^1). Overall, we have ($x^0 + x^1$).
- Element 3 can be placed in three locations. If 3 is placed in the last location then the number of inversions does not change (x^0). If 3 is placed in between 1 and 2 then the number of inversions increases by 1 (x^1). If 3 is placed at the front then the number of inversions increases by 2 (x^2). Overall, we have ($x^0 + x^1 + x^2$).
- $\vdots$
- Element n can be placed in n locations. If n is placed in the last position then the number of inversions does not change (x^0). If n is placed second to last then the number of inversions increases by 1 (x^1). If n is placed at the front then the number of inversions increases by $n-1$ (x^{n-1}). Overall, we have ($x^0 + x^1 + \dots + x^{n-1}$).

Since we need to place all n elements we get

$$(x^0)(x^0 + x^1)(x^0 + x^1 + x^2) \dots (x^0 + x^1 + \dots + x^{n-1}) = (1+x)(1+x+x^2) \dots (1+x+\dots+x^{n-1})$$

where the coefficient of x^k is the number of n -permutations with k inversions. □

Problem

Deduce that the number of even n -permutations is the same as the number of odd n -permutations.

Solution

Let $x = -1$ in the previous result to get

$$\sum_{k=0}^{\binom{n}{2}} I(n, k)(-1)^k = 0$$

Notice that we are summing over all the possible number of inversions of an n -permutation. As this sum is zero, it follows that the sum of the numbers of permutations with an even number of inversions (which corresponds to the terms with an even value of k) is equal to the sum of the numbers of permutations with an odd number of inversions (which corresponds to the terms with an odd value of k). Therefore, the number of even n -permutations is the same as the number of odd n -permutations.

As a closed formula for $I(n, k)$ may not exist, let us prove some identities for some special cases.

- $I(0, 0) = 1$ since there is only one way for a permutation of zero objects to have zero inversions.
- If $n \geq 1$, then $I(n, 0) = 1$ since there is only one n -permutation that has 0 inversions, which is the identity permutation (i.e. there is only one way for all the numbers to be in the correct order).
- If $n \geq 1$, then $I\left(n, \binom{n}{2}\right) = 1$ since there is only one n -permutation that has $\binom{n}{2}$ inversions, which is the permutation $f : [n] \rightarrow [n]$ defined by $f(i) = n - i + 1$ (i.e. there is only one way for all the numbers to be in the reverse order).
- Let $x = 1$ in the previous result to get

$$\sum_{k=0}^{\binom{n}{2}} I(n, k) = 2 \cdot 3 \cdot \dots \cdot n = n!$$

which says that summing over all the possible number of inversions of an n -permutation is equal to the number of n -permutations which is $n!$.

Now our goal is to find formulas for $I(n, 1)$, $I(n, 2)$, and $I(n, 3)$. To do so, we will recall the main idea used in showing

$$\sum_{k=0}^{\binom{n}{2}} I(n, k)x^k = (1+x)(1+x+x^2)\dots(1+x+\dots+x^{n-1})$$

which was for each $i \in [n]$, in increasing order, insert it anywhere in the line of the previously made numbers. The i^{th} number can be inserted in i places, and the number of inversions that insertion will add is anywhere between 0 and $i - 1$.

Problem

Find an explicit formula for $I(n, 1)$ where $n \geq 1$.

Solution

Consider the number of ways to end up with exactly one inversion using our construction above. There must have been a step which created one inversion (and every other step created zero). Hence, we simply need to choose which step created the single inversion which can be chosen in $\binom{n-1}{1}$ ways (since the first step, namely the insertion of 1, cannot create any inversions).

Therefore, for $n \geq 1$, we have

$$I(n, 1) = \binom{n-1}{1} = n - 1$$

Problem

Find an explicit formula for $I(n, 2)$ where $n \geq 2$.

Solution

Consider the number of ways to end up with exactly two inversions using our construction above. There are two ways in which this can happen

- there must have been two steps which created one inversion each (and every other step created zero). Hence, we simply need to choose which two steps created the two single inversions which can be chosen in $\binom{n-1}{2}$ ways.
- there must have been a step which created two inversions (and every other step created zero). Hence, we simply need to choose which step created the two inversions which can be chosen in $\binom{n-2}{1}$ ways (since the first two steps, namely the insertion of 1 and 2, cannot create two inversions).

Therefore, for $n \geq 2$ we have $I(n, 2) = \binom{n-1}{2} + \binom{n-2}{1} = \frac{1}{2}(n^2 - n - 2)$.

Problem

Find an explicit formula for $I(n, 3)$ where $n \geq 3$.

Solution

Consider the number of ways to end up with exactly three inversions using our construction above. There are three ways in which this can happen

- there must have been three steps which created one inversion each (and every other step created zero). Hence, we simply need to choose which three steps created the three single inversions which can be chosen in $\binom{n-1}{3}$ ways.
- there must have been a step which created two inversions and a step which created one inversion (and every other step created zero). We can pick a step that will create two inversions in $\binom{n-2}{1}$ ways (as we need to exclude the numbers 1 and 2) followed by a step that will create one inversion in $\binom{n-2}{1}$ ways (as we need to exclude the number 1 and the number that created the two inversions in the first step).
- there must have been a step which created three inversions (and every other step created zero). Hence, we simply need to choose which step created the three inversions which can be chosen in $\binom{n-3}{1}$ ways (since the first three steps, namely the insertion of 1, 2, and 3, cannot create three inversions).

Therefore, for $n \geq 3$ we have $I(n, 3) = \binom{n-1}{3} + (n-2)^2 + \binom{n-3}{1} = \frac{1}{6}(n^3 - 7n)$.

Problem

- Find an explicit formula for $I(n, \binom{n}{2} - 1)$.
- Find an explicit formula for $I(n, \binom{n}{2} - 2)$.
- Find an explicit formula for $I(n, \binom{n}{2} - 3)$.

Solution

We previously established that $I(n, k) = I(n, \binom{n}{2} - k)$. It now follows that

- $I(n, \binom{n}{2} - 1) = I(n, 1) = n - 1$
- $I(n, \binom{n}{2} - 2) = I(n, 2) = \frac{1}{2}(n^2 - n - 2)$
- $I(n, \binom{n}{2} - 3) = I(n, 3) = \frac{1}{6}(n^3 - 7n)$

Remark. The numbers $I(n, k)$ are called the Mahonian numbers. As a result of the identity $I(n, k) = I(n, \binom{n}{2} - k)$, the Mahonian numbers can be arranged in a symmetric “triangle” that resembles Pascal’s triangle.

$$\begin{array}{ccccccccccc}
 & & & & & & 1 & & & & & \\
 & & & & & & 1 & & 1 & & & \\
 & & & & & 1 & & 2 & & 2 & & 1 \\
 & & & 1 & & 3 & & 5 & & 6 & & 5 & & 3 & & 1 \\
 & 1 & & 4 & & 9 & & 14 & & 17 & & 17 & & 14 & & 10 & & 6 & & 3 & & 1 \\
 & & & & & & & & & \vdots & & & & & & & & & & & &
 \end{array}$$

The k^{th} entry row in n is the number $I(n, k)$ (indexing begins with zero).

7.3 100 Students Riddle

Problem

A course instructor has set up a challenge for his 100 students.

- Assume all the students have different names.
- There are 100 boxes inside of a room and each box is labelled with the name of a student.
- On 100 different pieces of paper, the instructor writes the names of each student and randomly distributes the names into the boxes (one paper per box) and closes the box.
- The students enter the room one by one and can open 50 boxes to search for the box containing their name. After, they must leave the room.
- The whole class wins if every single student finds their own name in a box.
- The whole class loses if one student fails to find their own name.

The students may only discuss their strategy before they enter the room. Once a student enters, they can no longer communicate with any students. Find a strategy that gives the class the best chance of winning the game.

Naive Approach

The probability that a student finds the box that contains their name is $\frac{1}{2}$. Hence, if every student that enters the room randomly opens 50 boxes, then the probability that all 100 students find the box that contains their name is $\left(\frac{1}{2}\right)^{100}$. So, it is extremely unlikely that they will win.

Smarter Approach

Each student should begin by opening the box that has their name on it. If that box contains their name, then they are done. Otherwise, that box has the name of another student and so the next box they should open is the box with that student's name. They should continue in this fashion until either they find the box that contains their name or they have opened 50 boxes.

Probability of Success

There are a couple of things to note with this new strategy:

- The main idea is that when the 100 names are randomly distributed in the 100 boxes, it corresponds to a permutation of the 100 names. This permutation will contain cycles of certain lengths.
- Every box contains a single, unique name. In other words, either a name is in the correct box or it points to another box. Furthermore, there is only one name pointing to each box and hence only one way to get to any box.
- The boxes are forming cycles. A box is a part of exactly one cycle as there is only one name pointing to it and only one name pointing out of it.

- When a student starts at the box labeled with their name, they are on the cycle that contains the box with their name (as there is only one name that points to that box). By following the cycle they are guaranteed to find the box containing their name.
- What determines whether or not a student finds the box containing their name is the length of the cycle. If the length of the cycle is at most 50 then the student will find their name. If the cycle is of length 51 or greater then the student will not find the box containing their name. Furthermore, if there is a cycle of length 51 or greater then all students on that cycle will fail to find their name.

Therefore, the probability that all 100 students find the box containing their name is equal to the probability that there are no cycles of length greater than 50. We can compute this as follows

$$P(\text{Success}) = 1 - P(\text{a cycle of length greater than 50})$$

Note that if there is a cycle of length greater than 50, then it is the only cycle of length greater than 50 (as we only have 100 names).

Consider forming a cycle of length k where $50 < k \leq 100$. We first need to choose k boxes that will participate in the cycle of length k . This can be done in $\binom{100}{k}$ ways. Now the k names within the boxes can be permuted in $\frac{k!}{k} = (k-1)!$ ways (accounting for cyclic symmetry). The remaining tickets can be permuted in $(100-k)!$ ways. Hence there are $\binom{100}{k} \cdot (k-1)! \cdot (100-k)! = \frac{100!}{k}$ permutations that contain a cycle of length k where $50 < k \leq 100$.

Therefore,

$$P(\text{a cycle of length greater than 50}) = \frac{\frac{100!}{51} + \frac{100!}{52} + \dots + \frac{100!}{100}}{100!} = \frac{1}{51} + \frac{1}{52} + \dots + \frac{1}{100}$$

and so

$$\begin{aligned} P(\text{Success}) &= 1 - P(\text{a cycle of length greater than 50}) \\ &= 1 - \left(\frac{1}{51} + \frac{1}{52} + \dots + \frac{1}{100} \right) \\ &= 1 - (H_{100} - H_{50}) \\ &= 1 - 0.68817 \\ &= 0.311828 \end{aligned}$$

where $H_n = \sum_{k=1}^n \frac{1}{k}$ is the n^{th} Harmonic number.

Therefore, the probability of all 100 students finding the box that contains their name is 31.18%.

Generalization

Suppose we had $2n$ students and each student was allowed to open n boxes in an attempt to find their name. Once again,

$$\begin{aligned}
 P(\text{Success}) &= 1 - P(\text{a cycle of length greater than } n) \\
 &= 1 - \frac{1}{(2n)!} \left[\frac{(2n)!}{n+1} + \frac{(2n)!}{n+1} + \dots + \frac{(2n)!}{2n} \right] \\
 &= 1 - \left(\frac{1}{n+1} + \frac{1}{n+1} + \dots + \frac{1}{2n} \right) \\
 &= 1 - (H_{2n} - H_n)
 \end{aligned}$$

Let us compute $H_{2n} - H_n$ for some values:

n	1	2	3	4	5	50
$H_{2n} - H_n$	$\frac{1}{2}$	$\frac{7}{12}$	$\frac{37}{60}$	$\frac{533}{840}$	$\frac{1627}{2520}$	
approximation	0.5	0.583333	0.616667	0.634524	0.645635	0.688172

So it appears that $H_{2n} - H_n$ may be approaching a limiting value L when $n \rightarrow \infty$.

$$\begin{aligned}
 L &= \lim_{n \rightarrow \infty} (H_{2n} - H_n) \\
 &= \int_1^{2n} \frac{1}{x} dx - \int_1^n \frac{1}{x} dx \\
 &= \int_n^{2n} \frac{1}{x} dx \\
 &= \ln(2) \\
 &\approx 0.693147
 \end{aligned}$$

Thus,

$$P(\text{Success}) = 1 - \ln(2) \approx 0.306853$$

which means that even for an arbitrary large number of students, the probability that each one finds their name is still above 30%.

CHAPTER 8

THE PRINCIPLE OF INCLUSION-EXCLUSION

Let S be a set of elements and let $P_1, P_2, \dots, P_n$ be a list of properties that elements of S may or may not have. For each property P_i , define the subset $A_i \subseteq S$ of elements that satisfy property P_i .

We can use the Principle of Inclusion-Exclusion to calculate the number of elements of S that satisfy at least one of the properties.

Let $A_1, A_2, \dots, A_n$ be finite sets. Then

$$\begin{aligned} \left| \bigcup_{i=1}^n A_i \right| &= \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \\ &= \sum_{k=1}^n (-1)^{k+1} \left(\sum_{1 \leq i_1 < \dots < i_k \leq n} |A_{i_1} \cap \dots \cap A_{i_k}| \right) \end{aligned}$$

In words, to count the number of elements in a finite union of finite sets, we begin by summing the cardinalities of the individual sets, but now we need to adjust for the elements that appear in more than one set. Hence, we subtract the number of elements that appear in at least two sets, then add back the number of elements that appear in at least three sets, then subtract the number of elements that appear in at least four sets, and so on. This process ends after we account the term for the intersection of all sets.

We can also use the Principle of Inclusion-Exclusion to calculate the number of elements of S that satisfy none of the properties $P_1, P_2, \dots, P_n$.

To do this, calculate the number of elements of S that satisfy at least one of the properties and then subtract this number from the total number of elements:

$$\left| \bigcap_{i=1}^n A_i^c \right| = |S| - \left| \bigcup_{i=1}^n A_i \right|$$

Problem

How many positive integers are there that are divisible by at least one of 5, 6, and 7, and are at most 1000?

Solution

Let $S = [1000]$, P_1 be the property that an integer is divisible by 5, P_2 be the property that an integer is divisible by 6, and P_3 be the property that an integer is divisible by 7. Now define A_i to be the subset of S whose elements satisfy property P_i . Then

$$\begin{aligned} |A_1 \cup A_2 \cup A_3| &= |A_1| + |A_2| + |A_3| - |A_1 \cap A_2| - |A_1 \cap A_3| - |A_2 \cap A_3| + |A_1 \cap A_2 \cap A_3| \\ &= \left\lfloor \frac{1000}{5} \right\rfloor + \left\lfloor \frac{1000}{6} \right\rfloor + \left\lfloor \frac{1000}{7} \right\rfloor - \left\lfloor \frac{1000}{5 \cdot 6} \right\rfloor - \left\lfloor \frac{1000}{5 \cdot 7} \right\rfloor - \left\lfloor \frac{1000}{6 \cdot 7} \right\rfloor + \left\lfloor \frac{1000}{5 \cdot 6 \cdot 7} \right\rfloor \\ &= 200 + 166 + 142 - 33 - 28 - 23 + 4 \\ &= 428 \end{aligned}$$

Therefore, by the Principle of Inclusion-Exclusion, there are 428 positive integers in $[1000]$ that are divisible by at least one of 5, 6, and 7.

Problem

How many positive integers are not divisible by 5, 6, or 7, and are at most 1000?

Solution

In the previous problem we found that there are 428 positive integers in $[1000]$ that are divisible by at least one of 5, 6, and 7.

Hence the number of integers in $[1000]$ that are not divisible by 5, 6, or 7 is

$$|A_1^c \cap A_2^c \cap A_3^c| = |(A_1 \cup A_2 \cup A_3)^c| = |S| - |A_1 \cup A_2 \cup A_3| = 1000 - 428 = 572$$

Therefore, by the Principle of Inclusion-Exclusion, there are 572 positive integers in $[1000]$ that are not divisible by 5, 6, or 7.

8.1 Derangements

Let $f : [n] \rightarrow [n]$ such that $f(i) = i$. We say that i is a fixed point of f . In other words, fixed points are inputs that get mapped to themselves.

Problem

Consider the permutations of the set $[n]$. How many of these permutations have no fixed points?

A permutation with no fixed points is called a derangement. The number of derangements of the set $[n]$ is denoted by $D(n)$ or $!n$.

We can use the Principle of Inclusion-Exclusion to find a formula for $D(n)$ by observing that the number of permutations with zero fixed points is equal to the number of all permutations minus the number of permutations with at least one fixed point.

Let $S = [n]$ and let P_i be the property that a permutation has element i fixed (for $i \in [n]$). Now define A_i to be the set of permutations of S that satisfy property P_i .

If k elements are fixed, then there are $\binom{n}{k}$ ways to pick the k elements to fix and the remaining $n - k$ elements can be permuted in $(n - k)!$ ways. Thus, the number of permutations with at least one fixed point is

$$\begin{aligned}
 \left| \bigcup_{i=1}^n A_i \right| &= \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \\
 &= \binom{n}{1}(n-1)! - \binom{n}{2}(n-2)! + \binom{n}{3}(n-3)! - \dots + (-1)^{n+1} \binom{n}{n} 0! \\
 &= \frac{n!}{1!(n-1)!}(n-1)! - \frac{n!}{2!(n-2)!}(n-2)! + \frac{n!}{3!(n-3)!}(n-3)! - \dots + (-1)^{n+1} \frac{n!}{n!} 0! \\
 &= \frac{n!}{1!} - \frac{n!}{2!} + \frac{n!}{3!} - \dots + (-1)^{n+1} \frac{n!}{n!} \\
 &= n! \sum_{k=1}^n \frac{(-1)^{k+1}}{k!}
 \end{aligned}$$

Hence, the number of permutations with zero fixed points is

$$D(n) = |S| - \left| \bigcup_{i=1}^n A_i \right| = n! - n! \sum_{k=1}^n \frac{(-1)^{k+1}}{k!} = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$$

Therefore, the set $[n]$ has

$$D(n) = n! \sum_{k=0}^n \frac{(-1)^k}{k!}$$

derangements.

Recall that $e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$. Setting $x = -1$ gives

$$\begin{aligned} e^{-1} &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k!} \\ &= \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k}{k!} \\ &= \lim_{n \rightarrow \infty} \frac{D(n)}{n!} \end{aligned}$$

This says that for arbitrarily large n , the probability that a permutation of the set $[n]$ is a derangement is approximately $\frac{1}{e}$. In fact, for any positive integer n , $D(n) = \left[\frac{n!}{e} \right]$, where the square brackets are the nearest integer function.

8.1.1 Extending Derangements

We considered the permutations of the set $[n]$ and found a formula for the number of permutations that map no element to itself (i.e. $f(k) \neq k$ for all $k \in [n]$). What if instead we wanted to know the number of permutations that satisfy $f(k) \neq k+1$ for all $k \in [n-1]$. Or the number of permutations that satisfy $f(k) \neq k+2$ for all $k \in [n-2]$. We want to come up with a general formula for the number of permutations that satisfy $f(k) \neq k+c$ for all $k \in [n-c]$.

Before tackling the general formula, let us see a concrete example for when $c = 2$.

Problem

How many permutations of the set $[4]$ satisfy $f(k) \neq k+2$ for all $k \in [2]$?

Solution

Here we are counting the number of permutations from $[4]$ to $[4]$ such that $f(1) \neq 3$ and $f(2) \neq 4$. To do this, we will consider three cases depending on $f(1) = 1, f(1) = 2$ or $f(1) = 4$ and then consider sub-cases depending on $f(2) = 1, f(2) = 2$, or $f(2) = 3$. Here are the 14 possible permutations of $[4]$ that satisfy the condition.

$1 \mapsto 1$	$1 \mapsto 2$	$1 \mapsto 4$
1234	2134	4132
1243	2143	4123
1324	2314	4231
1342	2341	4213
		4321
		4312

Now we will derive the main formula and check afterwards that it does indeed agree with our data above.

Problem

Let c be a non-negative integer. Find a formula for $M(n, c)$ which is the number of permutations of the set $[n]$, where $n \geq c + 1$, satisfying $f(k) \neq k + c$ for all $k \in [n - c]$.

Solution

Let $S = [n]$ where $n \geq c + 1$ and let P_i be the property that a permutation maps element i to $i + c$ (for $1 \leq i \leq n - c$). Now define A_i to be the set of permutations of S that satisfy property P_i .

If k elements are mapped c spots down from their original position, then there are $\binom{n-c}{k}$ ways to pick those k elements and the remaining $n - k$ elements can be permuted in $(n - k)!$ ways. Thus, the number of permutations mapping at least one element c spots down is

$$\begin{aligned}
 \left| \bigcup_{i=1}^n A_i \right| &= \sum_{i=1}^n |A_i| - \sum_{1 \leq i < j \leq n} |A_i \cap A_j| + \sum_{1 \leq i < j < k \leq n} |A_i \cap A_j \cap A_k| - \dots + (-1)^{n+1} |A_1 \cap A_2 \cap \dots \cap A_n| \\
 &= \binom{n-c}{1} (n-1)! - \binom{n-c}{2} (n-2)! + \binom{n-c}{3} (n-3)! - \dots + (-1)^{n+1} \binom{n-c}{n-c} (n - (n-c))! \\
 &= \frac{(n-c)!}{1!(n-c-1)!} (n-1)! - \frac{(n-c)!}{2!(n-c-2)!} (n-2)! + \frac{(n-c)!}{3!(n-c-3)!} (n-3)! - \dots + (-1)^{n+1} \frac{(n-c)!}{(n-c)!} c! \\
 &= (n-c)! \left[\frac{(n-1)(n-2) \dots (n-c)}{1!} - \frac{(n-2)(n-3) \dots (n-c-1)}{2!} + \right. \\
 &\quad \left. + \frac{(n-3)(n-4) \dots (n-c-2)}{3!} - \dots + (-1)^{n+1} \frac{c!}{(n-c)!} \right] \\
 &= (n-c)! \sum_{k=1}^{n-c} (-1)^{k+1} \frac{((n-k))_c}{k!}
 \end{aligned}$$

Hence, the number of permutations satisfying $f(k) \neq k + c$ for all $k \in [n - c]$ is

$$M(n, c) = |S| - \left| \bigcup_{i=1}^n A_i \right| = n! - (n-c)! \sum_{k=1}^{n-c} (-1)^{k+1} \frac{((n-k))_c}{k!} = (n-c)! \sum_{k=0}^{n-c} (-1)^k \frac{((n-k))_c}{k!}$$

Sanity Checks.

Setting $c = 2$ gives

$$M(n, 2) = (n-2)! \sum_{k=0}^{n-2} (-1)^k \frac{((n-k))_2}{k!} = (n-2)! \sum_{k=0}^{n-2} (-1)^k \frac{(n-k)(n-k-1)}{k!}$$

which gives the number of n -permutations satisfying $f(k) \neq k+2$ for all $k \in [n-2]$.

If $n = 4$ then

$$\begin{aligned} M(4, 2) &= (4-2)! \sum_{k=0}^{4-2} (-1)^k \frac{(4-k)(4-k-1)}{k!} \\ &= 2! \sum_{k=0}^2 (-1)^k \frac{(4-k)(3-k)}{k!} \\ &= 2! \left[\frac{4 \cdot 3}{0!} - \frac{3 \cdot 2}{1!} + \frac{2 \cdot 1}{2!} \right] \\ &= 2(12 - 6 + 1) \\ &= 14 \end{aligned}$$

as expected by the concrete example.

Setting $c = 0$ gives

$$M(n, 0) = (n-0)! \sum_{k=0}^{n-0} (-1)^k \frac{((n-k))_0}{k!} = n! \sum_{k=0}^n \frac{(-1)^k}{k!} = D(n)$$

which gives the number of n -permutations satisfying $f(k) \neq k$ for all $k \in [n]$.

As expected, we obtained the formula for the number of derangements.

8.2 Formula for Stirling Numbers of the Second Kind

To derive a formula for $S(n, k)$, we will count the number of surjective functions from $[n]$ to $[k]$. We know one answer is $k! \cdot S(n, k)$. Another way would be to count the number of functions that miss at least one element from the co-domain $[k]$, then subtract those functions from the total number of functions which is k^n . Finally, we can equate both answers and get an expression for $S(n, k)$.

Let F be the set of all functions from $[n]$ to $[k]$ and let P_i be the property that a function fails to hit i where $i \in [k]$. Now define A_i to be the set of functions from F that satisfy property P_i .

Consider $|A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_j}|$, that is, the number of functions that fail to hit $i_1, i_2, \dots$, and i_j . There are $\binom{k}{j}$ ways to choose the j elements will be missed by the function, and there are $(k-j)^n$ ways to map each element of the domain to an element of the co-domain (any element of the domain can be mapped to any one of the remaining $k-j$ elements of the co-domain). Hence, the number of functions that miss at least one element of $[k]$ in their image is

$$\begin{aligned} \left| \bigcup_{i=1}^k A_i \right| &= \sum_{i=1}^k |A_i| - \sum_{1 \leq i < j \leq k} |A_i \cap A_j| + \sum_{1 \leq i < j < r \leq k} |A_i \cap A_j \cap A_r| - \dots + (-1)^{k+1} |A_1 \cap A_2 \cap \dots \cap A_k| \\ &= \binom{k}{1} (k-1)^n - \binom{k}{2} (k-2)^n + \binom{k}{3} (k-3)^n - \dots + (-1)^{k+1} \binom{k}{k} (k-k)^n \\ &= \sum_{i=1}^k (-1)^{i+1} \binom{k}{i} (k-i)^n \end{aligned}$$

Hence, the number of surjective functions is

$$|F| - \left| \bigcup_{i=1}^k A_i \right| = k^n - \sum_{i=1}^k (-1)^{i+1} \binom{k}{i} (k-i)^n = \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n$$

Therefore, we must have

$$k! \cdot S(n, k) = \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n$$

from which it follows that

$$S(n, k) = \frac{1}{k!} \sum_{i=0}^k (-1)^i \binom{k}{i} (k-i)^n = \sum_{i=0}^k (-1)^i \frac{1}{i!(k-i)!} (k-i)^n$$

CHAPTER 9

GENERATING FUNCTIONS

A generating function is a way of encoding an infinite sequence of numbers by treating them as the coefficients of a formal power series. The coefficient of x^n is the n^{th} term of the sequence.

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of real numbers. Then the formal power series $A(x) = \sum_{n \geq 0} a_n x^n$ is called the ordinary generating function of the sequence $\{a_n\}_{n=0}^{\infty}$.

This series is called the generating function of the sequence $\{a_n\}_{n=0}^{\infty}$.

Encoding a sequence within a function allows us to analyze the sequence with tools from other branches of Mathematics such as Calculus.

Note that the generating function of a sequence is thought of as a formal power series, which means that we are treating it as an algebraic object rather than a function that can be evaluated for specific values of x . This means that we do not need to worry about the interval of convergence.

Examples.

Consider the sequence $a_n = 1$ for all $n \geq 0$. Then its ordinary generating function is

$$A(x) = 1 \cdot x^0 + 1 \cdot x^1 + 1 \cdot x^2 + \dots = 1 + x + x^2 + \dots = \sum_{n \geq 0} 1 \cdot x^n = \frac{1}{1-x}$$

Let n be a non-negative integer. Consider the sequence $a_i = \binom{n}{i}$ for all $i \geq 0$ (note that $\binom{n}{k} = 0$ if $k > n$). Then its ordinary generating function is

$$A(x) = \binom{n}{0} \cdot x^0 + \binom{n}{1} \cdot x^1 + \binom{n}{2} \cdot x^2 + \dots = \sum_{k=0}^n \binom{n}{k} x^k = (x+1)^n$$

Consider the sequence $a_n = n!$ for all $n \geq 0$. Then its ordinary generating function is

$$A(x) = 1 \cdot x^0 + 1 \cdot x^1 + 2 \cdot x^2 + 6 \cdot x^3 + 24 \cdot x^4 + \dots = 1 + x + 2x^2 + 6x^3 + 24x^4 + \dots = \sum_{n \geq 0} n! x^n$$

which has no closed form since it grows too quickly. This example motivates another type of generating functions, namely exponential generating function which will slow down the growth of a sequence.

Let $\{a_n\}_{n=0}^{\infty}$ be a sequence of real numbers. Then the formal power series $A(x) = \sum_{n \geq 0} a_n \frac{x^n}{n!}$ is called the exponential generating function of the sequence $\{a_n\}_{n=0}^{\infty}$.

Examples.

Consider the sequence $a_n = 1$ for all $n \geq 0$ once again. Its exponential generating function is

$$A(x) = 1 \cdot \frac{x^0}{0!} + 1 \cdot \frac{x^1}{1!} + 1 \cdot \frac{x^2}{2!} + \dots = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots = \sum_{n \geq 0} 1 \cdot \frac{x^n}{n!} = e^x$$

Consider the sequence $a_n = n!$ for all $n \geq 0$ once again. Its exponential generating function is

$$A(x) = 1 \cdot \frac{x^0}{0!} + 1 \cdot \frac{x^1}{1!} + 2 \cdot \frac{x^2}{2!} + 6 \cdot \frac{x^3}{3!} + 24 \cdot \frac{x^4}{4!} + \dots = 1 + x + x^2 + x^3 + x^4 + \dots = \sum_{n \geq 0} x^n = \frac{1}{1-x}$$

We can also take the product of generating functions.

Let $\{a_i\}_{i=0}^{\infty}$ and $\{b_j\}_{j=0}^{\infty}$ be two sequences of real numbers. Let $A(x) = \sum_{i \geq 0} a_i x^i$ and $B(x) = \sum_{j \geq 0} b_j x^j$ be their respective ordinary generating functions. Then

$$A(x) \cdot B(x) = \left(\sum_{i \geq 0} a_i x^i \right) \left(\sum_{j \geq 0} b_j x^j \right) = \sum_{n \geq 0} \left(\sum_{k=0}^n a_k b_{n-k} \right) x^n$$

Let $\{a_i\}_{i=0}^{\infty}$ and $\{b_j\}_{j=0}^{\infty}$ be two sequences of real numbers. Let $A(x) = \sum_{i \geq 0} a_i \frac{x^i}{i!}$ and $B(x) = \sum_{j \geq 0} b_j \frac{x^j}{j!}$ be their respective exponential generating functions. Then

$$A(x) \cdot B(x) = \left(\sum_{i \geq 0} a_i \frac{x^i}{i!} \right) \left(\sum_{j \geq 0} b_j \frac{x^j}{j!} \right) = \sum_{n \geq 0} \left(\sum_{k=0}^n \binom{n}{k} a_k b_{n-k} \right) \frac{x^n}{n!}$$

The two products above are quite similar. There is a nice combinatorial difference between the two.

The first one applies when we split $[n]$ into two parts so that one part is $[i]$. In other words, our objects are linearly ordered and we are cutting that linear order somewhere to get two subsets.

The second one applies when we split $[n]$ into two parts with no restrictions (the subsets do not have to be consecutive objects in a previously ordered line).

9.1 Basic Problems

Problem

Find a closed form for the generating function of the sequence defined by $a_n = n^2$.

Solution

Our goal is to find a closed form for $A(x) = \sum_{n \geq 0} a_n x^n$.

Recall that

$$\sum_{n \geq 0} x^n = \frac{1}{1-x}$$

Take the derivative with respect to x of both sides to get

$$\sum_{n \geq 1} n x^{n-1} = \frac{1}{(1-x)^2}$$

Multiply both sides by x to get

$$\sum_{n \geq 0} n x^n = \frac{x}{(1-x)^2}$$

Take the derivative once again to get

$$\sum_{n \geq 1} n^2 x^{n-1} = \frac{x+1}{(1-x)^3}$$

Multiply both sides by x to get

$$\sum_{n \geq 0} n^2 x^n = \frac{x(x+1)}{(1-x)^3}$$

Therefore, $A(x) = \frac{x(x+1)}{(1-x)^3}$.

Problem

Find an explicit formula for a_k if $a_0 = 0$ and $a_{k+1} = a_k + 2^k$ for $k \geq 0$.

Solution

Let $A(x) = \sum_{k \geq 0} a_k x^k$. Multiplying the recurrence relation by x^{k+1} and summing over all $k \geq 0$ gives

$$\begin{aligned}\sum_{k \geq 0} a_{k+1} x^{k+1} &= \sum_{k \geq 0} a_k x^{k+1} + \sum_{k \geq 0} 2^k x^{k+1} \\ \sum_{k \geq 0} a_{k+1} x^{k+1} &= \sum_{k \geq 0} a_k x^{k+1} + x \sum_{k \geq 0} (2x)^k\end{aligned}$$

The term on the LHS is $A(x) - a_0 = A(x)$, the first term on the RHS is $xA(x)$, and the second term on the RHS is $\frac{x}{1-2x}$. Hence,

$$A(x) = xA(x) + \frac{x}{1-2x}$$

and solving for $A(x)$ gives

$$A(x) = \frac{x}{(1-x)(1-2x)}$$

Now we will use the method of partial fractions to find constants A and B such that

$$\frac{x}{(1-x)(1-2x)} = \frac{A}{1-x} + \frac{B}{1-2x}$$

Multiplying both sides of the equation by $(1-x)(1-2x)$ gives

$$\begin{aligned}x &= A(1-2x) + B(1-x) \\ x &= (-2A - B)x + (A + B)\end{aligned}$$

For LHS = RHS, we must have

$$\begin{aligned}-2A - B &= 1 \\ A + B &= 0\end{aligned}$$

Solving this system gives $A = -1$ and $B = 1$. Thus,

$$\begin{aligned}A(x) &= \frac{1}{1-2x} - \frac{1}{1-x} \\ &= \sum_{k \geq 0} (2x)^k - \sum_{k \geq 0} x^k \\ &= \sum_{k \geq 0} (2^k - 1)x^k\end{aligned}$$

from which it follows that $a_k = 2^k - 1$ for all $k \geq 0$.

Problem

Let $a_0 = a$, let $a_1 = b$, and let $a_{n+2} = pa_{n+1} + qa_n + r$ for $n \geq 0$, where p and q are fixed nonzero real numbers, and r is any real number. Find $A(x) = \sum_{n \geq 0} a_n x^n$.

Solution

Multiplying both sides of the recurrence relation by x^{n+2} and summing over all $k \geq 0$ gives

$$\begin{aligned}\sum_{n \geq 0} a_{n+2} x^{n+2} &= \sum_{n \geq 0} p a_{n+1} x^{n+2} + \sum_{n \geq 0} q a_n x^{n+2} + \sum_{n \geq 0} r x^{n+2} \\ \sum_{n \geq 0} a_{n+2} x^{n+2} &= p x \sum_{n \geq 0} a_{n+1} x^{n+1} + q x^2 \sum_{n \geq 0} a_n x^n + r x^2 \sum_{n \geq 0} x^n\end{aligned}$$

In terms of $A(x)$ we get

$$\begin{aligned}A(x) - a_1 x - a_0 &= p x (A(x) - a_0) + q x^2 A(x) + r x^2 \left(\frac{1}{1-x} \right) \\ A(x)(1 - p x - q x^2) &= a_1 x + a_0 - p x a_0 + \frac{r x^2}{1-x}\end{aligned}$$

Solving for $A(x)$ gives

$$A(x) = \frac{a_1 x + a_0 - p x a_0 + \frac{r x^2}{1-x}}{1 - p x - q x^2} = \frac{(a_1 - a_0 p)x + a_0 + \frac{r x^2}{1-x}}{1 - p x - q x^2}$$

Since $a_0 = a$ and $a_1 = b$, we get

$$A(x) = \frac{(b - ap)x + a + \frac{r x^2}{1-x}}{1 - p x - q x^2}$$

9.2 Making Change

Problem

Let a_n be the number of ways to pay n dollars using ten-dollar bills, five-dollar bills, and one-dollar bills only. Find the ordinary generating function $A(x) = \sum_{n \geq 0} a_n x^n$.

Solution

Let $f(n)$ be the number of ways to pay n dollars using ten-dollar bills only. Then $f(n) = 1$ if n is a multiple of 10, and $f(n) = 0$ otherwise. Therefore, the ordinary generating function for $f(n)$ is

$$F(x) = \sum_{n \geq 0} f(n)x^n = 1 + x^{10} + x^{20} + x^{30} + \dots$$

Notice that $F(x)$ is a geometric series with a common ratio of x^{10} . Hence, $F(x) = \frac{1}{1-x^{10}}$.

Similarly, let $g(n)$ be the number of ways to pay n dollars using five-dollar bills only. Then $g(n) = 1$ if n is a multiple of 5, and $g(n) = 0$ otherwise. Therefore, the ordinary generating function for $g(n)$ is

$$G(x) = \sum_{n \geq 0} g(n)x^n = 1 + x^5 + x^{10} + x^{15} + \dots$$

Notice that $G(x)$ is a geometric series with a common ratio of x^5 . Hence, $G(x) = \frac{1}{1-x^5}$.

Finally, let $h(n)$ be the number of ways to pay n dollars using one-dollar bills only. Then $h(n) = 1$ for all $n \geq 0$. Therefore, the ordinary generating function for $h(n)$ is

$$H(x) = \sum_{n \geq 0} h(n)x^n = 1 + x^1 + x^2 + x^3 + \dots$$

Notice that $H(x)$ is a geometric series with a common ratio of x^1 . Hence, $H(x) = \frac{1}{1-x}$. Now consider

$$\begin{aligned} F(x) \cdot G(x) \cdot H(x) &= \underbrace{(1 + x^{10} + x^{20} + x^{30} + \dots)}_{\text{ten-dollar bills}} \cdot \underbrace{(1 + x^5 + x^{10} + x^{15} + \dots)}_{\text{five-dollar bills}} \cdot \underbrace{(1 + x^1 + x^2 + x^3 + \dots)}_{\text{one-dollar bills}} \\ &= \frac{1}{1-x^{10}} \cdot \frac{1}{1-x^5} \cdot \frac{1}{1-x} \end{aligned}$$

We claim that

$$A(x) = F(x) \cdot G(x) \cdot H(x) = \frac{1}{(1-x)(1-x^5)(1-x^{10})}$$

Indeed, we count the number of ways to pay n dollars using ten-dollar bills, five-dollar bills, and one-dollar bills only. One answer is a_n which is the coefficient in front of x^n in $A(x)$. Another way is to choose k_1 ten-dollar bills, k_2 five-dollar bills, and k_3 one-dollar bills such that $10 \cdot k_1 + 5 \cdot k_2 + 1 \cdot k_3 = n$. This corresponds to picking one term, x^{i_1} , from $(1 + x^{10} + x^{20} + x^{30} + \dots)$, one term, x^{i_2} , from $(1 + x^5 + x^{10} + x^{15} + \dots)$, and one term, x^{i_3} , from $(1 + x^1 + x^2 + x^3 + \dots)$ such that $x^{i_1} \cdot x^{i_2} \cdot x^{i_3} = x^n$. Hence, the coefficient of x^n on the right-hand side is precisely the number of ways we can pick three such terms, and that number is exactly a_n .

9.2.1 Reverse Engineering

Say we are given the ordinary generating function

$$C(x) = \frac{1}{(1-x)^2} = \sum_{n \geq 0} c_n x^n$$

Consider the following

$$\frac{1}{(1-x)^2} = \frac{1}{1-x} \cdot \frac{1}{1-x} = (1+x+x^2+x^3+\dots) \cdot (1+x+x^2+x^3+\dots)$$

which says that c_n is the number of ways for two non-negative integers to sum to n . There are $n+1$ ways this can happen $(0+n, 1+(n-1), \dots, n+0)$, hence $c_n = n+1$ which implies

$$C(x) = \frac{1}{(1-x)^2} = \sum_{n \geq 0} (n+1)x^n$$

In the general case, we have

$$C(x) = \frac{1}{(1-x)^k} = \sum_{n \geq 0} c_n x^n$$

Consider

$$\begin{aligned} \frac{1}{(1-x)^k} &= \frac{1}{1-x} \cdot \frac{1}{1-x} \cdot \dots \cdot \frac{1}{1-x} \\ &= (1+x+x^2+x^3+\dots) \cdot (1+x+x^2+x^3+\dots) \cdot \dots \cdot (1+x+x^2+x^3+\dots) \end{aligned}$$

which says that c_n is the number of ways for k non-negative integers to sum to n . In other words, c_n is the number of weak compositions of n into k parts. Hence, $c_n = \binom{n+k-1}{k-1}$ which implies

$$C(x) = \frac{1}{(1-x)^k} = \sum_{n \geq 0} \binom{n+k-1}{k-1} x^n$$

9.3 Partitions

Let $p(n)$ be the number of partitions of n . We will derive the generating function $\sum_{n \geq 0} p(n)x^n$.

Let $f_1(n)$ be the number of partitions of n using only 1's. Then $f_1(n) = 1$ for all $n \geq 0$ since there is only one way to express n as a sum of 1's. Therefore, the generating function for $f_1(n)$ is

$$F_1(x) = \sum_{n \geq 0} f_1(n)x^n = 1 + x^1 + x^2 + x^3 + \dots$$

Notice that $F_1(x)$ is a geometric series with a common ratio of x^1 . Hence, $F_1(x) = \frac{1}{1-x}$.

Similarly, let $f_2(n)$ be the number of partitions of n using only 2's. Then $f_2(n) = 1$ if n is a multiple of 2, and $f_2(n) = 0$ otherwise. This is because there is only one way of writing n as a sum of 2's if n is even, and zero ways if n is odd. Therefore, the generating function for $f_2(n)$ is

$$F_2(x) = \sum_{n \geq 0} f_2(n)x^n = 1 + x^2 + x^4 + x^6 + \dots$$

Notice that $F_2(x)$ is a geometric series with a common ratio of x^2 . Hence, $F_2(x) = \frac{1}{1-x^2}$.

In general, let $f_k(x)$ be the number of partitions of n using only k 's. Then $f_k(n) = 1$ if n is a multiple of k , and $f_k(n) = 0$ otherwise. Therefore, the generating function for $f_k(n)$ is

$$F_k(x) = \sum_{n \geq 0} f_k(n)x^n = 1 + x^k + x^{2k} + x^{3k} + \dots$$

Notice that $F_k(x)$ is a geometric series with a common ratio of x^k . Hence $F_k(x) = \frac{1}{1-x^k}$.

Consider the product

$$\begin{aligned} F_1(x) \cdot F_2(x) \cdot F_3(x) \cdot \dots &= \underbrace{(1 + x^1 + x^2 + x^3 + \dots)}_{\text{number of 1's}} \cdot \underbrace{(1 + x^2 + x^4 + x^6 + \dots)}_{\text{number of 2's}} \cdot \underbrace{(1 + x^3 + x^6 + x^9 + \dots)}_{\text{number of 3's}} \cdot \dots \\ &= \frac{1}{1-x} \cdot \frac{1}{1-x^2} \cdot \frac{1}{1-x^3} \cdot \dots = \prod_{k=1}^{\infty} \frac{1}{1-x^k} \end{aligned}$$

We claim that

$$\sum_{n \geq 0} p(n)x^n = \prod_{k=1}^{\infty} \frac{1}{1-x^k}$$

Count the number of partitions of n . One answer is $p(n)$ which is the coefficient in front of x^n on the LHS.

Another way is to

- choose k_1 1's
- choose k_2 2's
- choose k_3 3's
- and so on

so that $1 \cdot k_1 + 2 \cdot k_2 + 3 \cdot k_3 + \dots = n$. This corresponds to picking exactly one term from each factor $(1 + x^k + x^{2k} + x^{3k})$ such that the product of the chosen terms is x^n . Hence, the coefficient of x^n on the RHS is precisely the number of ways we can pick such terms, and that number is exactly $p(n)$.

Problem

Let $p_{\text{odd}}(n)$ be the number of partitions of n into odd parts. Find the ordinary generating function $\sum_{n \geq 0} p_{\text{odd}}(n)x^n$.

Solution

Recall the observation that led us to the generating function for $p(n)$ (the number of partitions of n):

$$\sum_{n \geq 0} p(n)x^n = \underbrace{(1 + x^1 + x^2 + x^3 + \dots)}_{\text{number of 1's}} \cdot \underbrace{(1 + x^2 + x^4 + x^6 + \dots)}_{\text{number of 2's}} \cdot \underbrace{(1 + x^3 + x^6 + x^9 + \dots)}_{\text{number of 3's}} \cdot \dots$$

Hence, to obtain the generating function for p_{odd} , we can only use odd parts:

$$\begin{aligned} \sum_{n \geq 0} p_{\text{odd}}(n)x^n &= \underbrace{(1 + x^1 + x^2 + x^3 + \dots)}_{\text{number of 1's}} \cdot \underbrace{(1 + x^3 + x^6 + x^9 + \dots)}_{\text{number of 3's}} \cdot \underbrace{(1 + x^5 + x^{10} + x^{15} + \dots)}_{\text{number of 5's}} \cdot \dots \\ &= \frac{1}{1-x} \cdot \frac{1}{1-x^3} \cdot \frac{1}{1-x^5} \cdot \dots \\ &= \prod_{k=0}^{\infty} \frac{1}{1-x^{2k+1}} \end{aligned}$$

Problem

Let $p_{\text{distinct}}(n)$ be the number of partitions of n into distinct parts. Find the ordinary generating function $\sum_{n \geq 0} p_{\text{distinct}}(n)x^n$.

Solution

Recall the observation that led us to the generating function for $p(n)$ (the number of partitions of n):

$$\sum_{n \geq 0} p(n)x^n = \underbrace{(1 + x^1 + x^2 + x^3 + \dots)}_{\text{number of 1's}} \cdot \underbrace{(1 + x^2 + x^4 + x^6 + \dots)}_{\text{number of 2's}} \cdot \underbrace{(1 + x^3 + x^6 + x^9 + \dots)}_{\text{number of 3's}} \cdot \dots$$

Hence, to obtain the generating function for p_{distinct} , we cannot take multiple copies of the same part. In other words, we either we pick a part of size k or we do not:

$$\begin{aligned} \sum_{n \geq 0} p_{\text{distinct}}(n)x^n &= \underbrace{(1 + x^1)}_{\text{pick 1 or not}} \cdot \underbrace{(1 + x^2)}_{\text{pick 2 or not}} \cdot \underbrace{(1 + x^3)}_{\text{pick 3 or not}} \cdot \dots \\ &= \prod_{k=1}^{\infty} (1 + x^k) \end{aligned}$$

Problem

Prove that the number of partitions of n into odd parts is equal to the number of partitions of n into distinct parts. That is, prove $p_{\text{odd}}(n) = p_{\text{distinct}}(n)$.

Proof

We will prove $p_{\text{odd}}(n) = p_{\text{distinct}}(n)$ by showing

$$\sum_{n \geq 0} p_{\text{odd}}(n)x^n = \sum_{n \geq 0} p_{\text{distinct}}(n)x^n$$

In other words, we will show that their generating functions agree on all coefficients.

Recall that for any real number a we have

$$1 - a^2 = (1 + a)(1 - a)$$

which can be rearranged as follows

$$(1 + a) = \frac{1 - a^2}{1 - a}$$

Now consider,

$$\begin{aligned} \sum_{n \geq 0} p_{\text{distinct}}(n)x^n &= \prod_{k=1}^{\infty} (1 + x^k) \\ &= \prod_{k=1}^{\infty} \frac{1 - x^{2k}}{1 - x^k} && (\text{let } a = x^k \text{ in the above fact}) \\ &= \frac{(1 - x^2) \cdot (1 - x^4) \cdot (1 - x^6) \cdot \dots}{(1 - x) \cdot (1 - x^2) \cdot (1 - x^3) \cdot (1 - x^4) \cdot \dots} \\ &= \frac{1}{(1 - x) \cdot (1 - x^3) \cdot (1 - x^5) \cdot \dots} \\ &= \prod_{k=0}^{\infty} \frac{1}{1 - x^{2k+1}} \\ &= \sum_{n \geq 0} p_{\text{odd}}(n)x^n \end{aligned}$$

which shows that two two generating functions are equal. Therefore, $p_{\text{odd}}(n) = p_{\text{distinct}}(n)$, as required. $\square$

Problem

Let $p_o(n)$ denote the number of partitions of n into an odd number of parts, let $p_e(n)$ denote the number of partitions of n into an even number of parts, and let $p_{d,odd}(n)$ denote the number of partitions of n into distinct odd parts. Prove that $p_{d,odd} = |p_e(n) - p_o(n)|$.

Solution

The generating function for $p_{d,odd}$ is

$$\begin{aligned} \sum_{n \geq 0} p_{d,odd}(n)x^n &= \underbrace{(1+x^1)}_{\text{pick 1 or not}} \cdot \underbrace{(1+x^3)}_{\text{pick 3 or not}} \cdot \underbrace{(1+x^5)}_{\text{pick 5 or not}} \cdot \dots \\ &= \prod_{k=0}^{\infty} (1+x^{2k+1}) \end{aligned}$$

To find the generating function for $p_o(n)$ and $p_e(n)$, we need to keep track of the number of parts. Recall the generating function for the number of partitions of n is

$$\begin{aligned} \sum_{n \geq 0} p(n)x^n &= \underbrace{(1+x^1+x^2+x^3+\dots)}_{\text{number of 1's}} \cdot \underbrace{(1+x^2+x^4+x^6+\dots)}_{\text{number of 2's}} \cdot \underbrace{(1+x^3+x^6+x^9+\dots)}_{\text{number of 3's}} \cdot \dots \\ &= \prod_{k=1}^{\infty} \frac{1}{1-x^k} \end{aligned}$$

We will add an extra parameter to keep track of the number of parts as follows

$$\prod_{k=1}^{\infty} \frac{1}{1-ux^k} = (1+ux+u^2x^2+u^3x^3+\dots) \cdot (1+ux^2+u^2x^4+u^3x^6+\dots) \cdot (1+ux^3+u^2x^6+u^3x^9+\dots)$$

So $u^k x^{m \cdot k}$ means that we have picked m to be in our partition k times. Setting $u = -1$ gives

$$\prod_{k=1}^{\infty} \frac{1}{1-(-1)x^k} = \prod_{k=1}^{\infty} \frac{1}{1+x^k} = (1-x^1+x^2-x^3+\dots) \cdot (1-x^2+x^4-x^6+\dots) \cdot (1-x^3+x^6-x^9+\dots)$$

from which we see that if k is even then $(-1)^k$ is positive and if k is odd then $(-1)^k$ negative. So the terms with positive coefficients correspond to $p_e(n)$ and the terms with negative coefficients correspond to $p_o(n)$. It follows that,

$$\prod_{k=1}^{\infty} \frac{1}{1+x^k} = \sum_{n=0}^{\infty} p_e(n)x^n - \sum_{n=0}^{\infty} p_o(n)x^n = \sum_{n=0}^{\infty} (p_e(n) - p_o(n))x^n$$

Replacing x with $-x$ gives

$$\prod_{k=1}^{\infty} \frac{1}{1+(-1)^k x^k} = \sum_{n=0}^{\infty} (-1)^n (p_e(n) - p_o(n))x^n$$

Now we wish to show

$$\prod_{k=0}^{\infty} (1 + x^{2k+1}) = \prod_{k=1}^{\infty} \frac{1}{1 + (-1)^k x^k}$$

which is equivalent to showing (replacing x with $-x$)

$$\prod_{k=0}^{\infty} (1 - x^{2k+1}) = \prod_{k=1}^{\infty} \frac{1}{1 + x^k}$$

Indeed,

$$\begin{aligned} \prod_{k=1}^{\infty} \frac{1}{1 + x^k} &= \prod_{k=1}^{\infty} \frac{1}{\frac{1-x^{2k}}{1-x^k}} \\ &= \prod_{k=1}^{\infty} \frac{1-x^k}{1-x^{2k}} \\ &= \frac{(1-x) \cdot (1-x^2) \cdot (1-x^3) \cdot (1-x^4) \cdot \dots}{(1-x^2) \cdot (1-x^4) \cdot (1-x^6) \cdot \dots} \\ &= (1-x) \cdot (1-x^3) \cdot (1-x^5) \cdot \dots \\ &= \prod_{k=0}^{\infty} (1 - x^{2k+1}) \end{aligned}$$

which shows that two generating functions are equal. Therefore, $p_{d,odd} = |p_e(n) - p_o(n)|$, as required.

9.4 Fibonacci

Problem

A child wants to walk up a stairway. At each step, she moves either one step up or two steps up. Let $f(n)$ be the number of ways she can reach the n^{th} stair. Find a closed explicit formula for $f(n)$.

Solution

Let us first find a recursive formula for $f(n)$.

The key observation is that before the child can reach the n^{th} stair, she must be at either the $(n-1)^{\text{st}}$ stair or the $(n-2)^{\text{nd}}$ stair. There are $f(n-1)$ ways for the child to reach the $(n-1)^{\text{st}}$ stair and there are $f(n-2)$ ways for the child to reach the $(n-2)^{\text{nd}}$ stair. Therefore,

$$f(n) = f(n-1) + f(n-2)$$

for $n \geq 2$. Since we want a formula for $f(n)$ for $n \geq 0$, we will work with the recurrence

$$f(n+2) = f(n+1) + f(n)$$

where $f(0) = 1$ and $f(1) = 1$.

Let $F(x) = \sum_{n \geq 0} f(n)x^n$ be the ordinary generating function for the numbers $f(n)$. Multiplying both sides of the recurrence relation by x^{n+2} and summing over all $n \geq 0$ gives

$$\sum_{n \geq 0} f(n+2)x^{n+2} = \sum_{n \geq 0} f(n+1)x^{n+2} + \sum_{n \geq 0} f(n)x^{n+2}$$

Notice that

$$\sum_{n \geq 0} f(n+2)x^{n+2} = F(x) - f(1)x - f(0) = F(x) - x - 1$$

and

$$\sum_{n \geq 0} f(n+1)x^{n+2} = x(F(x) - f(0)) = xF(x) - x$$

and

$$\sum_{n \geq 0} f(n)x^{n+2} = x^2 F(x)$$

Hence,

$$F(x) - x - 1 = xF(x) - x + x^2 F(x)$$

and solving for $F(x)$ gives

$$F(x) = \frac{1}{1-x-x^2} = \frac{1}{-(x^2+x-1)} = \frac{-1}{(x-\alpha)(x-\beta)}$$

where $\alpha = \frac{-1+\sqrt{5}}{2}$ and $\beta = \frac{-1-\sqrt{5}}{2}$.

Now we will use the method of partial fractions to find constants A and B such that

$$\frac{-1}{(x-\alpha)(x-\beta)} = \frac{A}{(x-\alpha)} + \frac{B}{(x-\beta)}$$

Multiplying both sides of the equation by $(x-\alpha)(x-\beta)$ gives

$$-1 = A(x-\beta) + B(x-\alpha)$$

Expanding gives

$$-1 = (A+B)x - (A\beta + B\alpha)$$

therefore, we must have

$$\begin{cases} A+B=0 \\ -(A\beta+B\alpha)=-1 \end{cases}$$

Solving this system yields, $A = -\frac{1}{\sqrt{5}}$ and $B = \frac{1}{\sqrt{5}}$. Now,

$$\begin{aligned} \frac{A}{(x-\alpha)} + \frac{B}{(x-\beta)} &= -\frac{1}{\sqrt{5}} \frac{1}{(x-\alpha)} + \frac{1}{\sqrt{5}} \frac{1}{(x-\beta)} \\ &= \frac{1}{\sqrt{5}} \frac{1}{(\alpha-x)} - \frac{1}{\sqrt{5}} \frac{1}{(\beta-x)} \\ &= \frac{1}{\sqrt{5}} \frac{\beta}{\beta(\alpha-x)} - \frac{1}{\sqrt{5}} \frac{\alpha}{\alpha(\beta-x)} \\ &= \frac{1}{\sqrt{5}} \frac{\beta}{(-1-\beta x)} - \frac{1}{\sqrt{5}} \frac{\alpha}{(-1-\alpha x)} && (\text{since } \alpha \cdot \beta = -1) \\ &= -\frac{1}{\sqrt{5}} \frac{\beta}{(1-(-\beta x))} + \frac{1}{\sqrt{5}} \frac{\alpha}{(1-(-\alpha x))} \\ &= \frac{1}{\sqrt{5}} (-\beta) \sum_{n \geq 0} (-\beta)^n x^n - \frac{1}{\sqrt{5}} (-\alpha) \sum_{n \geq 0} (-\alpha)^n x^n \\ &= \sum_{n \geq 0} \left[\frac{1}{\sqrt{5}} (-\beta)^{n+1} - \frac{1}{\sqrt{5}} (-\alpha)^{n+1} \right] x^n \end{aligned}$$

Therefore,

$$f(n) = \frac{1}{\sqrt{5}} \left(\frac{1+\sqrt{5}}{2} \right)^{n+1} - \frac{1}{\sqrt{5}} \left(\frac{1-\sqrt{5}}{2} \right)^{n+1}$$

9.5 Motzkin Numbers

Problem

Let M_n denote the number of paths from $(0, 0)$ to $(n, 0)$ which never go below $y = 0$ and are made up only of the steps $(1, 0)$, $(1, 1)$, and $(1, -1)$. Find the ordinary generating function $\sum_{n \geq 0} M_n x^n$.

Solution

We begin by finding a recurrence relation for M_n .

Consider a path satisfying the conditions. There are two cases

- The first step of the path is $(1, 0)$. There are M_{n-1} ways to complete the path from $(1, 0)$ to $(n, 0)$.
- The first step of the path is $(1, 1)$. Eventually, the path will come back down and touch the line $y = 0$. Let $(k, 0)$ be the point when it first happens ($2 \leq k \leq n$).

Consider the portion of the path from $(k, 0)$ to $(n, 0)$. There are M_{n-k} ways to complete this path.

Consider the portion of the path from $(0, 0)$ to $(k, 0)$. The number of ways to go from $(0, 0)$ to $(k, 0)$ without touching the line $y = 0$ is M_{k-2} . This is because the first step is $(1, 1)$ (to go from $(0, 0)$ to $(1, 1)$), and the last step is $(1, -1)$ (to go from $(k-1, 1)$ to $(k, 0)$). The number of paths from $(1, 1)$ to $(k-1, 1)$ which never go below $y = 1$ is M_{k-2} .

Therefore, we have the following recurrence relation

$$M_n = M_{n-1} + \sum_{k=2}^n M_{k-2} M_{n-k}$$

for $n \geq 2$, where $M_0 = 1$ and $M_1 = 1$. Since we want our generating function to be defined for all $n \geq 0$, we will work with the following recurrence relation

$$M_{n+2} = M_{n+1} + \sum_{k=2}^{n+2} M_{k-2} M_{n+2-k}$$

for $n \geq 0$, where $M_0 = 1$ and $M_1 = 1$. Let $M(x) = \sum_{n \geq 0} M_n x^n$. Multiplying both sides of the recurrence relation by x^{n+2} and summing over $n \geq 0$ gives

$$\sum_{n \geq 0} M_{n+2} x^{n+2} = \sum_{n \geq 0} M_{n+1} x^{n+2} + \sum_{n \geq 0} \left(\sum_{k=2}^{n+2} M_{k-2} M_{n+2-k} \right) x^{n+2}$$

$$(M_2 x^2 + M_3 x^3 + \dots) = (M_1 x^2 + M_2 x^3 + \dots) + [M_0 M_0 x^2 + (M_0 M_1 + M_1 M_0) x^3 + \dots]$$

In terms of $M(x)$, we get

$$M(x) - M_1 x - M_0 = x(M(x) - M_0) + x^2 M(x)^2$$

$$M(x) - x - 1 = x(M(x) - 1) + x^2 M(x)^2$$

$$x^2 M(x)^2 + (x - 1)M(x) + 1 = 0$$

Using the quadratic formula, we get

$$M(x) = \frac{-(x-1) \pm \sqrt{(x-1)^2 - 4x^2}}{2x^2} = \frac{1-x \pm \sqrt{1-2x-3x^2}}{2x^2}$$

Consider

$$\lim_{x \rightarrow 0} \frac{1-x + \sqrt{1-2x-3x^2}}{2x^2} = \infty$$

where we found the left-hand and right-hand limits. Moreover,

$$\lim_{x \rightarrow 0} \frac{1-x - \sqrt{1-2x-3x^2}}{2x^2} = \lim_{x \rightarrow 0} \frac{(3x+1)(1-2x-3x^2)^{-\frac{1}{2}} - 1}{4x} = \lim_{x \rightarrow 0} \frac{4(1-2x-3x^2)^{-\frac{3}{2}}}{4} = 1$$

where we applied L'Hopital's rule twice.

As we must have $\lim_{x \rightarrow 0} M(x) = 1$ to satisfy the initial condition, it follows that

$$M(x) = \frac{1-x - \sqrt{1-2x-3x^2}}{2x^2}$$

9.6 Schroder Numbers

Problem

Let f_n denote the number of paths from $(0,0)$ to (n,n) which never go above the diagonal $y = x$ and are made up only of the steps $(1,0)$, $(1,1)$, and $(0,1)$. Find the ordinary generating function $F(x) = \sum_{n \geq 0} f_n x^n$.

Solution

We begin by finding a recurrence relation for f_n .

Consider a path satisfying the conditions. There are two cases

- The first step of the path is $(1,1)$. There are f_{n-1} ways to complete the path from $(1,1)$ to (n,n) .
- The first step of the path is $(1,0)$. Eventually, the path will touch the line $y = x$ (at the very least, it has to touch (n,n) at the end). Let (k,k) be the point on the diagonal when it first happens ($1 \leq k \leq n$).

Consider the portion of the path from (k,k) to (n,n) . There are f_{n-k} ways to complete this path.

Consider the portion of the path from $(0,0)$ to (k,k) . The number of ways to go from $(0,0)$ to (k,k) without touching the diagonal $y = x$ is f_{k-1} . This is because the first step is $(1,0)$ (to go from $(0,0)$ to $(1,0)$), and the last step is $(0,1)$ (to go from $(k,k-1)$ to (k,k)). The number of paths from $(1,0)$ to $(k,k-1)$ which never go above $y = x-1$ is f_{k-1} .

Therefore, we have the following recurrence relation

$$f_n = f_{n-1} + \sum_{k=1}^n f_{k-1}f_{n-k}$$

for $n \geq 1$ where $f_0 = 1$. Since we want our generating function to be defined for all $n \geq 0$, we will work with the following recurrence relation

$$f_{n+1} = f_n + \sum_{k=1}^{n+1} f_{k-1}f_{n+1-k}$$

for $n \geq 0$, where $f_0 = 1$. Let $F(x) = \sum_{n \geq 0} f_n x^n$. Multiplying both sides of the recurrence relation by x^{n+1} and summing over $n \geq 0$ gives

$$\begin{aligned} \sum_{n \geq 0} f_{n+1} x^{n+1} &= \sum_{n \geq 0} f_n x^{n+1} + \sum_{n \geq 0} \left(\sum_{k=1}^{n+1} f_{k-1} f_{n+1-k} \right) x^{n+1} \\ (f_1 x^1 + f_2 x^2 + \dots) &= (f_0 x^1 + f_1 x^2 + \dots) + [f_0 f_0 x + (f_0 f_1 + f_1 f_0) x^2 + \dots] \end{aligned}$$

In terms of $F(x)$, we get

$$\begin{aligned} F(x) - f_0 &= xF(x) + xF(x)^2 \\ F(x) - 1 &= xF(x) + xF(x)^2 \\ xF(x)^2 + (x-1)F(x) + 1 &= 0 \end{aligned}$$

Using the quadratic formula, we get

$$F(x) = \frac{-(x-1) \pm \sqrt{(x-1)^2 - 4x}}{2x} = \frac{1-x \pm \sqrt{x^2 - 6x + 1}}{2x}$$

Consider

$$\lim_{x \rightarrow 0} \frac{1-x + \sqrt{x^2 - 6x + 1}}{2x} \text{ DNE}$$

where we found the left-hand and right-hand limits. Moreover,

$$\lim_{x \rightarrow 0} \frac{1-x - \sqrt{x^2 - 6x + 1}}{2x} = \lim_{x \rightarrow 0} \frac{(x^2 - 6x + 1)^{-\frac{1}{2}}(3-x) - 1}{2} = 1$$

where we applied L'Hopital's rule.

As we must have $\lim_{x \rightarrow 0} F(x) = 1$ to satisfy the initial condition, it follows that

$$F(x) = \frac{1-x - \sqrt{x^2 - 6x + 1}}{2x}$$

9.7 Bell Numbers

Problem

We proved that

$$B(n+1) = \sum_{k=0}^n \binom{n}{k} B(k)$$

for $n \geq 0$, where $B(0) = 1$. Use this recurrence relation to find an explicit formula for the generating function $B(x) = \sum_{n \geq 0} B(n) \frac{x^n}{n!}$.

Solution

Let $B(x) = \sum_{n \geq 0} B(n) \frac{x^n}{n!}$. Multiplying both sides of the recurrence relation by $\frac{x^n}{n!}$ and summing over $n \geq 0$ gives

$$\sum_{n \geq 0} B(n+1) \frac{x^n}{n!} = \sum_{n \geq 0} \left(\sum_{k=0}^n \binom{n}{k} B(k) \right) \frac{x^n}{n!}$$

Observe

$$B'(x) = \sum_{n \geq 1} B(n) \frac{n \cdot x^{n-1}}{n!} = \sum_{n \geq 1} B(n) \frac{x^{n-1}}{(n-1)!} = \sum_{n \geq 0} B(n+1) \frac{x^n}{n!}$$

and

$$B(x)e^x = \left(\sum_{n \geq 0} B(n) \frac{x^n}{n!} \right) \left(\sum_{n \geq 0} \frac{x^n}{n!} \right) = \sum_{n \geq 0} \left(\sum_{k=0}^n \binom{n}{k} B(k) \right) \frac{x^n}{n!}$$

Hence in terms of $B(x)$, we get the following first-order separable ODE

$$B'(x) = B(x)e^x$$

Rearranging and integrating both sides with respect to x gives,

$$\begin{aligned} \frac{B'(x)}{B(x)} &= e^x \\ \int \frac{B'(x)}{B(x)} dx &= \int e^x dx \end{aligned}$$

Let $u = B(x)$ then $du = B'(x) dx$ to get

$$\int \frac{du}{u} = e^x + C$$

$$\ln(|u|) = e^x + C$$

Since $u = B(x)$, we get

$$\ln(B(x)) = e^x + C$$

and using the initial condition $B(0) = 1$ yields

$$\ln(1) = 1 + C \implies C = -1$$

Therefore,

$$B(x) = e^{e^x - 1}$$

9.8 Permutation whose Square is the Identity

Problem

Let $r(n)$ be the number of n -permutations whose square is the identity permutation. We proved that

$$r(n+2) = r(n+1) + (n+1)r(n)$$

if $n \geq 0$, where $r(0) = r(1) = 1$. Use this recurrence relation to find an explicit formula for the generating function $R(x) = \sum_{n \geq 0} r(n) \frac{x^n}{n!}$.

Solution

Let $R(x) = \sum_{n \geq 0} r(n) \frac{x^n}{n!}$. Multiplying both sides of the recurrence relation by $\frac{x^n}{n!}$ and summing over all $n \geq 0$ gives

$$\sum_{n \geq 0} r(n+2) \frac{x^n}{n!} = \sum_{n \geq 0} r(n+1) \frac{x^n}{n!} + \sum_{n \geq 0} (n+1)r(n) \frac{x^n}{n!}$$

Expanding the second term on the RHS gives

$$\sum_{n \geq 0} r(n+2) \frac{x^n}{n!} = \sum_{n \geq 0} r(n+1) \frac{x^n}{n!} + \sum_{n \geq 0} nr(n) \frac{x^n}{n!} + \sum_{n \geq 0} r(n) \frac{x^n}{n!}$$

Now observe that

$$R'(x) = \sum_{n \geq 1} r(n) \frac{nx^{n-1}}{n!} = \sum_{n \geq 0} r(n+1) \frac{(n+1)x^n}{(n+1)!} = \sum_{n \geq 0} r(n+1) \frac{x^n}{n!}$$

and

$$R''(x) = \sum_{n \geq 2} r(n) \frac{n(n-1)x^{n-2}}{n!} = \sum_{n \geq 0} r(n+2) \frac{(n+2)(n+1)x^n}{(n+2)!} = \sum_{n \geq 0} r(n+2) \frac{x^n}{n!}$$

and

$$xR'(x) = x \sum_{n \geq 1} r(n) \frac{nx^{n-1}}{n!} = \sum_{n \geq 0} r(n) \frac{nx^n}{n!}$$

Hence in terms of $R(x)$ we get the following second-order ODE

$$R''(x) = R'(x) + xR'(x) + R(x) \iff R''(x) - (1+x)R'(x) - R(x) = 0$$

Using the observation that $-((1+x)R(x))' = -R(x) - (1+x)R'(x)$ we get

$$R''(x) - ((1+x)R(x))' = 0$$

Using the linearity of the derivative operator we get

$$[R'(x) - (1+x)R(x)]' = 0$$

Integrating both sides with respect to x yields

$$\int [R'(x) - (1+x)R(x)]' dx = \int 0 dx$$

$$R'(x) - (1+x)R(x) = C_1$$

where C_1 is some constant. We can solve this first-order ODE using the integrating factor method. Our integrating factor is

$$e^{\int -(1+x)dx} = e^{-\left(x+\frac{x^2}{2}\right)}$$

Multiplying both sides of the ODE by the integrating factor yields

$$e^{-\left(x+\frac{x^2}{2}\right)} R'(x) - e^{-\left(x+\frac{x^2}{2}\right)} (1+x)R(x) = e^{-\left(x+\frac{x^2}{2}\right)} C_1$$

$$\left(e^{-\left(x+\frac{x^2}{2}\right)} R(x) \right)' = C_1 e^{-\left(x+\frac{x^2}{2}\right)}$$

Integrating both sides with respect to x yields

$$\int \left(e^{-\left(x+\frac{x^2}{2}\right)} R(x) \right)' dx = \int C_1 e^{-\left(x+\frac{x^2}{2}\right)} dx$$

$$e^{-\left(x+\frac{x^2}{2}\right)} R(x) = \int C_1 e^{-\left(x+\frac{x^2}{2}\right)} dx + C_2$$

$$R(x) = e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 \int e^{-\left(x+\frac{x^2}{2}\right)} dx + C_2 \right)$$

Let $I(x) = \int e^{-\left(x+\frac{x^2}{2}\right)} dx$ then

$$R(x) = e^{\left(x+\frac{x^2}{2}\right)} (C_1 I(x) + C_2)$$

$$R(x) = C_1 e^{\left(x+\frac{x^2}{2}\right)} I(x) + C_2 e^{\left(x+\frac{x^2}{2}\right)}$$

Since $r(0) = 1$ and $r(1) = 1$, we must have $R(x)$ satisfy $R(0) = 1$ and $R'(0) = 1$.

Using $R(0) = 1$ we get

$$1 = C_1 I(0) + C_2$$

Using $R'(0) = 1$ and

$$\begin{aligned} R'(x) &= C_1(1+x)e^{\left(x+\frac{x^2}{2}\right)} I(x) + C_1e^{\left(x+\frac{x^2}{2}\right)} I'(x) + C_2(1+x)e^{\left(x+\frac{x^2}{2}\right)} \\ &= C_1(1+x)e^{\left(x+\frac{x^2}{2}\right)} I(x) + C_1e^{\left(x+\frac{x^2}{2}\right)} e^{-\left(x+\frac{x^2}{2}\right)} + C_2(1+x)e^{\left(x+\frac{x^2}{2}\right)} \quad (I'(x) = e^{-\left(x+\frac{x^2}{2}\right)}) \\ &= C_1(1+x)e^{\left(x+\frac{x^2}{2}\right)} I(x) + C_1 + C_2(1+x)e^{\left(x+\frac{x^2}{2}\right)} \end{aligned}$$

we get

$$1 = C_1 I(0) + C_1 + C_2$$

Substituting $C_2 = 1 - C_1 I(0)$ into the second equation gives

$$1 = C_1 I(0) + C_1 + 1 - C_1 I(0) \implies C_1 = 0$$

From which it follows that $C_2 = 1 - 0 \cdot I(0) = 1$.

Therefore,

$$R(x) = e^{x+\frac{x^2}{2}}$$

Remark. In the above solution we defined $I(x) = \int e^{-\left(x+\frac{x^2}{2}\right)} dx$ which is not a standard integral, but because of the nice initial conditions, terms containing $I(x)$ cancelled out. Here is a more general approach using the common Error Function:

$$\begin{aligned} R(x) &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 \int e^{-\left(x+\frac{x^2}{2}\right)} dx + C_2 \right) \\ R(x) &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 \int e^{-\frac{1}{2}(x+1)^2 + \frac{1}{2}} dx + C_2 \right) & \left(-\left(x+\frac{x^2}{2}\right) = -\frac{1}{2}(x+1)^2 + \frac{1}{2} \right) \\ &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 e^{\frac{1}{2}} \int e^{-\frac{1}{2}(x+1)^2} dx + C_2 \right) \\ &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 e^{\frac{1}{2}} \int e^{-\frac{1}{2}u^2} du + C_2 \right) & (\text{Letting } u = x+1, du = dx.) \\ &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 \sqrt{2} e^{\frac{1}{2}} \int e^{-t^2} dt + C_2 \right) & (\text{Letting } t = \frac{u}{\sqrt{2}}, dt = \frac{1}{\sqrt{2}} du.) \\ &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 \sqrt{2} e^{\frac{1}{2}} \frac{\sqrt{\pi}}{2} \frac{2}{\sqrt{\pi}} \int e^{-t^2} dt + C_2 \right) \\ &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 \sqrt{2} e^{\frac{1}{2}} \frac{\sqrt{\pi}}{2} \operatorname{erf}(t) + C_2 \right) & (\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int e^{-t^2} dt) \\ &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 \frac{\sqrt{2e\pi}}{2} \operatorname{erf}\left(\frac{u}{\sqrt{2}}\right) + C_2 \right) & (\text{Recall } t = \frac{u}{\sqrt{2}}) \\ &= e^{\left(x+\frac{x^2}{2}\right)} \left(C_1 \frac{\sqrt{2e\pi}}{2} \operatorname{erf}\left(\frac{x+1}{\sqrt{2}}\right) + C_2 \right) & (\text{Recall } u = x+1) \\ &= C_1 \frac{\sqrt{2e\pi}}{2} e^{\left(x+\frac{x^2}{2}\right)} \operatorname{erf}\left(\frac{x+1}{\sqrt{2}}\right) + C_2 e^{\left(x+\frac{x^2}{2}\right)} \end{aligned}$$

Therefore,

$$R(x) = C_1 \frac{\sqrt{2e\pi}}{2} e^{\left(x+\frac{x^2}{2}\right)} \operatorname{erf}\left(\frac{x+1}{\sqrt{2}}\right) + C_2 e^{\left(x+\frac{x^2}{2}\right)}$$

Since $r(0) = 1$ and $r(1) = 1$, we must have $R(x)$ satisfy $R(0) = 1$ and $R'(0) = 1$.

Using $R(0) = 1$ we get

$$1 = C_1 \frac{\sqrt{2e\pi}}{2} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) + C_2$$

Using the fact that $\frac{d}{dx} \operatorname{erf}\left(\frac{x+1}{\sqrt{2}}\right) = \frac{2}{\sqrt{\pi}} e^{-\left(\frac{x+1}{\sqrt{2}}\right)^2} \frac{1}{\sqrt{2}}$ we get

$$\begin{aligned} R'(x) &= C_1 \frac{\sqrt{2e\pi}}{2} (1+x) e^{\left(x+\frac{x^2}{2}\right)} \operatorname{erf}\left(\frac{x+1}{\sqrt{2}}\right) + C_1 \frac{\sqrt{2e\pi}}{2} e^{\left(x+\frac{x^2}{2}\right)} \frac{2}{\sqrt{\pi}} e^{-\left(\frac{x+1}{\sqrt{2}}\right)^2} \frac{1}{\sqrt{2}} + C_2 (1+x) e^{\left(x+\frac{x^2}{2}\right)} \\ &= C_1 \frac{\sqrt{2e\pi}}{2} (1+x) e^{\left(x+\frac{x^2}{2}\right)} \operatorname{erf}\left(\frac{x+1}{\sqrt{2}}\right) + C_1 + C_2 (1+x) e^{\left(x+\frac{x^2}{2}\right)} \end{aligned}$$

Using $R'(0) = 1$ and we get

$$1 = C_1 \frac{\sqrt{2e\pi}}{2} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) + C_1 + C_2$$

Substituting $C_2 = 1 - C_1 \frac{\sqrt{2e\pi}}{2} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right)$ into the second equation gives

$$1 = C_1 \frac{\sqrt{2e\pi}}{2} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) + C_1 + 1 - C_1 \frac{\sqrt{2e\pi}}{2} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) \implies C_1 = 0$$

From which it follows that $C_2 = 1 - 0 \cdot \frac{\sqrt{2e\pi}}{2} \operatorname{erf}\left(\frac{1}{\sqrt{2}}\right) = 1$.

Therefore,

$$R(x) = e^{x+\frac{x^2}{2}}$$

Remark. Another method to solve

$$R''(x) - (1+x)R'(x) - R(x) = 0$$

with initial conditions $R(0) = 1$ and $R'(0) = 1$ would be to use the power series method. Indeed, suppose $R(x) = \sum_{k \geq 0} a_k x^k$. Substitution into the differential equation gives,

$$\begin{aligned} \sum_{k \geq 2} a_k k(k-1) x^{k-2} - (1+x) \sum_{k \geq 1} a_k k x^{k-1} - \sum_{k \geq 0} a_k x^k &= 0 \\ \sum_{k \geq 2} a_k k(k-1) x^{k-2} - \sum_{k \geq 1} a_k k x^{k-1} - \sum_{k \geq 1} a_k k x^k - \sum_{k \geq 0} a_k x^k &= 0 \\ \sum_{k \geq 0} a_{k+2}(k+2)(k+1) x^k - \sum_{k \geq 0} a_{k+1}(k+1) x^k - \sum_{k \geq 0} a_k k x^k - \sum_{k \geq 0} a_k x^k &= 0 \\ \sum_{k \geq 0} [a_{k+2}(k+2)(k+1) - a_{k+1}(k+1) - a_k k - a_k] x^k &= 0 \end{aligned}$$

Hence we must have

$$a_{k+2}(k+2)(k+1) - a_{k+1}(k+1) - a_k k - a_k = 0 \text{ for all } k \in \{0, 1, 2, \dots\}$$

In other words,

$$a_{k+2} = \frac{a_{k+1}(k+1) + a_k(k+1)}{(k+2)(k+1)}$$

$$a_{k+2} = \frac{a_{k+1} + a_k}{(k+2)}$$

for all $k \geq 0$.

This allows us to compute the coefficients of x^n in $R(x)$:

k	a_{k+2}
0	1
2	$\frac{2}{3}$
3	$\frac{5}{12}$
4	$\frac{13}{60}$
$\vdots$	$\vdots$

Therefore,

$$R(x) = 1 + x + x^2 + \frac{2}{3}x^3 + \frac{5}{12}x^4 + \frac{13}{60}x^5 + \dots$$

which is indeed the Taylor series expansion of $e^{x+\frac{x^2}{2}}$ at $x = 0$ (you can check this using a CAS such as WolframAlpha).

Hence, written in terms of an exponential generating function

$$R(x) = 1 \cdot \frac{x^0}{0!} + 1 \cdot \frac{x^1}{1!} + 2 \cdot \frac{x^2}{2!} + 4 \cdot \frac{x^3}{3!} + 10 \cdot \frac{x^4}{4!} + 26 \cdot \frac{x^5}{5!} + \dots$$

10.1 Trees

Trees are graphs which are “just barely” connected.

Result: Trees

Let G be a connected simple graph on n vertices. Then the following are equivalent.

1. G is minimally connected, that is, if we remove any edge of G , then the obtained graph G' will not be connected.
2. G does not contain a cycle.

Proof

Suppose for contradiction that G is minimally connected and contain a cycle C . Let u and v be two vertices in G . As G is connected, there exists a path P from u to v . Remove an edge $e = (a, b)$ from the cycle C to get a new graph G' . If path P did not use edge e to get from u to v then P is still a path from u to v in G' . If path P did use edge e to get from u to v then replace the edge e by the longer arc from a to b to get from u to v . This shows that G' is connected which contradicts that G is minimally connected.

We use contrapostive to show the other direction. That is, we show if G is not minimally connected then G must contain a cycle. Indeed, if G is not minimally connected then there exists an edge $e = (u, v)$ that can be removed to obtain a graph G' which is still connected. As G' is connected, there is a path P from u to v . Then adding back edge $e = (u, v)$ creates a cycle which means that G has a cycle. $\square$

Result

Let T be a tree on n vertices, where $n \geq 2$. Then T has at least two leaves (i.e. vertices whose degree is 1).

Proof

Take a path P of maximal length in T . The endpoints of P are vertices of degree 1 (if one of them is not a leaf then we can further extend the path P which contradicts its maximality). $\square$

Result

All trees on n vertices have $n - 1$ edges. Conversely, all connected graphs on n vertices with exactly $n - 1$ edges are trees.

Proof

To prove the first statement, we proceed by induction on the number of vertices, n . For the base case $n = 1$. A tree on 1 vertex has $0 = 1 - 1$ edges so the claim holds.

Suppose for some $k \geq 1$ that all trees on k vertices have $k - 1$ edges. Consider a tree T on $k + 1$ vertices. By the previous result, there exists a leaf vertex v in T . Remove v and the edge e incident to v from T to get a new graph T' . Then T' is also a tree (as it is connected and has no cycles) but on k vertices. By the induction hypothesis, T' has $k - 1$ edges. Adding back vertex v and the removed edge e give T with $(k - 1) + 1 = k$ edges.

Therefore, by the Principle of Mathematical Induction, all trees on n vertices have $n - 1$ edges.

To prove the second statement, we argue by contradiction. Suppose for contradiction that there exists a connected graph G on n vertices with exactly $n - 1$ edges such that G is not a tree.

In other words, G contains a cycle. Remove an edge e from the cycle then G will still be connected with exactly $n - 2$ edges. This is a contradiction as connected graphs have at least $n - 1$ edges (we proved that trees are minimally connected graphs with $n - 1$ edges and so if G is connected with $n - 2$ edges then it is even more minimally connected than a tree). $\square$

10.2 Forests

A forest is a union of trees. This means that a forest does not have any cycles and does not need to be connected (each connected component is a tree).

Result

Let F be a forest on n vertices with k connected components. Then F has $n - k$ edges.

Proof

Label the connected components of F as $T_1, T_2, \dots, T_k$. and let n_i be the number of vertices in connected component T_i . As each connected component T_i is a tree, it follows that T_i has $n_i - 1$ edges. Therefore, F has $\sum_{i=1}^k n_i - 1 = n - k$ edges, as required. $\square$

Result

Let F and F' be two forests on the same vertex set V , and let F have less edges than F' . Then F' has an edge e that can be added to F so that the obtained graph $F \cup e$ is still a forest.

Proof

Suppose for contradiction that F' has no edge e that can be added to F so that the obtained graph $F \cup e$ is still a forest. In other words, adding any edge e from F' to F creates a cycle in F . This means that every edge e in F' connects two vertices in the same connected component in F . Let k_1 and k_2 be the number of connected components of F and F' , respectively. It follows that $k_1 \leq k_2$.

By the previous result, F has $n - k_1$ edges and F' has $n - k_2$ edges. Since F has less edges than F' it follows that $n - k_1 < n - k_2 \iff k_1 > k_2$, a contradiction.

Therefore, F' has an edge e that can be added to F so that the obtained graph $F \cup e$ is still a forest. $\square$

10.3 Minimum Weight Spanning Trees

If G is a connected graph, we say that T is a spanning tree of G if G and T have the same vertex set, and each edge of T is also an edge of G .

Result

Let G be a connected graph then G has a spanning tree.

Proof

If G is a tree, then G is its own spanning tree. If G is not a tree, then G is not minimally connected which means that we can remove an edge from G to obtain a new graph G' that is still connected. If G' is a tree then we are done, otherwise we can continue removing edges until the resulting graph becomes a tree. $\square$

If a connected graph G has a cycle, then there is more than one spanning tree for G , but each spanning tree has the same number of edges.

Let G be a simple connected graph on n vertices and with m edges. Assign each edge e of G a nonnegative real number weight (or cost) $w(e)$. We are interested in finding a spanning tree T of G with minimum cost (i.e. $\sum_{e \in T} w(e)$ is a minimum). That is, if H is another a spanning tree of G , then $\sum_{e \in T} w(e) \leq \sum_{e \in H} w(e)$.

It turns out that there is a rather simple greedy algorithm, called Kruskal's algorithm, that always returns a minimum spanning tree. Here is the algorithm:

Result: Kruskal's Algorithm

- Sort the edges of the graph G in non-decreasing order by their weight

$$w(e_1) \leq w(e_2) \leq \dots \leq w(e_m)$$

- Initially begin with an empty solution $F = \{\}$
- Go through the sorted edges one at a time and add edge e_i to F if $F \cup \{e_i\}$ does not create a cycle in F (if e_i does create a cycle, skip over it and consider the next edge e_{i+1}).
- Return F once all the edges have been examined. (Technically, we can return F once $n - 1$ edges have been added, but the way we have written it will make the proof of correctness easier).

Result

Kruskal's algorithm always finds a minimum weight spanning tree.

First note that the solution F that we obtain using Kruskal's algorithm after i steps is not necessarily connected. All we know is that F is a cycle free graph which means that F is a forest. However, when the algorithm terminates, F will be a forest with $n - 1$ edges which means that F will be a tree.

Now we show that the tree F obtained is indeed a minimum spanning tree.

Proof

Let T be the spanning tree obtained by running Kruskal's algorithm on some graph G and suppose for contradiction that G has a spanning tree H such that $\sum_{e \in H} w(e) < \sum_{e \in T} w(e)$.

Order the edges of T and H in non-decreasing order

$$w(t_1) \leq w(t_2) \leq \dots \leq w(t_{n-1})$$

$$w(h_1) \leq w(h_2) \leq \dots \leq w(h_{n-1})$$

Observe that Kruskal's algorithm begins by adding the absolute minimum weighted edge t_1 to T which means that $w(t_1) \leq w(h_1)$. Hence initially, the weight of T is smaller than or equal to the weight of H . Moreover, by assumption, we know that at the end, the total weight of T is greater than the total weight of H . Let k be the first point where the weight of T becomes greater than the weight of H . In other words, k is the smallest integer such that

$$\sum_{i=1}^{k-1} w(t_i) \leq \sum_{i=1}^{k-1} w(h_i) \wedge \sum_{i=1}^k w(t_i) > \sum_{i=1}^k w(h_i)$$

Since k is the first index at which the weight of T exceeds the weight of H , we must have $w(t_k) > w(h_k)$. Now the idea is to show that Kruskal's algorithm could not have chosen edge t_k at step k .

Let F be the forest created by Kruskal's algorithm in the first $k - 1$ steps. In other words, $F = t_1 \cup t_2 \cup \dots \cup t_{k-1}$. Let F' be the forest formed by the union of the edges $h_1, h_2, \dots, h_k$. Since F and F' are two forests on the same vertex set, and F has less edges than F' , it follows that F' has an edge h_i (where $i \leq k$) that can be added to F so that the obtained graph $F \cup h_i$ is still a forest. However, $w(h_i) \leq w(h_k) < w(t_k)$.

We know that Kruskal's algorithm at each step adds the minimum weighted edge that does not create a cycle. Hence, it could not have added edge t_k at step k because edge h_i exists with a smaller weight and does not form a cycle.

This proves that no spanning tree H of G can have a smaller total weight than the spanning tree T obtained by Kruskal's algorithm. $\square$

The next problem shows that the property of forests that was key in proving that Kruskal's algorithm finds the minimum cost spanning tree can be found in other structures as well.

Problem

Let $B_1 = \{v_1, v_2, \dots, v_k\}$ and $B_2 = \{w_1, w_2, \dots, w_{k+1}\}$ be two linearly independent sets of vectors from the same vector space U . Prove that there exists an index $i \in [k+1]$ so that the set $B_1 \cup w_i$ of vectors is linearly independent.

Proof

Suppose for contradiction that for all $i \in [k+1]$, $B_1 \cup w_i$ is linearly dependent.

This means that there exists scalars $c_1, c_2, \dots, c_{k+1}$ not all zero such that

$$c_1 \cdot v_1 + c_2 \cdot v_2 + \dots + c_k \cdot v_k + c_{k+1} \cdot w_i = 0$$

We claim that $c_{k+1} \neq 0$. Suppose $c_{k+1} = 0$ then there exists some index $i \in [k]$ such that $c_i \neq 0$. Solving in terms of v_i gives

$$v_i = -\frac{c_1}{c_i} \cdot v_1 - \dots - \frac{c_k}{c_i} \cdot v_k$$

which means that $v_i \in \text{span}(B_1)$. This contradicts the fact that B_1 is a linearly independent set of vectors. Hence, $c_{k+1} \neq 0$.

Now solving in terms of w_i gives

$$w_i = -\frac{c_1}{c_{k+1}} \cdot v_1 - \dots - \frac{c_k}{c_{k+1}} \cdot v_k$$

which means that $w_i \in \text{span}(B_1)$ for all $i \in [k+1]$. In other words, $B_2 \subseteq \text{span}(B_1)$. It follows that $\text{span}(B_2) \subseteq \text{span}(B_1)$ which is a contradiction as B_2 is a strictly larger set of linearly independent vectors than B_1 .

Therefore, there exists an index $i \in [k+1]$ so that the set $B_1 \cup w_i$ of vectors is linearly independent. $\square$

We now give an alternative proof that also shows the correctness of Kruskal's algorithm.

Let F_i denote the set F after i iterations of the loop where $i \in \{0, 1, \dots, m\}$. Initially, $F_0 = \{\}$ and we only add edges to F which means that $F_0 \subseteq F_1 \subseteq F_2 \subseteq \dots \subseteq F_m$. After i iterations, we have only considered the first i edges $e_1, e_2, \dots, e_i$ which means that $F_i \subseteq \{e_1, e_2, \dots, e_i\}$. At the end, F_m is returned.

We say F_i is promising if there exists a minimum spanning tree H of G such that

$$F_i = H \cap \{e_1, e_2, \dots, e_i\}$$

In other words, F_i made the same decisions as H for $e_1, e_2, \dots, e_i$.

We want to show for all $i \in \{0, 1, \dots, m\}$, F_i is promising.

Proof

We proceed by induction on the loop iteration.

For the base case we have $i = 0$. Thus far Kruskal's algorithm has not added any edges to F , hence $F_0 = \{\}$. Since G is connected, it has a minimum spanning tree H and

$$H \cap \{\} = \{\} = F_0$$

which shows that F_0 is promising.

Now suppose for some $i \geq 0$ that F_i is promising. That is, there exists a minimum spanning tree H of G such that $H \cap \{e_1, e_2, \dots, e_i\} = F_i$. We wish to show that F_{i+1} is promising. Consider the edge e_{i+1} . Either $F_i \cup \{e_{i+1}\}$ has a cycle or it does not:

- if $F_i \cup \{e_{i+1}\}$ has a cycle then $e_{i+1} \notin F_{i+1}$. We know that F_i is promising which means that there is a minimum spanning tree H such that $H \cap \{e_1, e_2, \dots, e_i\} = F_i$ from which it follows that $F_i \subseteq H$. This means that $e_{i+1} \notin H$ (if $e_{i+1} \in H$ then $F_i \cup \{e_{i+1}\} \subseteq H$ which means that H has a cycle which contradicts that H is a tree). Hence, $F_{i+1} = F_i = H \cap \{e_1, e_2, \dots, e_{i+1}\}$ which means that F_{i+1} is promising.
- if $F_i \cup \{e_{i+1}\}$ does not have a cycle then $e_{i+1} \in F_{i+1}$.
 - if $e_{i+1} \in H$ then F_{i+1} agrees with H on $\{e_1, e_2, \dots, e_{i+1}\}$. Hence, F_{i+1} is promising.
 - if $e_{i+1} \notin H$ then F_{i+1} and H do not agree on edge e_{i+1} . Hence, we need to find a new minimum spanning tree H' which does agree with F_{i+1} on $\{e_1, e_2, \dots, e_{i+1}\}$.

Let e_{i+1} be the edge (u, v) . Since H is a minimum spanning tree it follows that H has a path P from u to v . So $P \cup \{e_{i+1}\}$ is a cycle which means that $P \cup \{e_{i+1}\} \not\subseteq F_{i+1}$ as F_{i+1} has no cycles. So there exists an edge $e_j \in P$ such that $e_j \notin F_{i+1}$. As $F_i \subseteq F_{i+1}$, it follows that $e_j \notin F_i$. We know that F_i agrees with H on $\{e_1, e_2, \dots, e_i\}$ which means that $e_j \notin \{e_1, e_2, \dots, e_i\}$ hence $j > i$. Furthermore, $j \neq i + 1$ since $e_j \in H$ but $e_{i+1} \notin H$ so $j > i + 1$ from which it follows that $w(e_j) \geq w(e_{i+1})$.

Now build $H' = (H \setminus \{e_j\}) \cup \{e_{i+1}\}$.

Now H' is a spanning tree since $H \cup \{e_{i+1}\}$ has a unique cycle and we remove e_j from that cycle which gives back a spanning tree. Moreover, since H is a minimum spanning tree, it follows that $w(H) \leq w(H')$. Furthermore, since we build H' from H by replacing edge e_j with edge e_{i+1} where $w(e_{i+1}) \leq w(e_j)$, it follows that $w(H') \leq w(H)$. Therefore, $w(H') = w(H)$ which shows that H' is a minimum spanning tree.

Hence, H' and F_{i+1} agree on e_{i+1} which means that H' and F_{i+1} agree on $\{e_1, e_2, \dots, e_{i+1}\}$.

In other words, F_{i+1} is promising.

Therefore, by the Principle of Mathematical Induction, F_i is promising for all $i \in \{0, 1, \dots, m\}$. □

At the end when all the edges have been considered, we have that F_m is promising. In other words, there exists a minimum spanning tree H such that $F_m = H \cap \{e_1, e_2, \dots, e_m\} = H$ (both F_m and H picked the same $n - 1$ edges). Hence, F_m is a minimum spanning tree.

10.4 Mr. and Mrs. Smith

Problem

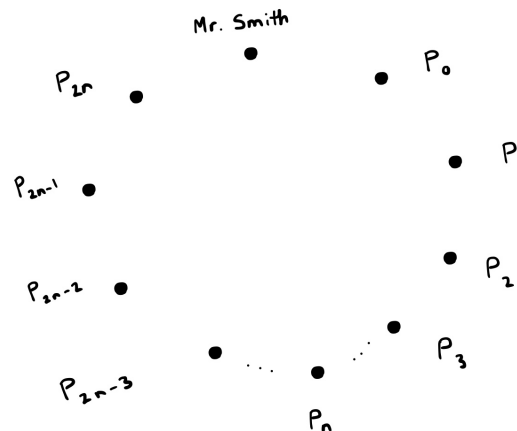
Mr. and Mrs. Smith invited n couples to their home. Some guests were friends of Mr. Smith, and some were friends of Mrs. Smith. When the guests arrived, people who knew each other beforehand shook hands, those who did not know each other just greeted each other. After all this took place, the observant Mr. Smith said “How interesting. If you disregard me, there are no two people present who shook hands the same number of times”. How many times did Mrs. Smith shake hands?

Solution

We will model this problem as a graph. The graph will have $2n + 2$ vertices representing the $2n + 2$ people present at the house. Moreover, we will label each vertex with the number of hands shaken by that person (except for Mr. Smith). Furthermore, we will draw an edge between two vertices if and only if the people corresponding to those vertices have shaken hands.

Since a person cannot shake their own hand nor the hand of their spouse, it follows that each person can shake $0, 1, 2, \dots$, or $2n$ hands. Additionally, disregarding Mr. Smith, we know that no two people have shaken hands the same number of times which means that each possibility $0, 1, 2, \dots, 2n$ occurs exactly once.

Consider the initial graph



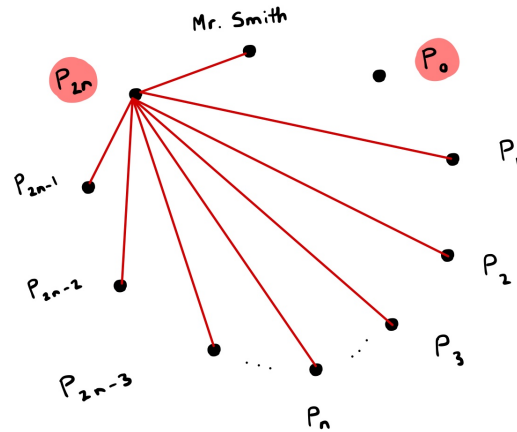
where P_i is the person who has shaken i people's hands.

Our task is to determine the number of hands shaken by Mrs. Smith. This corresponds to determining which vertex P_k corresponds to Mrs. Smith.

Consider the following cases:

- Who is the spouse of P_{2n} ?

P_{2n} is the person who shook hands with everyone else present except one person, who must be their spouse. We know P_{2n} did not shake P_0 's hand (in fact nobody did). Hence, P_{2n} and P_0 must be a couple.

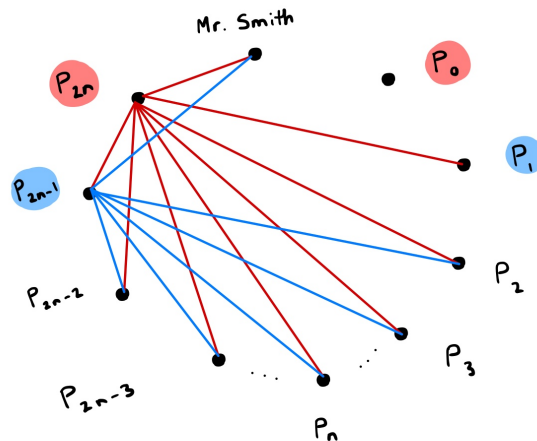


- Who is the spouse of P_{2n-1} ?

P_{2n-1} is the person who shook hands with everyone else present except two people. Hence, one of those two people must be P_{2n-1} 's spouse. We have two potential candidates

- P_0 did not shake anyone's hand and hence did not shake P_{2n-1} 's hand.
- P_1 did not shake P_{2n-1} 's hand as they only shook one person's hand which was P_{2n} .

We previously determined that P_0 is married to P_{2n} . Therefore, P_{2n-1} and P_1 must be a couple.



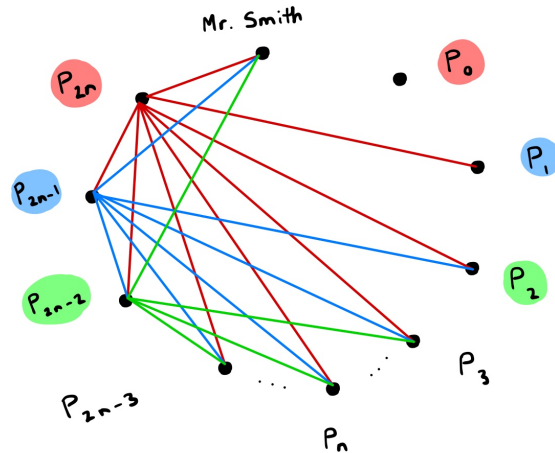
- Who is the spouse of P_{2n-2} ?

P_{2n-2} is the person who shook hands with everyone else present except three people. Hence, one of those three people must be P_{2n-2} 's spouse. We have three potential candidates

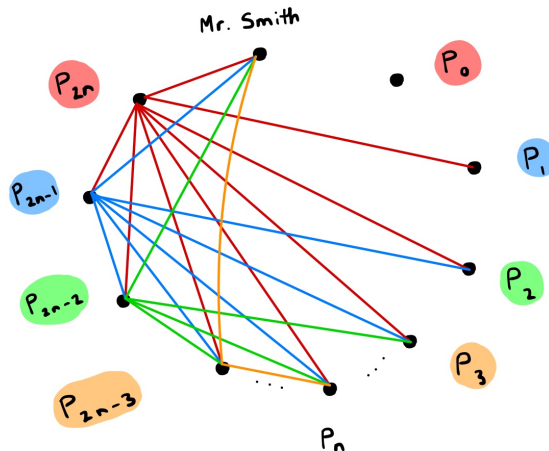
- P_0 did not shake anyone's hand and hence did not shake P_{2n-2} 's hand.
- P_1 did not shake P_{2n-2} 's hand as they only shook one person's hand which was P_{2n} .

- P_2 did not shake P_{2n-2} 's hand as they only shook two people's hands which were P_{2n} and P_{2n-1} .

We previously determined that P_0 is married to P_{2n} and P_1 is married to P_{2n-1} . Therefore, P_{2n-2} and P_2 must be a couple.



Continuing in this fashion we see that P_{2n-k} is married to P_k for $k \in \{0, 1, \dots, n-1\}$.



Now only Mr. Smith and P_n are left which means that P_n must be Mrs. Smith. Therefore, Mrs. Smith shook hands n times.

11.1 Catalan Numbers

In this section we take a look at a couple of problems which, at first glance, look quite different from each other but end up being *related*.

Super Sequences

Define a sequence of numbers $(a_1, a_2, \dots, a_n)$ to be *super* if $a_i = \pm 1$ for all $i \in [n]$, $\sum_{i=1}^n a_i = 0$, and $\sum_{i=1}^k a_i \geq 0$ for all $k \in [n]$. How many super sequences are there of length 6?

Solution. There are 5 valid sequences:

$$\begin{aligned} &+1 - 1 + 1 - 1 + 1 - 1 \\ &+1 - 1 + 1 + 1 - 1 - 1 \\ &+1 + 1 - 1 - 1 + 1 - 1 \\ &+1 + 1 - 1 + 1 - 1 - 1 \\ &+1 + 1 + 1 - 1 - 1 - 1 \end{aligned}$$

(Notice that a super sequence must have an even length otherwise the sum of its terms will not be zero.)

Balanced Parentheses

Define a sequence of parentheses to be *balanced* if the parentheses can be properly matched (each opening parenthesis should be matched to a closing one that lies further to its right). How many sequences of balanced parentheses are there using 3 pairs of parentheses?

Solution. There are 5 valid sequences:

$$\begin{aligned} &()()() \\ &()(()) \\ &(())() \\ &(())() \\ &((())) \end{aligned}$$

Jelly Bean Sequences

Start with an empty jar. For n days, you can either put one jelly bean in the jar or take one jelly bean out of the jar and then you must record the total number of jelly beans in the jar to generate a sequence. At the end of the n days the jar must be empty. How many jelly bean sequences are there of length 6?

Solution. There are 5 valid sequences:

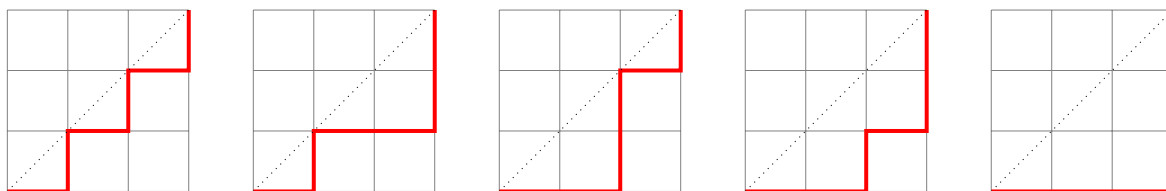
1, 0, 1, 0, 1, 0
 1, 0, 1, 2, 1, 0
 1, 2, 1, 0, 1, 0
 1, 2, 1, 2, 1, 0
 1, 2, 3, 2, 1, 0

(Notice that a jelly bean sequence must have an even length otherwise the jar will not be empty after the last day.)

Lattice Paths Below Main Diagonal

How many North-East lattice paths are there from $(0,0)$ to $(3,3)$ that never go above the main diagonal $y = x$?

Solution. There are 5 valid sequences:



They are all Connected

Notice that the answer to each question was 5 ways. This is no coincidence. All four problems are actually the same problem.

To see this, we will reduce all problems to the Super Sequence problem.

Connection Between Super Sequences and Balanced Parentheses

Given a super sequence, generate a sequence of balanced parentheses by replacing $+1$ s with $($ s and -1 s with $)$ s.

Given a sequence of balanced parentheses, generate a super sequence by replacing $($ s with $+1$ s and $)$ s with -1 s.

Connection Between Super Sequences and Jelly Bean Sequences

Given a super sequence $a_1, a_2, \dots, a_n$, generate a jelly bean sequence by computing $\sum_{i=1}^k a_i$ for all $k \in [n]$.

Given a jelly bean sequence, generate a super sequence by recording $+1$ if a jelly bean is added and -1 if a jelly bean is taken out (instead of recording the total number of jelly beans in the jar).

Connection Between Super Sequences and Lattice Paths Below Main Diagonal

Given a super sequence, generate a lattice path by travelling EAST for each $+1$ and NORTH for each -1 . Given a lattice path that never goes above the main diagonal, generate a super sequence by recording $+1$ for EAST moves and -1 for NORTH moves.

This means that if we solve one of them in full generality, we will have ended up solving all four (and possibly even more problems).

11.1.1 Deriving the Formula by Counting

We will solve the following lattice path problem

Problem

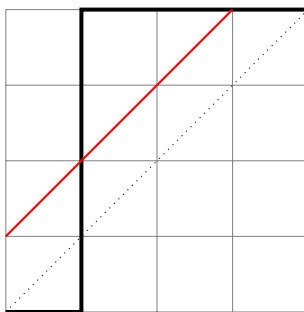
How many North-East lattice paths are there from $(0, 0)$ to (n, n) that never go above the main diagonal $y = x$?

Solution

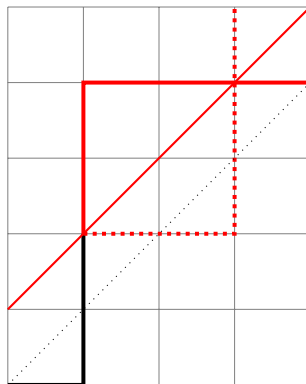
A (NE) lattice path from $(0, 0)$ to (n, n) is a sequence of n NORTH and n EAST steps. The number of ways to permute the string $\underbrace{N, N, \dots, N}_{n \text{ of them}}, \underbrace{E, E, \dots, E}_{n \text{ of them}}$ is given by $\frac{(n+n)!}{n!n!} = \binom{2n}{n}$. Therefore, the total number of (NE) lattice paths from $(0, 0)$ to (n, n) is $\binom{2n}{n}$.

Now from this total, we want to subtract the number of paths that DO go above the diagonal $y = x$. Define an illegal path to be a (NE) lattice path that goes above the diagonal $y = x$ and touches the diagonal $y = x + 1$.

Consider the following illegal path which crosses $y = x$ (dotted diagonal) and touches $y = x + 1$ (red diagonal):



Take the part of the illegal path after the crossing the diagonal $y = x + 1$ and reflect it about $y = x + 1$. (The part of the path after crossing $y = x + 1$ is in red and its reflection is in dotted red.)



This ended up swapping all the NORTH steps to EAST steps and vice versa. The part of the path that was not reflected (in black) has one more NORTH step than EAST steps, hence the remaining section of the path (in red) has one more EAST step than NORTH steps. When this (red) portion of the path is reflected, it now has one more NORTH step than EAST steps.

Consider the path consisting of the black line and the red dotted line. There are still $2n$ total steps, but there are now $n+1$ NORTH steps and $n-1$ EAST steps. Hence, instead of reaching (n, n) , all illegal paths after reflection end at $(n-1, n+1)$. So for any illegal path, we have constructed a path from $(0, 0)$ to $(n-1, n+1)$. Furthermore, notice that every path in the $(n-1) \times (n+1)$ grid touches the diagonal $y = x + 1$, and because the reflection process is reversible, it follows that the number of illegal paths are exactly the number of (NE) lattice paths from $(0, 0)$ to $(n-1, n+1)$ which is $\frac{[(n-1)+(n+1)]!}{(n-1)!(n+1)!} = \binom{2n}{n+1}$.

Hence the number of legal paths is

$$\begin{aligned}
 \binom{2n}{n} - \binom{2n}{n+1} &= \frac{2n!}{n! \cdot n!} - \frac{2n!}{(n+1)!(2n-n-1)!} \\
 &= \frac{2}{n!} - \frac{2n!}{(n+1) \cdot n! \cdot (n-1)!} \\
 &= \frac{2}{n!} - \frac{2}{(n+1)(n-1)!} \\
 &= \frac{2(n+1)(n-1)! - 2n!}{(n!)(n+1)(n-1)!} \\
 &= \frac{2[(n+1)(n-1)! - n(n-1)!]}{(n+1)!(n-1)!} \\
 &= \frac{2}{(n+1)!} \\
 &= \frac{1}{n+1} \cdot \frac{2n!}{n! \cdot n!} \\
 &= \frac{1}{n+1} \binom{2n}{n}
 \end{aligned}$$

Therefore, there are $\frac{1}{n+1} \binom{2n}{n}$ lattice paths from $(0, 0)$ to (n, n) that never go above the main diagonal $y = x$.

It turns out that the quantity $\frac{1}{n+1} \binom{2n}{n}$ has a special name. We call $C_n = \frac{1}{n+1} \binom{2n}{n}$ the n^{th} Catalan number. The first few Catalan numbers are

$$1, 2, 5, 14, 42, 132, 429, \dots$$

and show up in many counting problems.

Moreover, the idea in the previous solution can be used to show

Result

Let $a \geq b$ be two positive integers. The number of northeastern lattice paths from $(0, 0)$ to (a, b) that never go above the main diagonal $y = x$ is $\binom{a+b}{b} - \binom{a+b}{a+1}$.

The previous problem was the special case when $a = b$. Since we did not really use the equality of these two parameters in the derivation, a very similar proof will establish the result.

So now we have the answer to the other three problems as well

Problem

- a) How many super sequences are there of length $2n$?
- b) How many sequences of balanced parentheses are there using n pairs of parentheses?
- c) How many jelly bean sequences are there of length $2n$?

Solution

The answer for parts a), b), and c) is $C_n = \frac{1}{n+1} \binom{2n}{n}$.

And C_n turns out to be the answer to hundreds of more counting problems!

11.1.2 Deriving the Formula by Generating Functions

Consider the following jelly bean sequence

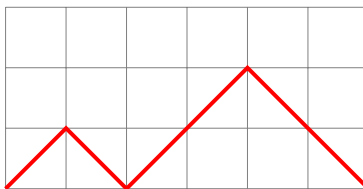
$$1, 0, 1, 2, 1, 0$$

or equivalently the following super sequence,

$$+1 - 1 + 1 + 1 - 1 - 1$$

We can visualize these sequences on a grid in a very nice way.

For each jelly bean added, take a $(1, 1)$ step and for each jelly bean removed, take a $(1, -1)$ step to get the following path



Such paths are called Dyck Paths. Hence, we have yet another formulation of the Catalan numbers.

Result

Let C_n denote the number of paths from $(0,0)$ to $(2n,0)$ which never go below $y = 0$ and are made up only of the steps $(1,1)$ and $(1,-1)$.

Problem

Find a recurrence relation for C_n .

Solution

Consider an arbitrary path satisfying the conditions. Since such paths cannot dip below the line $y = 0$, it follows that the first step of the path must be $(1,1)$ (or put another way, we cannot take out a jelly bean from an empty jar). Eventually, the path will come back down and touch the line $y = 0$ (the jar must be empty on the last day at the very least). Let $(2k,0)$, where $1 \leq k \leq n$, be the first day when the jar becomes empty again, disregarding the initial day.

Consider the portion of the path from $(2k,0)$ to $(2n,0)$. There are C_{n-k} ways to complete this path. Consider the portion of the path from $(0,0)$ to $(2k,0)$. The number of ways to go from $(0,0)$ to $(2k,0)$ without touching the line $y = 0$ is C_{k-1} . This is because the first step is $(1,1)$ (to go from $(0,0)$ to $(1,1)$), and the last step is $(1,-1)$ (to go from $(2k-1,1)$ to $(2k,0)$). The number of paths from $(1,1)$ to $(2k-1,1)$ which never go below $y = 1$ is C_{k-1} .

Therefore, the Catalan numbers satisfy

$$C_n = \sum_{k=1}^n C_{k-1} C_{n-k}$$

for $n \geq 1$, where $C_0 = 1$.

Our next goal is to derive the generating function for the Catalan numbers.

Since we want our generating function to be defined for all $n \geq 0$, we will work with the following recurrence relation

$$C_{n+1} = \sum_{k=1}^{n+1} C_{k-1} C_{n+1-k}$$

for $n \geq 0$, where $C_0 = 1$.

Problem

Find the ordinary generating function $C(x) = \sum_{n \geq 0} C_n x^n$.

Solution

Let $C(x) = \sum_{n \geq 0} C_n x^n$. Multiplying both sides of the recurrence relation by x^{n+1} and summing over $n \geq 0$ gives

$$\begin{aligned} \sum_{n \geq 0} C_{n+1} x^{n+1} &= \sum_{n \geq 0} \left(\sum_{k=1}^{n+1} C_{k-1} C_{n+1-k} \right) x^{n+1} \\ (C_1 x^1 + C_2 x^2 + C_3 x^3 + \dots) &= [C_0 C_0 x + (C_0 C_1 + C_1 C_0) x^2 + \dots] \end{aligned}$$

In terms of $C(x)$, we get

$$\begin{aligned} C(x) - C_0 &= xC(x)^2 \\ C(x) - 1 &= xC(x)^2 \\ xC(x)^2 - C(x) + 1 &= 0 \end{aligned}$$

Using the quadratic formula, we get

$$C(x) = \frac{1 \pm \sqrt{1-4x}}{2x}$$

Consider

$$\lim_{x \rightarrow 0} \frac{1 + \sqrt{1-4x}}{2x} \text{ DNE}$$

where we found the left-hand and right-hand limits. Moreover,

$$\lim_{x \rightarrow 0} \frac{1 - \sqrt{1-4x}}{2x} = \frac{2(1-4x)^{-\frac{1}{2}}}{2} = 1$$

where we applied L'Hopital's rule.

As we must have $\lim_{x \rightarrow 0} C(x) = 1$ to satisfy the initial condition, it follows that

$$C(x) = \frac{1 - \sqrt{1-4x}}{2x}$$

Recall that we computed

$$\sqrt{1-4x} = 1 - 2x - 2 \sum_{n \geq 2} \frac{1}{n} \binom{2n-2}{n-1} x^n$$

in the Binomial Theorem chapter. We will use this to find a closed form for C_n .

Problem

Deduce a closed formula for the sequence C_n from the generating function.

Solution

Consider

$$\begin{aligned}
 C(x) &= \frac{1 - \sqrt{1 - 4x}}{2x} \\
 &= \frac{1}{2x} - \frac{1}{2x} \sqrt{1 - 4x} \\
 &= \frac{1}{2x} - \frac{1}{2x} \left(1 - 2x - 2 \sum_{n \geq 2} \frac{1}{n} \binom{2n-2}{n-1} x^n \right) \\
 &= \frac{1}{2x} - \frac{1}{2x} + 1 + \frac{1}{x} \sum_{n \geq 2} \frac{1}{n} \binom{2n-2}{n-1} x^n \\
 &= 1 + \sum_{n \geq 2} \frac{1}{n} \binom{2n-2}{n-1} x^{n-1} \\
 &= 1 + \sum_{n \geq 1} \frac{1}{n+1} \binom{2n}{n} x^n \\
 &= \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} x^n
 \end{aligned}$$

from which it follows that

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

11.2 Bounding a General Sequence

Problem

Define $\{a_n\}_{n=0}^{\infty}$ where $a_0 = 0$, $a_1 = 1$, and $a_n = A \cdot a_{n-1} + B \cdot a_{n-2}$ for all $n \geq 2$ where $A, B \in \mathbb{R}$. Find a constant M such that $a_n \leq M^n$ for all $n \geq 0$.

Claim. Suppose $A \geq 1$ and $B \geq 1$. Let $M = A + B$ then $a_n \leq M^n$ for all n .

Proof

We proceed by strong induction.

For the base cases we have,

- if $n = 0$ then $a_0 = 0$ (given) and $a_0 = 0 \leq 1 = (A + B)^0$.
- if $n = 1$ then $a_1 = 1$ (given) and $a_1 = 1 \leq A + B = (A + B)^1$.

Now suppose for all $n \in \{0, 1, 2, \dots, k\}$ where $k \geq 1$, we have that $a_n \leq (A + B)^n$. Consider $n = k + 1$,

$$\begin{aligned}
 a_{k+1} &= A \cdot a_k + B \cdot a_{k-1} \\
 &\leq A \cdot (A + B)^k + B \cdot (A + B)^{k-1} && \text{(by I.H.)} \\
 &= A(A + B) \cdot (A + B)^{k-1} + (A + B)^{k-1} \\
 &= (A(A + B) + 1) \cdot (A + B)^{k-1} \\
 &= (A^2 + AB + 1) \cdot (A + B)^{k-1} \\
 &\leq (A + B)^2 \cdot (A + B)^{k-1} && (*) \\
 &= (A + B)^{k+1}
 \end{aligned}$$

(*) $A^2 + AB + 1 \leq A^2 + 2AB + B^2 \iff 0 \leq B^2 + AB - 1 \iff B(A + B) \geq 1$.

which shows $a_{k+1} \leq (A + B)^{k+1}$.

Therefore, by the Principle of Mathematical Induction, $a_n \leq (A + B)^n$ for all $n \geq 0$. $\square$

There are still some issues.

- $M = A + B$ is an upper bound if $A \geq 1$ and $B \geq 1$. What if $A < 1$ or $B < 1$ or both?
- $M = A + B$ is not the tightest upper bound. Can we find a better bound?

Maybe if we can find a closed form for $\{a_n\}_{n=0}^{\infty}$ then tackling these problems may become easier.

11.2.1 Finding a Closed Form

Suppose $a_n = c\lambda^n$ for some $\lambda \in \mathbb{R}$. Substituting this guess gives,

$$c\lambda^n = A(c\lambda^{n-1}) + B(c\lambda^{n-2})$$

Dividing both sides by $c\lambda^{n-2}$ gives the characteristic equation

$$\lambda^2 - A\lambda - B = 0$$

Using the quadratic formula we get

$$\lambda = \frac{A \pm \sqrt{A^2 + 4B}}{2}$$

There are three cases to consider:

- If $A^2 + 4B = 0$ then $\lambda = \frac{A}{2}$ which gives the general form

$$a_n = c_1 \left(\frac{A}{2}\right)^n + c_2 n \left(\frac{A}{2}\right)^n$$

Using the initial condition $a_0 = 0$, we get

$$0 = c_1$$

and using $a_1 = 1$, we get

$$1 = c_2 \left(\frac{A}{2}\right) \iff c_2 = \frac{2}{A}$$

Therefore,

$$a_n = \frac{2n}{A} \left(\frac{A}{2}\right)^n = n \left(\frac{A}{2}\right)^{n-1}$$

- If $A^2 + 4B > 0$ then $\lambda_1 = \frac{A + \sqrt{A^2 + 4B}}{2}$ and $\lambda_2 = \frac{A - \sqrt{A^2 + 4B}}{2}$ which gives the general form

$$a_n = c_1 \left(\frac{A + \sqrt{A^2 + 4B}}{2}\right)^n + c_2 \left(\frac{A - \sqrt{A^2 + 4B}}{2}\right)^n$$

Using the initial condition $a_0 = 0$, we get

$$0 = c_1 + c_2$$

and using $a_1 = 1$, we get

$$1 = c_1 \left(\frac{A + \sqrt{A^2 + 4B}}{2}\right) + c_2 \left(\frac{A - \sqrt{A^2 + 4B}}{2}\right)$$

and solving the system gives

$$c_1 = \frac{1}{\sqrt{A^2 + 4B}}, c_2 = -\frac{1}{\sqrt{A^2 + 4B}}$$

Therefore,

$$a_n = \frac{1}{\sqrt{A^2 + 4B}} \left(\frac{A + \sqrt{A^2 + 4B}}{2} \right)^n - \frac{1}{\sqrt{A^2 + 4B}} \left(\frac{A - \sqrt{A^2 + 4B}}{2} \right)^n$$

- If $A^2 + 4B < 0$ then $\lambda_1 = \frac{A + \sqrt{-(A^2 + 4B)}i}{2}$ and $\lambda_2 = \frac{A - \sqrt{-(A^2 + 4B)}i}{2}$ which gives the general form

$$a_n = c_1 \left(\frac{A + \sqrt{-(A^2 + 4B)}i}{2} \right)^n + c_2 \left(\frac{A - \sqrt{-(A^2 + 4B)}i}{2} \right)^n$$

There is a more compact way to write our solution in terms of real numbers.

Recall that $\alpha + \beta i = re^{i\theta}$ where $r = \sqrt{\alpha^2 + \beta^2}$ and $\theta = \arctan\left(\frac{\beta}{\alpha}\right)$. Hence

$$\frac{A}{2} + \frac{\sqrt{-(A^2 + 4B)}}{2}i = \sqrt{\left(\frac{A}{2}\right)^2 + \left(\frac{\sqrt{-(A^2 + 4B)}}{2}\right)^2} e^{\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)i} = \sqrt{-B} e^{\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)i}$$

and

$$\frac{A}{2} - \frac{\sqrt{-(A^2 + 4B)}}{2}i = \sqrt{\left(\frac{A}{2}\right)^2 + \left(\frac{\sqrt{-(A^2 + 4B)}}{2}\right)^2} e^{-\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)i} = \sqrt{-B} e^{-\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)i}$$

which gives

$$\begin{aligned} a_n &= c_1 \left(\frac{A + \sqrt{-(A^2 + 4B)}i}{2} \right)^n + c_2 \left(\frac{A - \sqrt{-(A^2 + 4B)}i}{2} \right)^n \\ &= c_1 \left(\sqrt{-B} e^{\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)i} \right)^n + c_2 \left(\sqrt{-B} e^{-\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)i} \right)^n \end{aligned}$$

Now apply DeMoivre's Theorem $(re^{i\theta})^n = r^n(\cos(n\theta) + i\sin(n\theta))$ to get

$$a_n = (\sqrt{-B})^n \left(C_1 \cos\left(\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)n\right) + C_2 \sin\left(\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)n\right) \right)$$

(Note we set $C_1 = c_1 + c_2$ and $C_2 = i(c_1 - c_2)$).

Using the initial condition $a_0 = 0$, we get

$$0 = C_1$$

and using $a_1 = 1$, we get

$$\begin{aligned} 1 &= (\sqrt{-B}) \left(C_2 \sin\left(\arctan\left(\frac{\sqrt{-(A^2 + 4B)}}{A}\right)\right) \right) \\ \frac{1}{(\sqrt{-B})} &= \left(C_2 \cdot \frac{\sqrt{-(A^2 + 4B)}}{2A\sqrt{-\frac{B}{A^2}}} \right) \\ C_2 &= \frac{2A}{\sqrt{-A^2(A^2 + 4B)}} \end{aligned}$$

Therefore,

$$\begin{aligned} a_n &= \frac{2A}{\sqrt{-A^2(A^2 + 4B)}} (\sqrt{-B})^n \sin \left(\arctan \left(\frac{\sqrt{-(A^2 + 4B)}}{A} \right) n \right) \\ &= \left(\frac{2A (\sqrt{-B})^n}{\sqrt{-A^2(A^2 + 4B)}} \right) \sin \left(\arctan \left(\frac{\sqrt{-(A^2 + 4B)}}{A} \right) n \right) \end{aligned}$$

The above three cases hold for all real A, B as long as $A \neq 0$. If $A = 0$ then $a_n = B \cdot a_{n-2}$ and

$$a_n \begin{cases} 0 & \text{if } n = 2k \\ B^k & \text{if } n = 2k + 1 \end{cases}$$

Now we can find the tightest upper bound for all real numbers A and B .

- If $A^2 + 4B = 0$ then let $M =$
- If $A^2 + 4B > 0$ then let $M = \frac{A + \sqrt{A^2 + 4B}}{2}$
- If $A^2 + 4B < 0$ then let $M =$
- If $A = 0$ then let $M = B$

11.3 Connection between Stirling Numbers

Recall that the Stirling Numbers of the Second Kind satisfy

$$\sum_{k=0}^n S(n, k)(x)_k = x^n$$

It turns out that signless Stirling Numbers of the First Kind satisfy

$$\sum_{k=0}^n c(n, k)x^k = x(x+1)\dots(x+n-1)$$

Replacing x by $-x$ and multiplying both sides by $(-1)^n$ gives

$$\sum_{k=0}^n (-1)^n c(n, k)(-x)^k = (-1)^n (-x)(-x+1)\dots(-x+n-1)$$

$$\sum_{k=0}^n (-1)^{n-k} c(n, k)x^k = (-1)^{2n} x(x-1)\dots(x-n+1)$$

which means that the Stirling Numbers of the First Kind satisfy

$$\sum_{k=0}^n s(n, k)x^k = (x)_n$$

Notice that $S(n, k)$ translates expressions from $(x)_k$ to expressions in x^n whereas $s(n, k)$ translates expressions from x^k to $(x)_n$. Hence, the two kinds of Stirling numbers $S(n, k)$ and $s(n, k)$ are in some sense inverses of one another. Let's make this a bit more concrete.

The set of all polynomials with real coefficients forms a vector space V . The standard basis for V is $B = \{1, x, x^2, \dots\}$. Another basis for V is $B' = \{1, (x)_1, (x)_2, \dots\}$ (since all the degrees are distinct).

Let S be the infinite matrix whose entry in position (n, k) is $S(n, k)$:

$$S = \begin{bmatrix} S(0,0) & S(0,1) & S(0,2) & S(0,3) & \dots \\ S(1,0) & S(1,1) & S(1,2) & S(1,3) & \dots \\ S(2,0) & S(2,1) & S(2,2) & S(2,3) & \dots \\ S(3,0) & S(3,1) & S(3,2) & S(3,3) & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots \\ 0 & 1 & 0 & 0 & \dots \\ 0 & 1 & 1 & 0 & \dots \\ 0 & 1 & 3 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Let s be the infinite matrix whose entry in position (n, k) is $s(n, k)$:

$$s = \begin{bmatrix} s(0,0) & s(0,1) & s(0,2) & s(0,3) & \dots \\ s(1,0) & s(1,1) & s(1,2) & s(1,3) & \dots \\ s(2,0) & s(2,1) & s(2,2) & s(2,3) & \dots \\ s(3,0) & s(3,1) & s(3,2) & s(3,3) & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & \dots \\ 0 & 1 & 0 & 0 & \dots \\ 0 & -1 & 1 & 0 & \dots \\ 0 & 2 & -3 & 1 & \dots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Then S is the change of basis matrix from B' to B whereas s is the change of basis matrix from B to B' . In

other words, the two matrices are inverses of one another which gives

$$sS = Ss = I$$

CHAPTER 12

REFLECTION

In this final chapter, I take some time to reflect on my walk through combinatorics.

12.1 Goals

At the start of my walk, I had the following three broad goals in mind:

1. Want to gain a strong grasp on the fundamental concepts in combinatorics.
2. Want to further develop as a professional mathematician.
3. Want to create a very impressive portfolio documenting my journey.

I am proud to say that I have accomplished a lot more than I expected.

12.1.1 Learning Strategy

Concepts related to combinatorics have shown up multiple times in my life, and I always found it hard to grasp this area of mathematics. For instance, the fact that a set with n elements has 2^n subsets has shown up in multiple of my computer science courses, and the use of basic counting formulas has shown up in multiple of my probability and statistics courses. Hence, I want to build a strong foundation in this area of mathematics with a clear picture of how the fundamental concepts connect with one another.

Fortunately, this course provided me with the opportunity to build this foundation by playing around and exploring concepts on my own.

My approach to gain a strong grasp on the fundamental concepts in combinatorics included:

- Attending lectures and tutorials each week in which I actively participated, collaborated, and asked questions in.
- Reading the “A Walk Through Combinatorics” by Miklós Bóna. This book was a bit challenging to read, but because of my efforts and willingness to read and understand difficult mathematics, I was able to read through it. I strongly believe that reading math textbooks is one the best ways to get a good grasp on the material (it’s also a great challenge to read a math textbook, and I love challenges).

Moreover, this textbook had many exercises with full solutions which was a huge opportunity for me to learn. I was always motivated to complete problems since I knew if I got stuck, I could get unstuck by simply looking at the solutions.

- Exploring concepts on my own in further detail. After learning the fundamentals, I would usually explore Wikipedia, Mathematics Stack Exchange, and YouTube to learn more about a concept. This helped me connect things together and also gave me good ideas for further problems to approach.

Overall, I feel confident in my understanding of several key concepts in combinatorics. For instance, I have a good grasp on the basics of set partitions and could explain the concept to someone who was unfamiliar with it.

Looking back on my learning process, I realize that I learn the best when I have genuine interest and passion for the material. When I find a concept particularly fascinating, I am more likely to ask follow-up questions and explore additional resources on my own. I plan to keep this in mind for my future courses.

12.1.2 Developing as a Mathematician

To further develop as a professional mathematician, I challenged myself to approach difficult problems and pushed myself to think creatively. Moreover, after solving a problem, I often asked follow-up questions or tried to generalize my results, which helped me deepen my understanding of the concepts.

Additionally, I put great amounts of effort into formulating well-written solutions by myself. This required me to reason through tough definitions and notation. Furthermore, I would double check my work with tools like WolframAlpha and OEIS to ensure precision and accuracy. I also polished all my solutions and presented them in a clear and presentable format, paying close attention to small details.

I also collaborated with other students to tackle problems and concepts which was a valuable experience as it allowed me to develop my communication and teamwork skills. I also proved results that were formulated by fellow classmates. This gave me an opportunity to practice and develop my problem solving skills on never-seen problems.

Overall, I have made good progress towards becoming a professional mathematician. I have developed critical skills such as problem-solving, reasoning, and collaboration, and I am more confident in my ability to approach complex mathematical problems and statements.

12.1.3 Documenting my Journey

Professional mathematicians need to be able to publish their work and findings in a clear and presentable format. Since one of my main goals was to further develop as a mathematician, I thought it would be a great idea to document my work as a book using using $\text{\LaTeX}$.

Creating, organizing, and managing such a large project was a very challenging task. However, I now have many $\text{\LaTeX}$ commands engraved in my brain.

When planning my portfolio, I had to think carefully about how to present the material in a clear and visually appealing way. I included detailed solutions to challenging problems, diagrams to help explain concepts, and even learned the basics of TikZ to create custom visuals. These efforts improved my ability to communicate mathematical ideas.

One of the reasons I chose to create the portfolio in $\text{\LaTeX}$ was to prepare myself for graduate school, where I know I will have to write reports and a thesis. By challenging myself to create a book early on, I gained valuable experience in organizing a large project and using technical tools to present information effectively.

12.2 Arising Themes

Looking back at the concepts I learned and the problems I worked on, I have noticed the following ideas emerge often

- To prove many identities or to derive many recurrence relations, we would single out a special element (e.g. the $n + 1$ element) and consider cases.
- Sometimes counting the size a set S directly would be too hard and so we would instead count the size of another set S' and argue why the two sets have the same size.
- Answering a problem in multiple ways would establish identities.
- Many times we would set $x = -1$ in a generating function expression and deduce that the sum of the terms corresponding to odd indices would be equal to or differ by ± 1 from the sum of the terms corresponding to even indices.

12.3 Conclusion

One thing that I appreciate even more now is that mathematics is not just about finding the correct answer to a problem, but it's about the process in which we arrive at a solution. The ability to solve complex problems using different approaches is a very fascinating quality of humans.

Finally, I would like to acknowledge the support I received from my instructor and classmates during this course. Their feedback, passion, and encouragement helped me to stay motivated and focused throughout the semester, and I am grateful for the opportunity to learn alongside such talented and dedicated mathematicians.

Overall, I am proud of the high-quality work that I produced during this course. I pushed myself to the fullest extent and tackled some of very challenging problems. Through this experience, I have gained confidence in my abilities as a mathematician and am excited to continue exploring the fascinating field of combinatorics.